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PROJECT REPORT

*ASSESSMENT OF POTENTIAL HEAVY METAL POLLUTION OF GROUNDWATER IN A
MINING COMMUNITY- A CASE STUDY FROM HIA, AMANSIE CENTRAL DISTRICT.*

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ABSTRACT

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1.1 BACKGROUND

Water is an essential, inorganic, and polar chemical compound (H_2O) composed of two hydrogen atoms bonded covalently to one oxygen atom. Water resources are essential to maintaining ecosystem balance and fostering a harmonious development of human society, the environment, and the economy (Sanad et al., 2024). 3% of the total water on earth is freshwater and from this 3%, 95% constitutes groundwater serving as a vital freshwater source valued for its broad accessibility, safety, and cleanliness. Approximately 2.5 billion people globally rely exclusively on groundwater for their daily drinking needs (Sanad et al., 2024). Moreover, groundwater plays an important role in providing water for agricultural and industrial purposes, with about 43% of agricultural water sourced from it (Sanad et al., 2024).

Groundwater is one of the most important freshwater resources for domestic purposes, agriculture, and industrial activities worldwide, particularly in regions facing surface water scarcity and rapid population growth (Kana, 2022; Rezaei et al., 2019). Although groundwater is commonly perceived as safer than surface water due to natural filtration processes within aquifer systems, its quality has deteriorated considerably over recent decades. This decline is largely attributed to intensified anthropogenic activities such as urbanization, industrialization, agriculture, mining, and inadequate waste management practices (Kana, 2022; Venkatramanan et al., 2015). The resulting degradation of groundwater quality poses serious risks to public health, ecosystem integrity, and sustainable socio-economic development (Sanad et al., 2024). Among the various contaminants affecting groundwater systems, heavy metals are of particular concern due to their toxicity, persistence, non-biodegradable nature, and capacity to bioaccumulate in living organisms (Kana, 2022; Venkatramanan et al., 2015).

Heavy metals are naturally occurring elements with high atomic weight and a density at least five times greater than water. Heavy metals frequently reported in groundwater include mercury (Hg), manganese (Mn), nickel (Ni), cadmium (Cd), iron (Fe), zinc (Zn), arsenic (As), and lead (Pb), originating from both natural and anthropogenic sources (Rezaei et al., 2019). Natural inputs primarily result from geological weathering, mineral dissolution, atmospheric precipitation, infiltration of surface runoff, and volcanic activity, whereas anthropogenic contributions are associated with agricultural runoff, industrial effluents, sewage discharge, landfill leachates, mining, and mineral processing activities (Kana, 2022; Venkatramanan et al., 2015).

Assessment of the activities that introduce heavy metals into the ground are of high importance because of its adverse effects on human life, animals and many others since the groundwater is tapped for domestic purposes, and by plants roots for growth. Although some of these heavy metals such as Cr, Pb, Hg, Cd, As, and Co have no useful effects in the body system, long time exposure may cause more acute interruptions in the normal operations of the human organ systems where the metals are deposited (Chaturvedi et al., 2018). Other metals like Zn and Fe for example, are considered micronutrients required for normal growth and functioning of the human body; but at higher concentrations, they become toxic (Dagdag et al., 2023; Kana, 2022).

The Amansie district (Central, West and South) in the Ashanti Region of Ghana are underlain by three geological formations, primarily dominated by the Birimian Supergroup and the Tarkwaian system.

In Ghana, groundwater resources are increasingly threatened by intensive anthropogenic activities, particularly those related to mining activities. The Amansie District, located within the mineral-rich Ashanti Region, has experienced widespread artisanal and small-scale gold mining activities in recent years. These activities, commonly referred to as illegal mining or *galamsey*, involve the excavation and processing of auriferous materials using methods that often lack systems that consider how these heavy metals affect the environment.

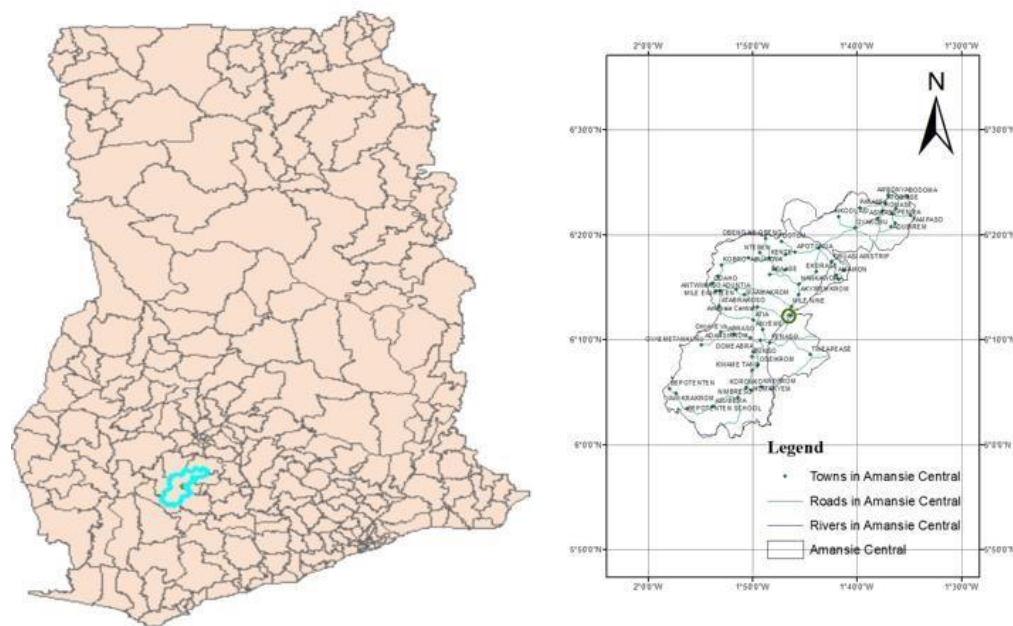


Figure 1: Study area of Hia, Amansie Central.

To evaluate groundwater contamination by heavy metals in the study area, integrated analytical, statistical, and index-based approaches will be used. In this study, pollution indices including the Heavy Metal Pollution Index (HPI), Heavy Metal Evaluation Index (HEI), Metal Index (MI), Degree of Contamination (Cd), and Pollution Index (PI) are employed to assess overall groundwater quality and contamination severity (Rezaei et al., 2019; Venkatramanan et al., 2015). In addition, multivariate statistical techniques such as correlation analysis, principal component analysis (PCA), and cluster analysis (CA) are used to elucidate relationships among heavy metals, differentiate between geogenic and anthropogenic sources, and identify controlling hydrogeochemical processes. Geographic Information System (GIS)-based spatial analysis is

further applied to enhance the interpretation of spatial contamination patterns and potential pollution hotspots across the Amansie District (Sanad et al., 2024; Sbai et al., 2024).

1.2 PROBLEM STATEMENT

Hia in the Amansie Central District is a galamsey dominated area and due to the chemicals involved such as arsenic, mercury, lead, cadmium, zinc, iron, manganese, and nickel this activity can potentially introduce heavy metals into groundwater. Due to the adverse effect of this activity on human life, there is a need to assess the potential extent of heavy metal pollution of groundwater in the Amansie District.

1.3 JUSTIFICATION

Groundwater is a vital water source for many communities in the Amansie District of Ghana. Mining (galamsey), industrialization, and agricultural activities are potential sources that introduce heavy metals into the environment including groundwater. Understanding the extent and sources of the heavy metals that potentially pollute the groundwater can help policies to be made to continually check the rate of groundwater contamination by these heavy metals.

4 AIM AND SPECIFIC OBJECTIVES

1.4.1 AIM

The objective of this project is to evaluate the potential for heavy metals (Mn, Hg, Ni, Cd, Fe, Zn, As, Pb) pollution of groundwater in Amansie.

1.4.2 SPECIFIC OBJECTIVES

The specific objectives are to

determine the levels and spatial distribution of heavy metals in Groundwater,

determine the degree of heavy metal pollution using various pollution indices and

(iii) Identify the potential sources of heavy metals through multivariate statistical analysis.

CHAPTER 2: LITERATURE REVIEW

STUDY AREA

Geographical location and size (Insert heading)

The study area, Hia, is located within the Amansie Central District in the Ashanti Region. The district lies approximately between 6° 00'N and 6° 30'N and 1° 15'W and 2° 00'W (Statistical Service, 2022; Tay et al., 2018). It covers a land area of about 710 km² with a current population density of 138.7 persons per km² and has about 206 settlements with Jacobu as the administrative capital (Statistical Service, 2022). Amansie Central District shares common boundaries with Amansie East to the northeast, Amansie West to the west, Obuasi District Assembly to the southeast, Adansi North to the east, Adansi South to the south, and Upper Denkyira in the Central Region to the south (Statistical Service, 2022). The population of the district was recorded as 90,741 which represents the inter-censal growth rate of 1.02% between the years 2000 and 2010. This constitutes 45,466 females and 45,275 males representing 50.2% and 49.8% of the population. The Amansie District is dominated by an agricultural sector which employs about 78% of the labour force in the district (Statistical Service, 2022). However, due to relatively small farm sizes and low yields, poverty among farmers is very high hence they cultivate only few crops and cash crops. In addition to agriculture, artisanal and small-scale mining activities are widespread due to the presence of gold-bearing formations within the district. Water supply in the district is mainly derived from groundwater sources, including small town water systems, boreholes, hand-dug wells and rivers/ streams which serve as the main source of water for both industrial and domestic purposes in the district. Among these sources, about 88% of people rely on boreholes and 9.5% rely on hand dug wells (Statistical Service, 2022). This high dependence on groundwater highlights the importance of assessing its quality, especially in areas affected by mining activities.

GEOLOGY

The District is predominantly underlain by the Birimian Supergroup which consists of metasedimentary and metavolcanic units (Tay et al., 2021, 2019). Even though site-specific geological data for Hia is limited, the geology of the area can be inferred from that of the Amansie Central District and surrounding regions. The Tarkwaian System also occurs locally, consisting mainly of sedimentary rocks such as quartzites and conglomerates (Boansi et al., 2021; Tay et al., 2021). The Birimian and the Tarkwaian are the two gold- bearing Lower Proterozoic stratigraphic units in West Africa.

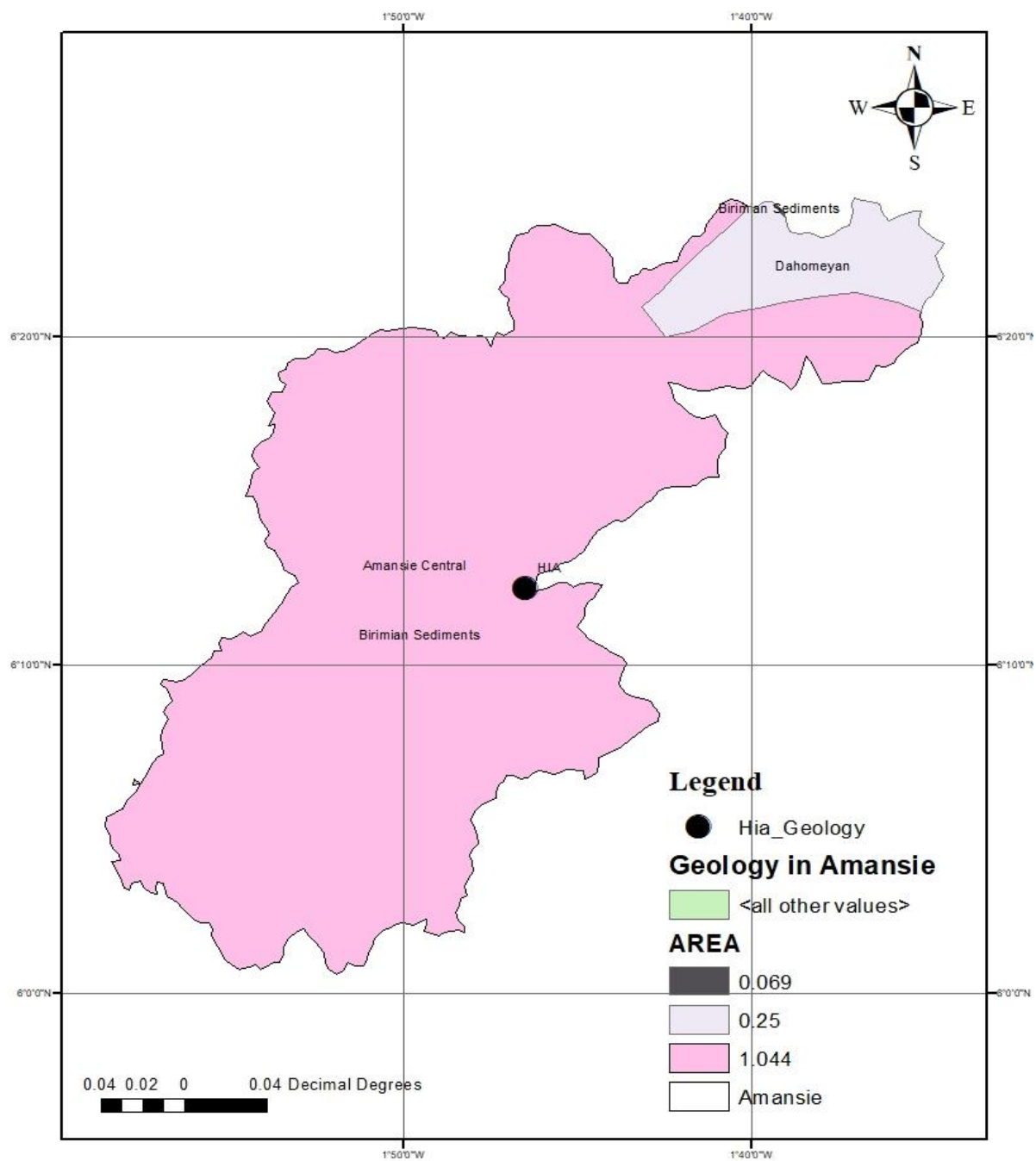


Figure 2: Map of geology of the study area.

The Birimian consists of two distinct lithologic associations that are generally referred to as the Lower and Upper Birimian (Tay et al., 2021). Lower Birimian rocks comprise an assemblage of fine-grained rocks with a large volcanoclastic component. Typical lithologies include tuffaceous shale, siltstone, greywacke and some chemical (Mn-rich) sediments. Upper Birimian rocks comprise mostly basalts with some interflow sediments. These rocks are predominantly volcanic and pyroclastic (Dzlgbodl-adjlmah, 1993; Leube et al., 1990). The rocks consist of a bedded group of tuffs, sediments and mafic lavas (greenstones), jointly with minor bands of phyllite that include a zone of manganiferous phyllites containing manganese ore. Batholithic masses of granite and gneiss intrude the sequence. The mostly argillaceous sediments in this series have been metamorphosed to schist, slate and phyllite jointly with some interbedded greywacke. A major unconformity separates the Birimian Supergroup from the Tarkwaian system. The term Tarkwaian refers to clastic sedimentary rocks representing erosion products of the Birimian provenance. The Tarkwaian consists of coarse clastic sedimentary rocks that include conglomerates, arkoses, sandstones and minor amounts of shale. Pebble clasts in the conglomerate include volcanic and sedimentary clasts derived from Birimian rocks and granitoid. The Tarkwaian is usually regarded as the detritus of Birimian rocks that were uplifted and eroded following the Eburnean tectono-thermal event (Dzlgbodl-adjlmah, 1993; Leube et al., 1990). The geological setting of the area is closely related to gold mineralization. The styles of gold mineralization differ. Birimian deposits are epigenetic gold quartz vein deposit-types, associated with disseminated sulfides in fractured and sheared zones. Tarkwaian gold, on the other hand, is in paleoplacer Banket conglomerates of the Witwatersrand-type (Dzlgbodl-adjlmah, 1993). Tarkwaian deposits are generally interpreted to be derived from the Birimian. This has led to widespread artisanal and small-scale mining activities in areas underlain by these formations. Such mining activities can result in the release of heavy metals into the environment, thereby influencing groundwater quality, particularly in crystalline basement terrains where groundwater occurs within weathered and fractured zones.

HYDROGEOLOGY AND DRAINAGE

Hia is predominantly underlain by the Birimian supergroup which consists of metavolcanic and metasedimentary aquifers as well as the Tarkwaian whereas the Crystalline Basement Granitoid complexes are made up of granitoid aquifer (Boansi et al., 2021). Both aquifers are dominated by low primary porosities and permeabilities. Hydrogeologically, both the Birimian and granitoid aquifers are characterized by low porosity and permeability; secondary porosities are also observed which arises from weathering and the activities associated with the eburnean orogeny (Boansi et al., 2021). This has led to an increase in the pores and fractures within the aquifers. Groundwater occurrence in the region is controlled by factors such as topography, lithology, and geologic structures. Groundwater occurs in the saprolite, saprock, and fractured bedrock in the Crystalline basement complex. The occurrence of groundwater in the Birimian province is more defined in the lower part of saprolite and the upper part of the saprock. These two regolith

profiles are hydraulically connected with their storage and permeability (Boansi et al., 2021). The Birimian exhibits a higher degree of weathering than the granitoids due to the lower jointing and fracturing contained in the granitoids. This has consequently resulted in lower groundwater yields and shallower ground water tables in granitic terrains. The presence of quartz veins or pegmatites in Birimian Province has been documented to possess higher quantities of groundwater. On the basis of long-term borehole drilling records collated from 1991-2011, the Birimian has an average well yield of 30–60 l/min and a higher drilling success rate (approximately 91%) than the granitoids aquifer with an average borehole yield of less than 30 l/min (Boansi et al., 2021; Osei-Nuamah, I., 2024). The hydrogeological characteristics influence the drainage and recharge processes within area. Hia lies in the Pra River Basin and is drained by a network of local streams that form tributaries of major rivers such the Offin River, which ultimately feeds into the Pra River.

VEGETATION

Hia lies in Ghana's (Moist Semi-Deciduous Forest Zone) forest ecological zone. This zone is critical to the country's biodiversity and ecosystem services (Donkor et al., 2023). Moreover, the landscape surrounding Hia has been profoundly altered by both large-scale mining, artisanal and small-scale mining (galamsey).

The regional vegetation context

The Ashanti region's natural vegetation is characterized by tall, multi-layered forest structures. In undisturbed areas, the flora includes high value timber and hardwood tree species such as Odum (*Milicia excelsa*), Wawa (*Triplochiton scleroxylon*), Mahogany, Edinam, and Sapele, which are common across the district (Asamoah, 2025). These forests support farming activities, especially cocoa, oil palm, and food crops, making the area largely agrarian and also used as medicines (Asamoah, 2025).

Mining activities in Amansie districts has led to a 304% escalation in land conversion and deforestation, with 30% and 86% of their original forest cover (Universitesi et al., 2026). As mining removes the top soil and introduces heavy metal like Mercury (Hg), Arsenic (As), Lead (Pd), which prevents the natural plants regrowth (Glazebrook et al., 2020).

CLIMATE

Hia is characterized by a wet semi-equatorial climate (Tay et al., 2019). This climatic zone is defined by a double rainfall maximum regime, which is critical for the region's socio-economic structure. The major rainy season typically occurs between March and July, followed by a minor rainy season from September to November (Tay et al., 2019).

Quantitative data indicates that the district receives a mean annual rainfall ranging from 855 mm to 1,500 mm (Tay et al., 2019). Other regional profiles suggest total annual rainfall in similar adjacent areas may fall within the 800 to 900 mm range during drier cycles, highlighting internal variability (Ghana Districts, 2025). The thermal regime is remarkably stable, with mean monthly temperatures averaging approximately 27°C (Tay et al., 2019).

Vegetation and Environmental Interaction

The climatic conditions directly support a moist semi-deciduous forest vegetation type, which facilitates the cultivation of major cash crops such as cocoa and citrus, as well as food crops like plantain and vegetables (Tay et al., 2019). However, the natural environment and its role in local climate regulation are under threat. Illegal mining (locally known as *galamsey*) have significantly degraded the natural forest cover, including vital reserves such as the Odaho, Apanprama, and Aboaboso Forest Reserves (Tay et al., 2019).

GROUNDWATER QUALITY AND CHEMISTRY

Groundwater quality comprises the physical, chemical, and biological qualities of groundwater. The physical qualities of groundwater include parameters such as temperature, turbidity, colour, taste, and odor whereas groundwater is mostly colourless and odourless hence its biological and chemical properties are of more importance (Harter, 2003).

The usefulness of groundwater for various purposes is determined by its chemical, physical and bacterial characteristics (Ojo et al., 2012). Chemical analysis of groundwater includes the determination of the concentration of inorganic constituents as well as the measurement of pH and specific electrical conductance (EC) (Ojo et al., 2012).

Groundwater contains ions which slowly dissolve from soil particles, sediments, and rocks as the water travels along mineral surfaces in the pores or fractures of the unsaturated aquifer (Harter, 2003). Naturally, groundwater is highly pure and frequently demands little or no treatment be safe (MacDonald & Calow, 2009). However, anthropogenic activities can change the natural composition of groundwater by injecting chemicals and waste into the soil which can significantly alter the natural composition of groundwater. (A. Edet, 2012; Harter, 2003).

GROUNDWATER PHYSICO-CHEMICAL PARAMETERS

Assessing the quality of groundwater for drinking and other essential purposes depends largely on a set of chemical indicators, namely pH, Total Dissolved Solids (TDS), electrical conductivity, and salinity, each of which offers insight into the suitability of water for human use.

pH is essentially a logarithmic measure of how acidic or basic a solution is. A reading of 7 marks the neutral point; values below this indicate acidity, while values above it indicates alkalinity.

Several factors can push groundwater away from this neutral state, including seasonal variation between wet and dry periods, as well as agricultural runoff containing fertilizers and pesticides that seep into the soil and eventually reach the water table (Abanyie et al., 2023; Gautam et al., 2022).

TDS represents the cumulative concentration of minerals, salts, and organic substances dissolved within a water sample, and it has a direct bearing on taste. This dissolved load is generally composed of two ion categories: cations such as calcium, magnesium, sodium, and potassium, and anions such as carbonate, bicarbonate, chloride, sulphate, and nitrate (Riaz et al., 2017). A higher TDS means that there are more cations and anions in the water. With more ions in the water, the water's electrical conductivity (EC) increases. By measuring the water's electrical conductivity, we can indirectly determine its TDS concentration. At a high TDS concentration, water becomes saline. Water with a TDS above 500 mg/l is not recommended for use as drinking water (EPA secondary drinking water guidelines). Water with a TDS above 1,500 to 2,600 mg/l (EC greater than 2.25 to 4 mmho/cm) is generally considered problematic for irrigation use on crops with low or medium salt tolerance (Harter, 2003)

Electrical conductivity (EC) gauges water's capacity to carry an electrical current and serves as a proxy for the overall concentration of dissolved ions, effectively mirroring TDS levels. Because of this relationship, EC is widely regarded as a dependable and convenient indicator of water purity (Kumar & Kumar, 2013; Sajitha & Vijayamma, 2016). EC is directly related to TDS as it measures ionic content of water sample which determines its ability to conduct an electric current. As TDS concentration of water sample increases, the ionic strength also increases (Taşan et al., 2022). It is typically measured in microsiemens per centimeter using a dedicated conductivity meter.

Salinization is a significant environmental problem that degrades soil and water resources, reduces agricultural productivity, and disrupts natural ecosystems. It is also associated with elevated groundwater concentrations of harmful elements such as sodium, sulphate, boron, fluoride, selenium, and arsenic, as well as increased radioactivity (Kulshrestha, 2019). Shallow aquifers are vulnerable to natural geogenic processes along with anthropogenic influences, and this is particularly evident in desert regions. Within arid and hyperarid climates, evaporation is a controlling hydrologic process causes a significant increase in the concentration of dissolved minerals of both surface water and groundwater. In groundwater, this increase is not only dependent on shallow water table depth, but also on the hydraulic properties of sediments present within the unsaturated zone of the aquifer itself (Medjani et al., 2021).

Turbidity reflects how clear or cloudy a liquid appears, influenced by temperature as well as the presence of suspended particles, dissolved substances, and organic or biological material (Kitchener et al., 2017). Measured in Nephelometric Turbidity Units (NTU) through nephelometric techniques, turbidity levels rise and fall in step with the concentration of

suspended matter present. From a treatment standpoint, elevated turbidity creates favorable conditions for bacterial growth, raises chlorine requirements, and can shield pathogens from disinfection processes, making this parameter important for guiding treatment decisions (World Health Organization, 2008). Higher turbidity affects life indirectly by cutting the light utilized by phytoplankton growth (Soceanu et al., 2021). Geological conditions such as steep terrain, karst formations, and soils prone to erosion can intensify turbidity, since heavy rainfall washes dissolved material through rock crevices and directly into groundwater systems (Ford & Williams, 1989; Kaufmann & Braun, 2000).

Rising temperatures also encourage faster microbial growth, and certain microorganisms produce by-products that give water an unpleasant taste. Temperature further influences how intensely odors are perceived, owing to its effect on vapor pressure, meaning that odor assessments are inherently temperature-sensitive (Ojo, 2012). It also sets the boundaries within which other inorganic contaminants remain tolerable (World Health Organization, 2008). As a general guideline, water intended for household consumption is most palatable when its temperature stays under 10°C (Water Temperature—Influential Factors, Field Measurement, and Data Presentation, 1975).

In terms of consumer acceptability, drinking water is expected to be free of any noticeable taste or smell that the public might find unpleasant. Because people rely on their own senses to judge water quality, the physical, chemical, and biological characteristics that shape taste and odor play a major role in perceived safety. Even when such aesthetic concerns pose no direct health risk, an unpleasant taste or smell is often enough to make consumers distrust the water and avoid using it (World Health Organization, 2008).

Finally, discoloration in water is usually attributed to colored organic compounds, most notably humic matter produced as surface vegetation decomposes. To quantify this, the true color of a sample is compared visually against a standard platinum-cobalt reference solution, with one color unit defined as the intensity produced by 1 mg/l of platinum in the form of chloroplatinate ion (Ojo, 2012).

2.2.2

QUALITY CONTROL

Quality control constitutes an essential pillar of any methodologically sound research endeavor, as the reliability of a study's conclusions is inherently contingent upon the precision and integrity of its underlying data. Quality assurance encompasses the proactive measures implemented prior to data collection, designed to ensure that established procedures conform to predefined standards from the beginning, whereas quality control involves the evaluative checks conducted during and immediately following data collection to verify that those standards have been effectively upheld (Taşan et al., 2022). Together, quality assurance and quality control (QA/QC) form indispensable components of contemporary analytical geochemistry. A well-structured QA/QC framework serves to identify sources of analytical error while simultaneously providing a basis for establishing confidence in, and recognizing the limitations of, analytical datasets. Such a program encompasses the continuous monitoring of precision, accuracy, and potential contamination throughout the entire process from sample collection to laboratory analysis. Precision is assessed through the systematic incorporation of sample, pulp, and analytical duplicates alongside certified reference materials, with the resultant data evaluated using scatterplots, statistical tests such as percentage relative standard deviation, Thompson-Howarth plots, and the average coefficient of variation (CV_{avg} (%)). Accuracy is determined through the submission of reference materials and assessed using statistical approaches including percentage relative difference, t-tests, and Shewhart control charts. Contamination is examined through the use of blanks, the outcomes of which are similarly tracked via Shewhart control charts (Piercey, 2014).

HEAVY METALS

Heavy metal pollution of groundwater is a serious problem for human health, as it causes toxicity and carcinogenic effects (Soceanu et al., 2021). Human health, industrialization, as well as the practices of modern agriculture, require increased attention and protection of groundwater quality but also a rational exploitation within the limits of exploitation reserves (Soceanu et al., 2021). For example, the presence of lead in water beyond the permissible level could result in hypertension, interference with Vitamin D and calcium metabolism, brain development hindrance in foetus and young children, damage to tissues and organs in human and many more (Popoola et al., 2019). In order to assess the groundwater quality, identify sources of contamination, and plan preventive measures to reduce the public health risks like, authorities will need to perform routine monitoring of groundwater (Soceanu et al., 2021).

HEAVY METAL POLLUTION IN GROUNDWATER

Heavy metal pollution of groundwater is a major environmental and public health concern, particularly in mining regions such as Hia in the Amansie District in the Ashanti Region of Ghana. Heavy metals are persistent, non-biodegradable, and can accumulate in water over a period of time, posing long-term risks to health (Hossain et al., 2025).

Studies carried out within the Amansie district show that groundwater contains significant concentrations of heavy metals such as iron (Fe), manganese (Mn), arsenic (As), zinc (Zn), copper (Cu), lead (Pb), cadmium (Cd), and mercury (Hg). The relative abundance of these metals has been reported in the order $Fe > Mn > As > Zn > Cu > Pb > Cd > Hg$, indicating dominance of iron and manganese in the groundwater system (C. K. Tay et al., 2019a). These elevated concentrations are attributed to both natural geological processes (such as weathering of mineral-rich Birimian rocks) and anthropogenic activities, particularly artisanal and small-scale mining (galamsey).

Similarly, studies in other Ghanaian mining areas such as Tarkwa have confirmed that groundwater contamination is strongly linked to mining activities, which introduce toxic elements like arsenic and lead into aquifers (Gwira et al., 2023). These metals are of particular concern due to their toxicity and potential to cause severe health effects, including organ damage and increased cancer risk.

In addition, research conducted within Amansie West indicates that mining activities significantly contribute to heavy metal contamination of environmental materials, including soils and groundwater systems (Baah et al., 2023). This suggests that groundwater pollution in the Amansie area is not only widespread but also closely tied to ongoing mining practices.

Looking at all these studies in nearby locations in the Region or country, it is indicated that heavy metal pollution in groundwater within the Amansie Central District is possibly influenced by both geogenic and anthropogenic factors, with mining activities playing a major role. Hence, there is a need for continuous monitoring and effective management strategies to safeguard water quality.

POLLUTION INDICES IN GROUNDWATER ASSESSMENT

To effectively evaluate groundwater pollution, researchers have developed various pollution indices that simplify complex hydro-chemical data into easily interpretable values. Among the most commonly used are the Water Quality Index (WQI), Heavy Metal Pollution Index (HPI), and Hazard Quotient (HQ) and Hazard Index (HI) for health risk assessment.

In the Amansie and Adansi districts, pollution evaluation indices such as HQ and HI have been used to assess the potential health risks associated with heavy metal pollution. These indices are based on standard methodologies, such as those developed by the United States Environmental Protection Agency (USEPA), and are used to determine whether groundwater consumption poses significant health risks. (C. K. Tay et al., 2019b) The application of these indices allows researchers to quantify both carcinogenic and non-carcinogenic risks associated with contaminated water.

Studies from other mining regions in Ghana, such as Tarkwa, have also applied similar indices to assess groundwater quality. Results show that although some heavy metal concentrations may fall within allowable limits, long-term exposure can still pose health risks, particularly due to bioaccumulation and long-term intake (Seidu & Ewusi, 2020). These findings highlight the importance of using pollution indices as tools for comprehensive water quality assessment.

Pollution indices are particularly useful because they combine multiple parameters into a single value, making it easier to classify water quality into categories such as safe, moderately polluted, or highly polluted. However, some studies suggest that these indices may simplify complex hydrochemical processes too much and should therefore be used alongside other analytical techniques.

MULTIVARIATE STATISTICAL METHODS IN GROUNDWATER STUDIES

Multivariate statistical techniques have become essential tools in groundwater quality assessment due to their effectiveness to analyse large and complex datasets. Commonly used methods include Principal Component Analysis (PCA) and Hierarchical Cluster Analysis (HCA).

In Ghana, these techniques have been widely applied to identify the sources and processes controlling groundwater chemistry. For instance, a study on groundwater systems in northern Ghana employed PCA and cluster analysis to determine that silicate weathering, agrochemical inputs, and domestic waste were the main factors influencing groundwater quality (Loh et al., 2020). These findings are consistent with studies conducted in the Amansie District, where groundwater chemistry is primarily controlled by silicate weathering and ion-exchange processes (C. Tay et al., 2021).

Principal Component Analysis is particularly useful in reducing data dimensionality and identifying the dominant factors influencing water quality. It helps distinguish between natural processes and anthropogenic influences such as mining. Cluster analysis, on the other hand, groups water samples based on similarities in chemical composition, allowing for the identification of spatial patterns and pollution zones.

The integration of multivariate statistical methods with hydrochemical analysis enhances the interpretation of groundwater data and improves the identification of contamination sources. These techniques are especially important in complex environments like Hia, where multiple factors possibly influence groundwater quality.

Despite their advantages, multivariate methods require careful interpretation and a solid understanding of the study area. When used alongside pollution indices and conventional hydrochemical techniques, they provide a comprehensive framework for groundwater quality assessment.

CHAPTER 3: METHODOLOGY

For the purpose of this report, secondary data will be utilized which was obtained from Kwame Nkrumah University of Science and Technology (KNUST). This data includes 19 borehole samples which were collected taking into consideration how close the inhabitants are close to the illegal mining sites.

1. DATASET SUMMARY

Table 11 to Table 12. Summary of the groundwater dataset.

Number of groundwater samples: 19

Heavy metals analysed: Cd, Fe, Mn, Ni, Pb, Zn, As, Hg

(In appendix)

2. EXPLORATORY DATA ANALYSIS TABLES

Table 1-3 Descriptive statistics of heavy metal concentrations.

Table 1.Descriptive Statistics

	Metal	Count	Mean	Median	Minimum	Maximum	Variance	Std Dev	CV (%)	Q1	Q3	IQR	Skewness	Interpretation
0	Cd	19	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	nan	
1	Fe	19	0.9334	0.3281	0.0000	7.9490	3.7416	1.9343	207.2324	0.1728	0.4654	0.2926	2.9507	The variable is highly skewed.
2	Mn	19	0.0813	0.0247	0.0073	0.8490	0.0371	0.1927	237.0003	0.0127	0.0477	0.0350	3.6107	The variable is highly skewed.
3	Ni	19	0.0083	0.0009	0.0000	0.0296	0.0001	0.0106	127.8669	0.0000	0.0171	0.0171	0.7444	The variable exhibits moderate skewness.
4	Pb	19	0.0166	0.0107	0.0013	0.0679	0.0003	0.0171	103.2249	0.0045	0.0187	0.0142	1.6668	The variable is highly skewed.
5	Zn	19	0.0251	0.0000	0.0000	0.2567	0.0035	0.0590	234.8702	0.0000	0.0222	0.0222	3.4314	The variable is highly skewed.
6	As	19	0.0029	0.0027	0.0015	0.0059	0.0000	0.0011	38.7088	0.0022	0.0031	0.0009	1.3755	The variable is highly skewed.
7	Hg	19	0.0050	0.0050	0.0041	0.0069	0.0000	0.0006	12.7541	0.0046	0.0052	0.0006	1.2268	The variable is highly skewed.

3. EXPLORATORY DATA ANALYSIS

Visualizations that are used to explain Statistical data description can go as far as from histogram, bar chart, Box plot, violin plots, pie chat, Q-Q plots etc. For the purpose of this report histograms, Boxplots and Q-Q plots, were used. Boxplots and Q-Q plot are placed in the Appendix with violin plots of some prominent metals in this study.

Histograms

Boxplots

Histograms

The histograms show the frequency distribution of the Heavy Metals' concentrations in groundwater samples. It helps identify concentration pattern, data spread, skewness, outliers, possible contamination behavior.

The smooth red curves in the “Before” and green in the “After” represent the density distribution (KDE), showing the overall trend of Heavy Metals' concentrations. The red curves in the “After” represents the (KDE) after normalizations.

X-axis represents:

Heavy Metals concentration values (mg/L).

Y-axis represents:

frequency (number of samples).

The curves represents:

the distribution trend (density curve) either after or before normalization.

This helps visualize skewness, concentration patterns, and data spread.

Boxplots

The boxplot summarizes the statistical distribution of Heavy Metals concentrations in groundwater samples. It shows median concentration, spread of the data, variability, outliers, skewness.

This plot is very important for identifying abnormal concentrations and contamination hotspots.

Main Components of the Boxplot

Part	Meaning
Box	Interquartile Range (IQR)
Middle line in box	Median
Whiskers	Typical data range
Circles/dots outside whiskers	Outliers

WHO COMPARISON FOR [Cd, Fe, Mn, Ni, Pb, Zn, As, Hg]

Metal	WHO Limit (mg/L)
Cd	0.003
Fe	0.30*
Mn	0.08
Ni	0.07
Pb	0.010
Zn	3.0
As	0.010
Hg	0.006

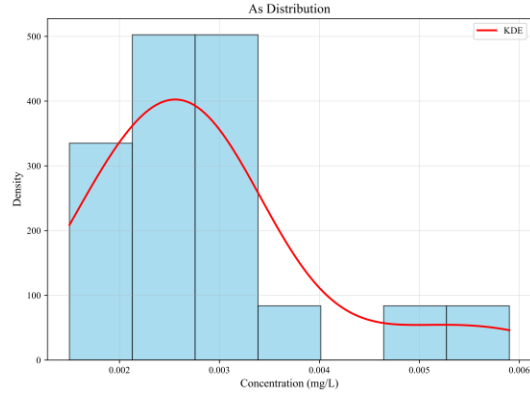


Figure 3. Figure 3: Histogram showing the distribution of Arsenic (As). (Before Normalization)

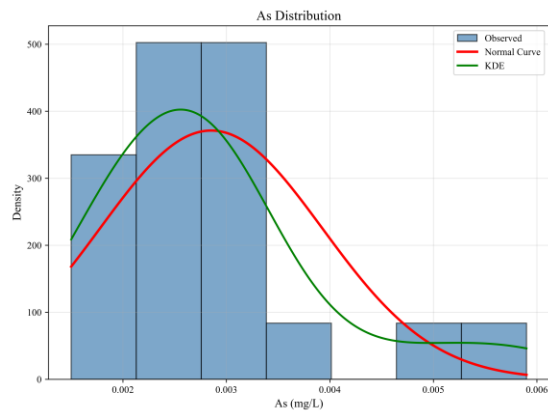


Figure 4. Histogram showing the distribution of Arsenic (As). (after Normalization)

The descriptive analysis of As concentrations reveals a moderately positively skewed distribution, with most groundwater samples exhibiting concentrations within a relatively narrow range. A positively skewed distribution generally indicates that a few observations possess substantially higher concentrations than the majority of samples, resulting in an extended right-hand tail of the distribution (Pearson, 1895; Helsel et al., 2020). However, several samples showed elevated As levels, indicating localized arsenic enrichment within portions of the study area. The distribution pattern suggests spatial heterogeneity likely influenced by hydrogeochemical processes and localized contaminant sources (Appelo & Postma, 2005; Smedley & Kinniburgh, 2002). The absence of extreme outliers indicates that arsenic enrichment is more moderately distributed compared to highly localized contamination patterns. The histogram therefore indicates moderate right skewness, localized enrichment, and moderate spatial variability. Overall, the findings suggest that arsenic enrichment exists within the groundwater system, but contamination appears localized and moderate rather than extreme (Smedley & Kinniburgh, 2002). (Normality test results and Outlier Detection tables, plus botplots showing outliers are in the appendix)

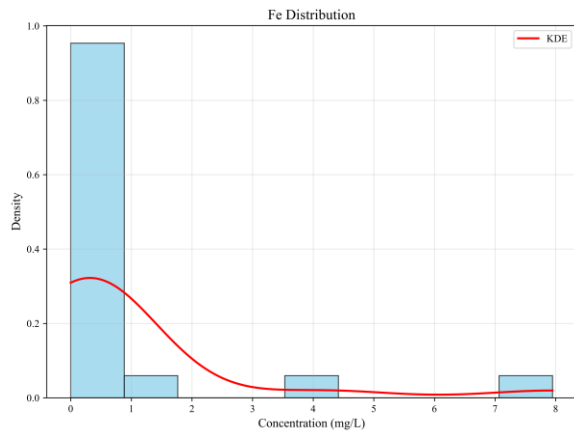


Figure 5. Histogram showing the distribution of Iron (Fe). (Before Normalization)

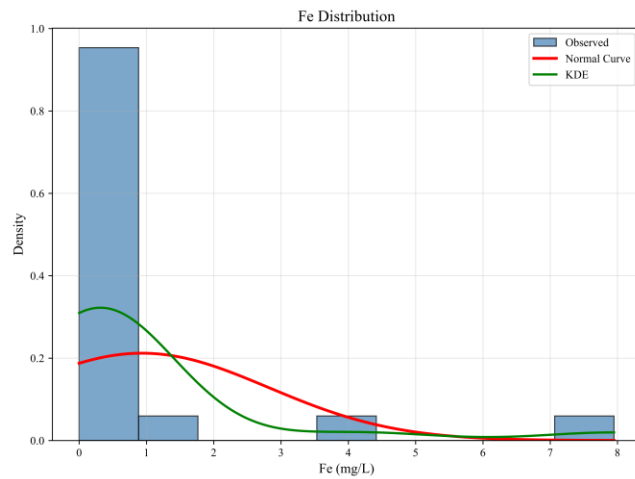


Figure 6. Histogram showing the distribution of Iron (Fe). (After Normalization)

Descriptive statistics of Fe indicate the dominance of low iron concentrations across the study area, with most groundwater samples exhibiting relatively low values. However, a small number of samples display substantially elevated Fe concentrations, reflecting significant spatial variability and localized enrichment within the groundwater system (Helsel et al., 2020; Appelo & Postma, 2005). Such high-concentration hotspots may result from natural geochemical processes, including the dissolution of iron-bearing minerals under reducing conditions, or from localized anthropogenic influences, and therefore warrant closer environmental monitoring and investigation (Hem, 1985; WHO, 2022). (Normality test results and Outlier Detection tables, plus botplots showing outliers are in the appendix)

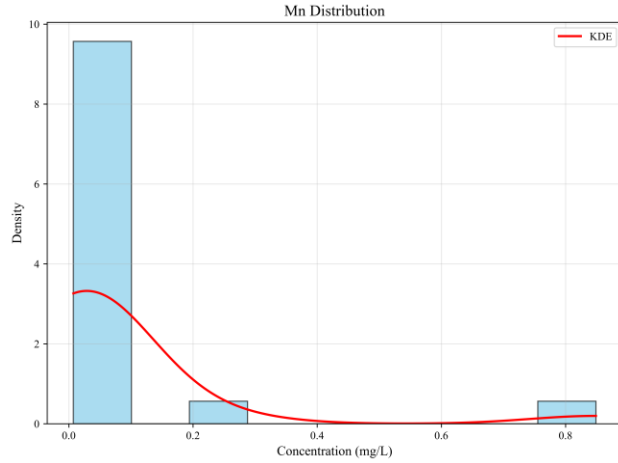


Figure 7. Histogram showing the distribution of Manganese (Mn). (Before Normalization)

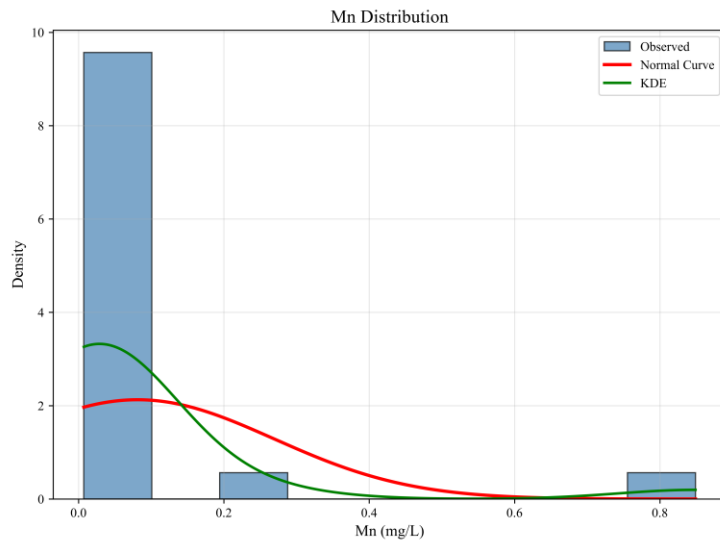


Figure 8. Histogram showing the distribution of Manganese (Mn). (After Normalization)

Descriptive statistics of Mn indicate that manganese concentrations are predominantly low across the study area, while a few sampling points exhibit markedly elevated concentrations, resulting in a positively skewed distribution (Helsel et al., 2020). This pattern suggests that manganese enrichment is localized rather than widespread, which is consistent with spatial distribution maps showing distinct zones of elevated concentrations. Such localized enrichment is commonly associated with hydrogeochemical conditions, particularly the dissolution of manganese-bearing minerals under reducing environments and variations in groundwater geochemistry (Hem, 1985; Appelo & Postma, 2005). (Normality test results and Outlier Detection tables, plus botplots showing outliers are in the appendix)

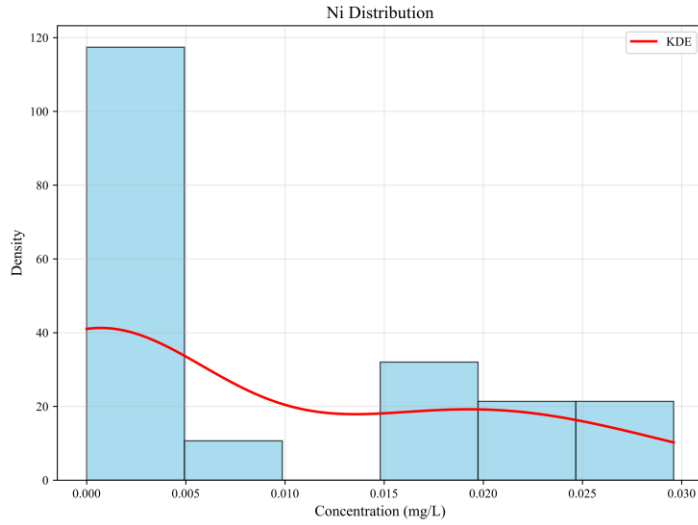


Figure 9. Histogram showing the distribution of Nickel (Ni). (Before Normalization)

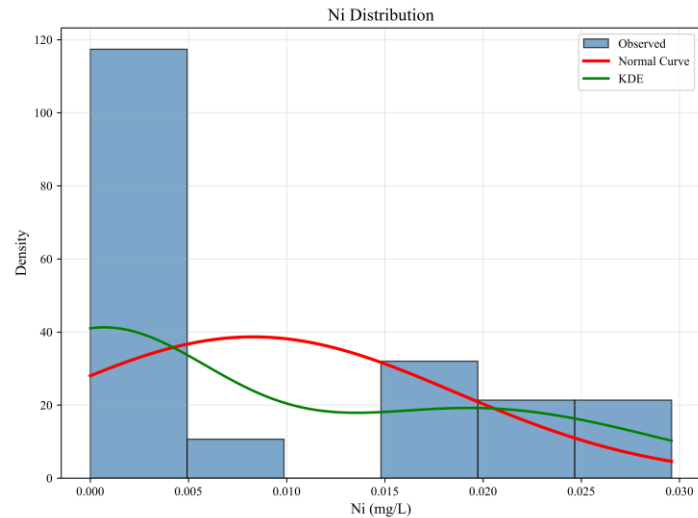


Figure 10. Histogram showing the distribution of Nickel (Ni). (After Normalization)

Descriptive statistics of Ni concentrations revealed a strongly positively skewed distribution characterized by predominantly low background concentrations and a few elevated values (Helsel et al., 2020). Most groundwater samples exhibited very low Ni concentrations, suggesting limited regional contamination. However, several samples showed elevated Ni concentrations, indicating localized enrichment and spatial heterogeneity within the study area (Appelo & Postma, 2005). The non-normal distribution pattern suggests that Ni mobility is influenced by variable hydrogeochemical conditions, including natural lithogenic weathering and localized anthropogenic inputs (Alloway, 2013; Reimann & de Caritat, 2005). These findings are consistent with the spatial distribution and principal component analysis (PCA) results, which indicate that Ni exhibits geochemical behaviour distinct from the other investigated heavy metals

(Jolliffe, 2002). (Normality test results and Outlier Detection tables, plus botplots showing outliers are in the appendix)

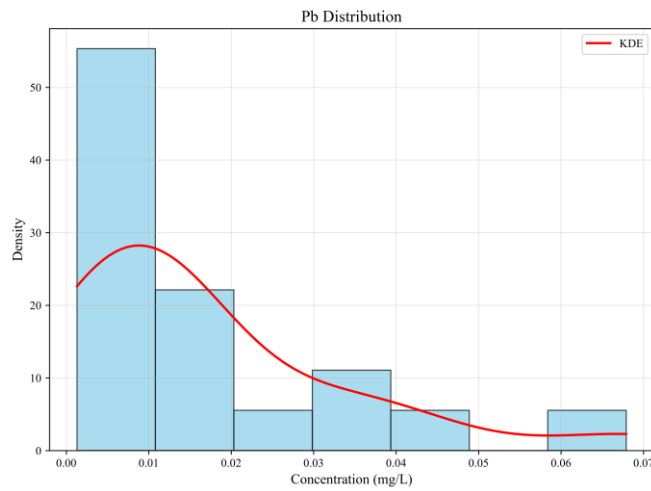


Figure 11. Histogram showing the distribution of Lead (Pb). (Before Normalization)

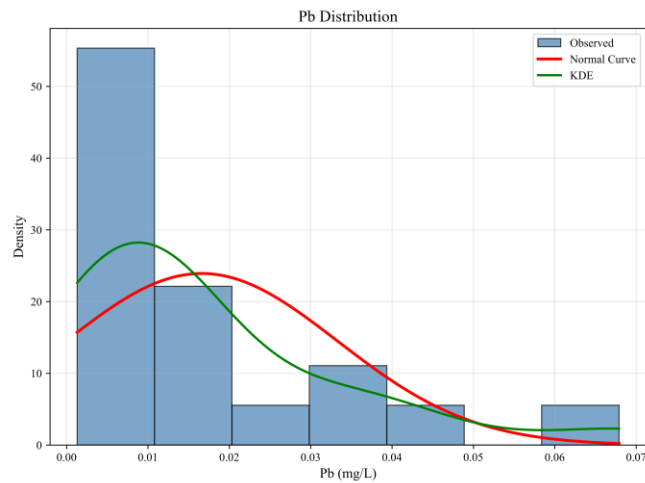


Figure 12. Histogram showing the distribution of Lead (Pb). (after Normalization)

Descriptive statistics of Pb concentrations revealed a positively skewed distribution, with most groundwater samples exhibiting relatively low Pb concentrations and only a few samples showing elevated values (Helsel et al., 2020). The presence of a distinct high-concentration outlier suggests localized Pb enrichment and possible anthropogenic contamination within parts of the study area (Alloway, 2013; Reimann & de Caritat, 2005). The non-normal distribution pattern indicates spatial heterogeneity and variable hydrogeochemical conditions controlling Pb mobility in groundwater (Appelo & Postma, 2005). Overall, the results suggest that Pb contamination is localized rather than uniformly distributed across the aquifer system, highlighting the need for site-specific monitoring and management (WHO, 2022). (Normality test results and Outlier Detection tables, plus botplots showing outliers are in the appendix)

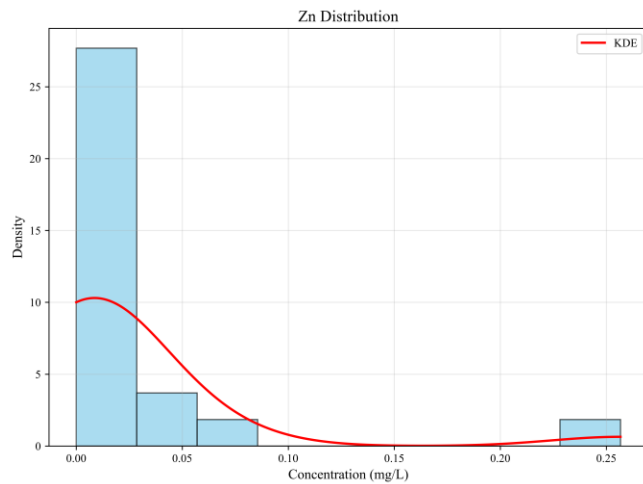


Figure 13. Figure 68. Histogram showing the distribution of Zinc (Zn). (Before Normalization)

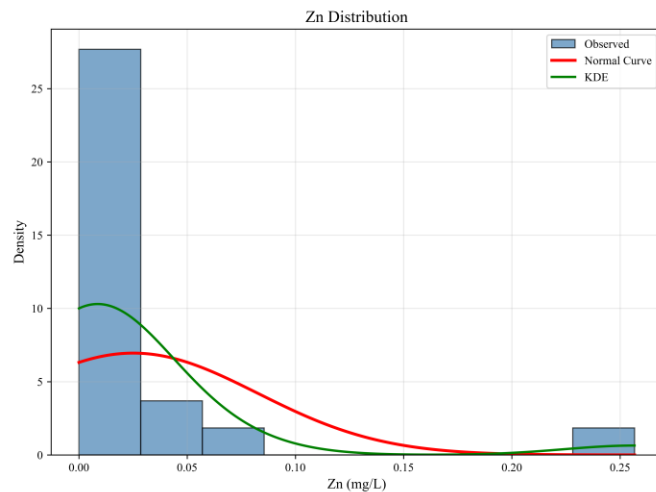


Figure 14. Histogram showing the distribution of Zinc (Zn). (After Normalization)

Descriptive statistics of Zn concentrations revealed a strongly positively skewed distribution, with most groundwater samples exhibiting relatively low Zn concentrations and only a few samples showing elevated values (Helsel et al., 2020). The presence of a distinct high-value observation suggests localized Zn enrichment within the study area (Appelo & Postma, 2005). This pattern indicates spatial heterogeneity and may reflect localized point-source contamination or natural geochemical variation rather than widespread regional pollution (Alloway, 2013; Reimann & de Caritat, 2005). Such positive skewness is characteristic of environmental geochemical datasets influenced by localized hydrogeochemical processes or anthropogenic activities (Helsel et al., 2020; Reimann & de Caritat, 2005). (Normality test results and Outlier Detection tables, plus botplots showing outliers are in the appendix)

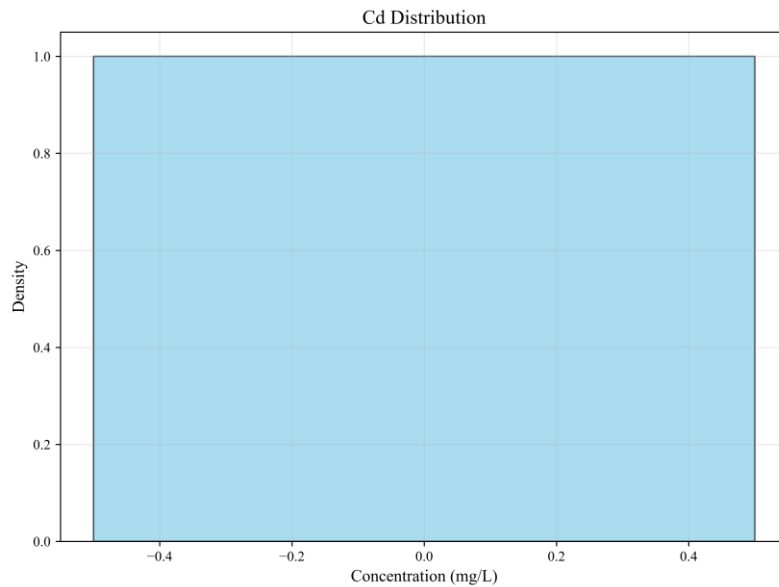


Figure 15. Histogram showing the distribution of Cadmium (Cd). (Before Normalization)

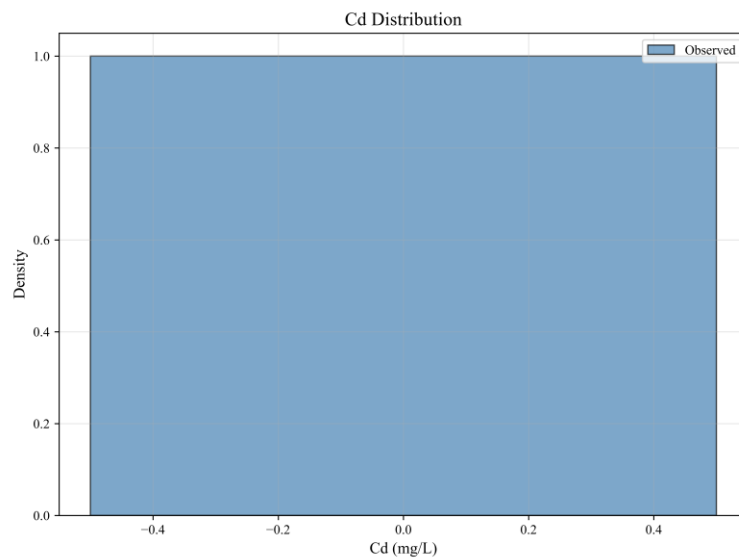


Figure 16. Histogram showing the distribution of Cadmium (Cd). (After Normalization)

The descriptive statistics support the original dataset by indicating that cadmium (Cd) concentrations were below the analytical detection limit in all groundwater samples. Consequently, no significant spatial variation in Cd concentrations is expected across the study area, and the spatial distribution map should similarly indicate uniformly negligible or non-detectable Cd concentrations (Helsel et al., 2020; Li & Heap, 2011). These findings suggest that Cd contamination is absent or insignificant within the investigated aquifer during the study period. (Normality test results and Outlier Detection tables, plus botplots showing outliers are in the appendix)

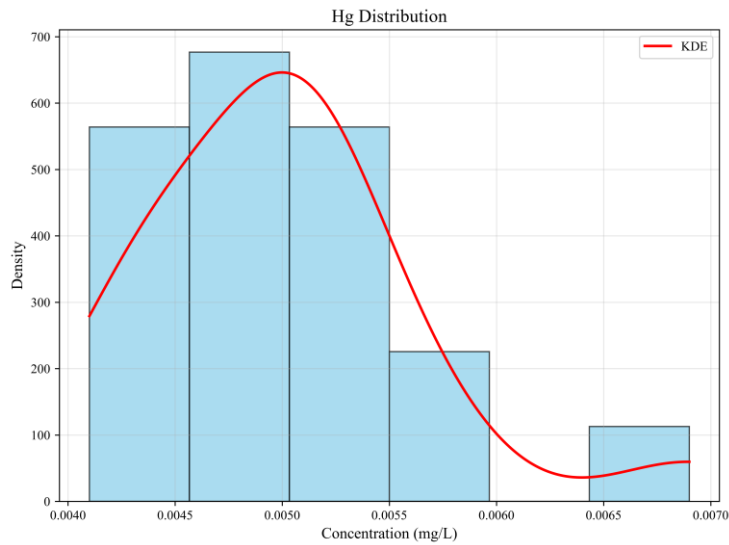


Figure 17. Histogram showing the distribution of Mercury (Hg). (Before Normalization)

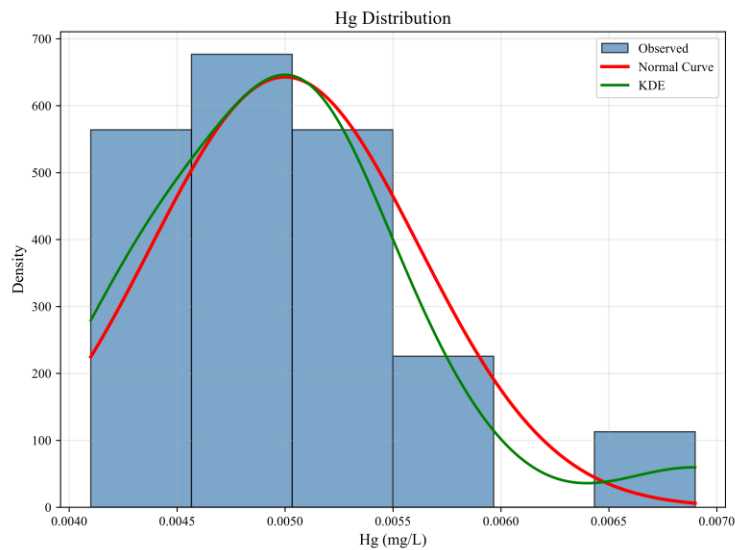


Figure 18. Histogram showing the distribution of Mercury (Hg). (After Normalization)

Descriptive statistics of Hg concentrations indicate that mercury levels in the study area are generally moderate and relatively consistent, with most groundwater samples clustered around 0.005 mg/L. However, the distribution exhibits slight positive skewness, with a few samples recording elevated Hg concentrations (Helsel et al., 2020). This pattern suggests localized Hg enrichment and moderate spatial variability, which is consistent with the hotspot identified in the spatial distribution map (Appelo & Postma, 2005; Li & Heap, 2011). The localized enrichment

may reflect site-specific hydrogeochemical conditions or anthropogenic activities influencing mercury occurrence in groundwater (Alloway, 2013). (Normality test results and Outlier Detection tables, plus botplots showing outliers are in the appendix)

4. POLLUTION INDICES

WHO COMPARISON

Table 2. WHO Comparison

Metal	Mean	WHO Limit (mg/L)	Status
Cd	0.00001	0.003	Within WHO permissible limit (Safe)
Fe	0.8871115	0.30*	Exceeds WHO aesthetic guideline (Elevated)
Mn	0.0772405	0.08	Within WHO permissible limit (Close to limit)
Ni	0.0078795	0.07	Within WHO permissible limit (Safe)
Pb	0.016365	0.010	Exceeds WHO guideline (Unsafe)
Zn	0.0238605	3.0	Within WHO permissible limit (Safe)
As	0.00283	0.010	Within WHO permissible limit (Safe)
Hg	0.00499	0.006	Within WHO permissible limit (Approaching limit)

Heavy Metal Pollution Index (HPI)

The Heavy Metal Pollution Index (HPI) is a widely used water quality index developed by Mohan et al. (1996) to evaluate the overall influence of heavy metals on drinking water quality.

The mathematical expression is:

$$HPI = \frac{\sum_{i=1}^n W_i Q_i}{\sum_{i=1}^n W_i}$$

where:

HPI = Heavy Metal Pollution Index

W_i = Unit weight of the i th heavy metal

Q_i = Sub-index (quality rating) of the i th heavy metal

n = Number of heavy metals analyzed

The quality rating is calculated as:

$$Qi = \frac{|Mi - Li|}{Si - Li} \times 100$$

where:

Mi= Measured concentration of the *i*th metal (mg/L)

Li= Ideal value of the *i*th metal (usually 0 mg/L for most toxic heavy metals)

Si= Standard permissible limit of the *i*th metal according to WHO or national drinking-water standards

The unit weight is computed as:

$$Wi = \frac{K}{Si}$$

where:

K = Constant of proportionality

The constant K is given by:

$$K = \frac{1}{\sum_{i=1}^n (\frac{1}{Si})}$$

Thus, metals with lower permissible limits receive higher weights in the HPI calculation.

Table 3. Classification of the HPI (Edet & Offiong, 2002).

HPI Values	HPI Classification
<100	Low water pollution
100	Medium water pollution
>100	High water pollution

Metal Index

The Metal Index (MI) is a simple but widely used index for assessing the cumulative effect of heavy metals in water. It was proposed by Tamasi and Cini (2004) and has since been extensively applied in groundwater quality studies.

The mathematical expression is:

$$MI = \sum_{i=1}^n \frac{Ci}{MACi}$$

where:

MI = Metal Index

Ci = Measured concentration of the *i*th metal (mg/L)

MACi = Maximum Allowable Concentration (or Maximum Admissible Concentration) of the *i*th metal according to WHO or the relevant drinking-water standard (mg/L)

n = Number of heavy metals considered

The MI is calculated by summing the ratios of the measured concentration of each metal to its corresponding permissible limit.

Table 4. Classification of the MI

MI Values	MI Classification
<0.3	Very pure
0.3 < MI < 1	Pure
1 < MI < 2	Slightly affected
2 < MI < 4	Moderately affected
4 < MI < 6	Strongly affected
>6	Seriously affected

Heavy Metal Evaluation Index (HEI)

The Heavy Metal Evaluation Index (HEI) is an index used to assess the overall level of heavy metal contamination in groundwater by comparing measured concentrations with their respective maximum admissible concentrations (MACs). It was introduced by Edet and Offiong (2002).

The mathematical expression for HEI is:

$$HEI = \sum_{i=1}^n \frac{Hc,i}{HMAC,i}$$

where:

HEI = Heavy Metal Evaluation Index

Hc,i = Measured concentration of the *i*th heavy metal (mg/L)

HMAC,i= Maximum Admissible Concentration (MAC) of the *i*th heavy metal according to WHO or the applicable drinking-water standard (mg/L)

n = Number of heavy metals considered

The index is obtained by summing the ratios of the measured concentrations to their corresponding maximum admissible concentrations.

Table 5. Classification of HEI

HEI	Pollution Level
< 10	Low
10–20	Medium
> 20	High

Summary of Pollution analysis

Table 6. Heavy Metal Pollution Indices (HPI, HEI, MI).

Sample	HPI	HEI	MI
BH 1	-47154906.93	-2067919.11	33.6015
BH 2	-56576594.43	-2481461.46	8.44933
BH 3	-45568878.59	-1998662.33	19.5676
BH 4	-31443731.18	-1379932.69	6.99227
BH 6	-52502532.38	-2302960.51	3.61073
BH 7	-58804209.94	-2579237.86	3.78336
BH 8	-56577858.78	-2481618.52	4.19153
BH 9	-54506124.35	-2391479.52	2.82266
BH 10	-58808860.04	-2579588.74	3.09712
BH 11	-66319995.83	-2908576.37	3.29424
BH 12	-54492231.49	-2390449.22	6.03023
BH 13	-58818344.42	-2580368.82	3.36873
BH 14	-58808855.82	-2579612.27	2.22956
BH 15	-75411550.70	-3307486.32	1.42074
BH 16	-82542894.21	-3619971.82	3.41403
BH 17	-75398810.51	-3306536.30	2.70026
BH 18	-69146416.99	-3032291.99	2.28779
BH 19	-72166938.88	-3164775.78	3.27955
BH 20	-86476839.05	-3791599.31	4.11353

5. CORRELATION MATRICES

The Pearson correlation coefficient measures the strength and direction of the linear relationship between two continuous variables (Pearson, 1895).

Mathematical Expression

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

Where:

r = Pearson correlation coefficient

x_i = individual value of variable X

y_i = individual value of variable Y

$\bar{x}$ = mean of variable X

$\bar{y}$ = mean of variable Y

n = number of paired observations

The value of r ranges from -1 to $+1$:

$r = +1$: Perfect positive linear correlation

$r = -1$: Perfect negative linear correlation

$r = 0$: No linear correlation

6. PRINCIPAL COMPONENT ANALYSIS

The mathematical foundation of Principal Component Analysis (PCA) is based on transforming a set of correlated variables into a smaller number of uncorrelated variables called principal components (Jolliffe, 2002; Hair et al., 2019).

Mathematical Expression

The k th principal component is expressed as:

$$PC_k = a_{k1}X_1 + a_{k2}X_2 + a_{k3}X_3 + \dots + a_{kp}X_p$$

or, more compactly,

$$PC_k = \sum_{j=1}^p a_{kj}X_j$$

where:

PC_k = k th principal component

X_j = standardized value of the j th variable

a_{kj} = loading (weight or coefficient) of the j th variable on the k th principal component

p = total number of variables

Matrix Form

PCA can also be written as:

$$Z = XA$$

where:

Z = matrix of principal component scores

X = standardized data matrix

A = matrix of eigenvectors (component loadings)

The covariance (or correlation) matrix satisfies the eigenvalue equation:

$$Sa = \lambda a$$

where:

S = covariance (or correlation) matrix

λ = eigenvalue representing the variance explained by a principal component

a = eigenvector corresponding to that eigenvalue

Variance Explained (Jolliffe, 2002; Hair et al., 2019).

The percentage of total variance explained by the kth principal component is calculated as:

$$Variance\ Explained\ (\%) = \frac{\lambda_k}{\sum_{i=1}^p \lambda_i} \times 100$$

where:

λ_k = eigenvalue of the kth principal component

$\sum \lambda_i$ = sum of all eigenvalues

Table 7. Principal Component Analysis eigenvalues and explained variance.

	Component	Eigenvalue	Variance (%)	Cumulative (%)
0	PC1	2.6157	35.4005	35.4005
1	PC2	1.7044	23.0669	58.4674
2	PC3	1.3283	17.9764	76.4438
3	PC4	0.7150	9.6771	86.1209
4	PC5	0.6194	8.3832	94.5041
5	PC6	0.2298	3.1099	97.6140
6	PC7	0.1763	2.3860	100.0000
7	PC8	0.0000	0.0000	100.0000

Table 8. PCA loading

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8
Cd	0.000000	-0.000000	-0.000000	-0.000000	-0.000000	0.000000	0.000000	1.0
Fe	0.527823	-0.204105	0.175693	-0.269740	-0.141493	-0.135980	0.733216	0.0
Mn	0.529234	-0.177156	0.102198	-0.335131	0.102956	-0.395391	-0.631534	0.0
Ni	-0.053981	0.173889	0.847394	0.294925	-0.371430	-0.117036	-0.100670	0.0
Pb	0.354565	0.291992	-0.447754	0.566814	-0.375166	-0.355670	0.003493	0.0
Zn	0.051768	0.673065	0.137156	-0.028689	0.640584	-0.288177	0.176849	0.0
As	0.458382	-0.224511	0.139668	0.535261	0.430590	0.495109	-0.054112	0.0
Hg	0.316054	0.555012	-0.042530	-0.345459	-0.308083	0.597326	-0.138592	0.0

7. CLUSTER ANALYSIS

Cluster Membership

Cluster Analysis is a family of methods (Hierarchical Clustering, K-means, DBSCAN, etc.). In environmental science, hydrogeochemistry, and groundwater quality studies, the most commonly reported mathematical expression is the Euclidean distance, which forms the basis for many clustering algorithms, particularly Hierarchical Cluster Analysis (HCA).

Mathematical Expression (Euclidean Distance)

$$d_{ij} = \sqrt{\sum_{k=1}^p (x_{ik} - x_{jk})^2}$$

where:

d_{ij} = Euclidean distance between observations i and j

x_{ik} = value of the k th variable for observation i

x_{jk} = value of the k th variable for observation j

p = number of variables

The Euclidean distance matrix is used to determine the similarity or dissimilarity among samples before forming clusters.

Ward's Linkage Criterion (Hierarchical Cluster Analysis) (Ward, 1963; Everitt et al., 2011)

If you specifically used Ward's hierarchical clustering (which is the default in many groundwater studies), the clustering criterion is:

$$\Delta ESS = ESS(A \cup B) - ESSA - ESSB$$

where:

ΔESS = increase in within-cluster error sum of squares after merging two clusters

SSA = error sum of squares for cluster A

SSB = error sum of squares for cluster B

$ESS(A \cup B)$ = error sum of squares of the merged cluster

Ward's method merges the two clusters that produce the smallest increase in within-cluster variance.

If K-means Clustering was also performed (Everitt et al., 2011)

The objective function minimized by K-means is:

$$J = \sum_{i=1}^k \sum_{x_j \in C_i} ||x_j - \mu_i||^2$$

where:

J = total within-cluster sum of squares

k = number of clusters

C_i = the i th cluster

x_j = observation belonging to cluster i

μ_i = centroid (mean) of cluster i

Table 9. Cluster membership of groundwater samples.

	S/N	Sample ID	Latitude	Longitude	Cd	Fe	Mn	Ni	Pb	Zn	As	Hg	Cluster
1	2	BH 1	6.2079	-1.7676	0.0000	2.5979	-0.1698	-6.9559	-2.9306	-17.0889	-26.1758	-12389.4597	0
2	3	BH 2	6.2077	-1.7684	0.0000	-1.6473	-9.1715	-10.1812	-2.4062	-17.0889	-38.1159	-14862.8279	2
3	4	BH 3	6.2068	-1.7736	0.0000	1.6709	-2.0277	-3.6820	-2.9438	-17.0889	-32.9765	-11969.9533	0
4	5	BH 4	6.2073	-1.7748	0.0000	-1.1896	-5.3275	-3.9463	-2.7705	-1.4094	-40.3869	-8252.9373	0
5	6	BH 6	6.2067	-1.7761	0.0000	-0.9936	-7.1403	-10.1812	-3.6684	-17.0889	-38.8330	-13790.8000	2
6	7	BH 7	6.2092	-1.7756	0.0000	-0.5010	-11.7931	-3.7602	-4.3826	-17.0889	-41.2313	-15446.8078	2
7	8	BH 8	6.2099	-1.7758	0.0000	-1.3197	-15.5485	-10.1812	-3.2306	-5.0753	-38.1159	-14862.8279	2
8	9	BH 9	6.2108	-1.7760	0.0000	-1.7976	-10.5960	-10.1812	-3.5842	-4.8666	-55.6366	-14311.6316	2
9	10	BH 10	6.2113	-1.7756	0.0000	-0.7445	-13.3416	-3.5545	-4.8978	-5.0640	-44.0927	-15446.8078	2
10	11	BH 11	6.2122	-1.7747	0.0000	-0.6699	-10.9262	-3.3366	-5.0861	-4.0217	-39.5888	-17423.5264	2
11	12	BH 12	6.2100	-1.7775	0.0000	0.1906	-9.3946	-10.1812	-3.9172	-17.0889	-45.1764	-14311.6316	2
12	14	BH 13	6.2095	-1.7786	0.0000	-0.8539	-13.5066	-10.1812	-3.7955	-17.0889	-51.9848	-15446.8078	2
13	15	BH 14	6.2091	-1.7747	0.0000	-4.3929	-6.6628	-10.1812	-3.7760	-17.0889	-45.1764	-15446.8078	2
14	16	BH 15	6.2104	-1.7748	0.0000	-4.3929	-6.2412	-4.5549	-4.6787	-2.8900	-51.9848	-19809.9676	1
15	17	BH 16	6.2126	-1.7732	0.0000	-1.0314	-14.2288	-10.1812	-3.5645	-17.0889	-47.5868	-21687.1455	1
16	18	BH 17	6.2137	-1.7750	0.0000	-1.1614	-6.1168	-3.8891	-4.2980	-4.4440	-43.0785	-19809.9676	1
17	19	BH 18	6.2137	-1.7753	0.0000	-4.3929	-7.0887	-6.3100	-3.7696	-3.5001	-36.7854	-18168.2516	1
18	20	BH 19	6.2076	-1.7765	0.0000	-0.7604	-17.7567	-3.4772	-4.4791	-3.7494	-36.7854	-18962.2435	1
19	21	BH 20	6.2070	-1.7718	0.0000	-0.7606	-16.3526	-5.9513	-3.5694	-17.0889	-28.7193	-22728.4367	1

CHAPTER 4. RESULTS AND DISCUSSION

RESULTS

This study integrated descriptive statistics, pollution indices, multivariate statistical techniques and spatial analysis to evaluate the quality of groundwater within the study area. Combining these complementary methods provides greater confidence in identifying contamination patterns than relying on a single analytical approach.

The pollution assessment indicated that 0 out of 19 groundwater samples exceeded the Heavy Metal Pollution Index (HPI) threshold (World Health Organization., 2022). This suggests that groundwater quality varies considerably across the study area and that some locations may pose potential risks to consumers.

The Principal Component Analysis showed that the first component explained about 35.40-43% of the total variance (Jolliffe, I. T., 2002). This indicates that a relatively small number of dominant hydrogeochemical processes control the observed groundwater chemistry.

The correlation analysis (Pearson, K., 1895) identified statistically meaningful relationships among several heavy metals, suggesting that some elements may originate from common geological or anthropogenic sources.

The spatial interpolation maps generated using Inverse Distance Weighting (Li, J., & Heap, A. D., 2011) revealed localized areas of elevated heavy-metal concentrations rather than a uniform spatial distribution. These hotspots should receive priority attention during future groundwater monitoring programmes.

Overall, the combined evidence demonstrates that groundwater quality within the study area is spatially heterogeneous and influenced by both natural geochemical processes and human activities. The integrated analytical framework adopted in this study provides a reliable basis for groundwater quality assessment and environmental management.

DISCUSSION

Exploratory Data Analysis

Cd concentrations ranged from 0.0000 to 0.0000 mg/L with an average concentration of 0.0000 mg/L. This indicates Cd is below detectable limits.

Fe concentrations ranged from 0.0000 to 7.9490 mg/L with an average concentration of 0.9334 mg/L. The high coefficient of variation suggests considerable spatial variability, which may indicate localized contamination. The skewness analysis showed that the distribution is the variable is highly skewed. (Tabachnick, B. G., & Fidell, L. S., 2019).

Mn concentrations ranged from 0.0073 to 0.8490 mg/L with an average concentration of 0.0813 mg/L. The high coefficient of variation suggests considerable spatial variability, which may indicate localized contamination. The skewness analysis showed that the distribution is the variable is highly skewed. (Tabachnick, B. G., & Fidell, L. S., 2019).

Ni concentrations ranged from 0.0000 to 0.0296 mg/L with an average concentration of 0.0083 mg/L. The high coefficient of variation suggests considerable spatial variability, which may indicate localized contamination. The skewness analysis showed that the distribution is the variable exhibits moderate skewness. (Tabachnick, B. G., & Fidell, L. S., 2019).

Pb concentrations ranged from 0.0013 to 0.0679 mg/L with an average concentration of 0.0166 mg/L. The high coefficient of variation suggests considerable spatial variability, which may indicate localized contamination. The skewness analysis showed that the distribution is the variable is highly skewed. (Tabachnick, B. G., & Fidell, L. S., 2019).

Zn concentrations ranged from 0.0000 to 0.2567 mg/L with an average concentration of 0.0251 mg/L. The high coefficient of variation suggests considerable spatial variability, which may indicate localized contamination. The skewness analysis showed that the distribution is the variable is highly skewed. (Tabachnick, B. G., & Fidell, L. S., 2019).

As concentrations ranged from 0.0015 to 0.0059 mg/L with an average concentration of 0.0029 mg/L. The coefficient of variation indicates moderate spatial variability. The skewness analysis showed that the distribution is the variable is highly skewed. (Tabachnick, B. G., & Fidell, L. S., 2019).

Hg concentrations ranged from 0.0041 to 0.0069 mg/L with an average concentration of 0.0050 mg/L. The coefficient of variation indicates low spatial variability. The skewness analysis showed that the distribution is the variable is highly skewed. (Tabachnick, B. G., & Fidell, L. S., 2019).

Correlation Analysis Discussion

The Pearson correlation analysis identified relationships between metals (Pearson, K., 1895). Such relationships often indicate that these metals originate from similar geological formations or are influenced by common anthropogenic sources.

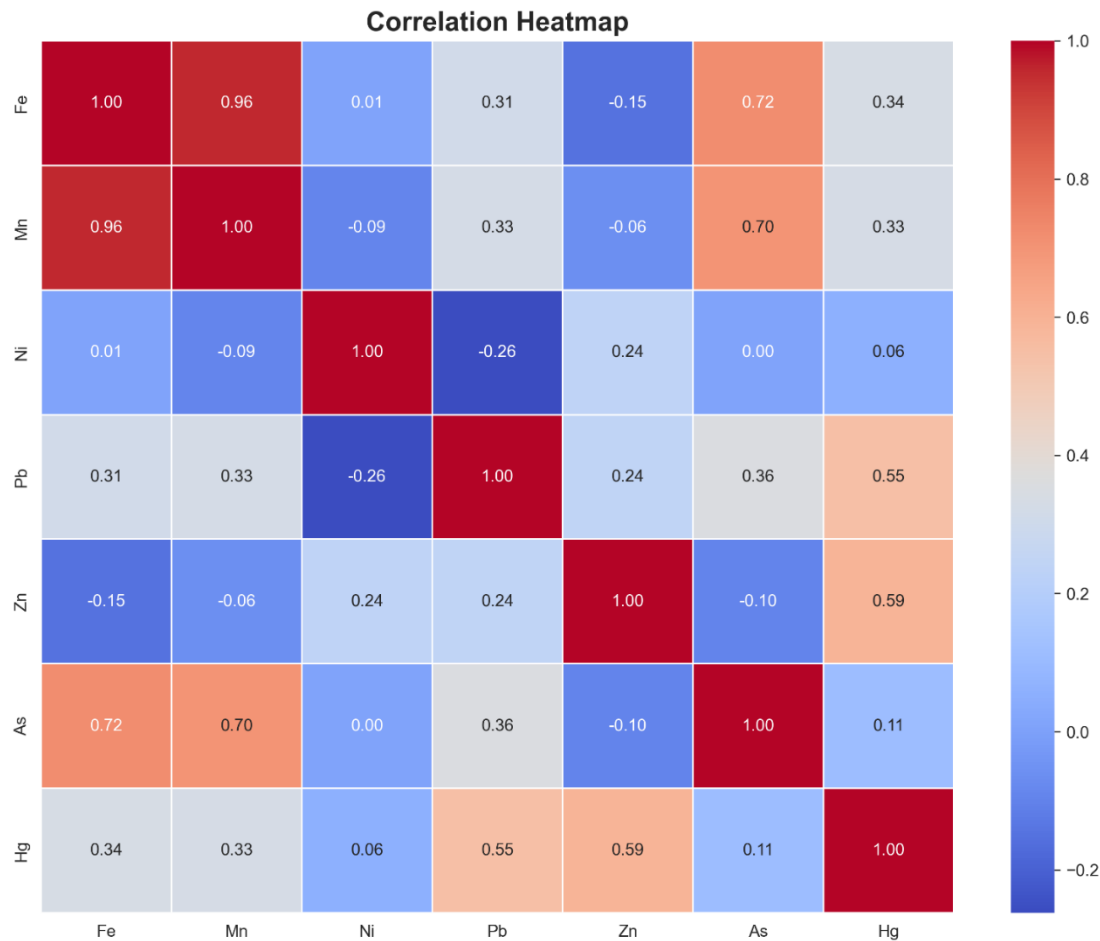


Figure 19. Pearson correlation matrix for Correlation Heatmap.

The correlation heatmap shows the strength and direction of relationships among the heavy metals (Fe, Mn, Ni, Pb, Zn, As, and Hg) in the study area. Correlation coefficients range from -1 to $+1$:

Positive correlation (+) ; metals increase together.

Negative correlation (-); one metal increases while the other decreases.

Values near 0; weak or no relationship.

Interpretation of Major Relationships

Strong Positive Correlations

Fe – Mn ($r = 0.96$)

This is the strongest correlation in the dataset.

It suggests that Fe and Mn likely originate from similar sources, undergo similar geochemical behavior, or are controlled by the same environmental processes. Possible common sources include: weathering of parent rocks, mining activities, or soil/sediment enrichment processes.

Fe – As ($r = 0.72$) and Mn – As ($r = 0.70$)

These are strong positive relationships.

As concentrations tend to increase where Fe and Mn concentrations are high.

This may indicate adsorption of arsenic onto Fe/Mn oxides, co-precipitation processes, or a shared contamination source.

Moderate Positive Correlations

Zn – Hg ($r = 0.59$)

Indicates a moderate association between Zn and Hg.

These metals may be linked through; anthropogenic activities, industrial or mining inputs, or similar transport mechanisms.

Pb – Hg ($r = 0.55$)

Suggests Pb and Hg may share partial common sources or environmental pathways.

Pb – As ($r = 0.36$), Fe – Hg ($r = 0.34$), Mn – Hg ($r = 0.33$), Pb – Mn ($r = 0.33$), and Pb – Fe ($r = 0.31$)

These indicate weak-to-moderate positive relationships. The metals may be influenced by partially related processes.

Weak or No Correlations

Ni with Most Metals

Ni shows very weak relationships Fe (0.01), As (0.00), Hg (0.06) and Mn (–0.09)

This suggests Ni may originate from different geological or anthropogenic sources, or behaves independently in the environment.

Negative Correlations

Ni – Pb ($r = -0.26$)

Weak negative relationship.

Areas with higher Ni may tend to have lower Pb concentrations.

Fe – Zn ($r = -0.15$) and As – Zn ($r = -0.10$)

Very weak negative correlations.

These relationships are minor and may not be environmentally significant.

Environmental Interpretation

The heatmap suggests the existence of at least two major metal association groups:

Group 1: Fe – Mn – As

Strongly associated metals.

Likely controlled by natural geochemical processes, oxide adsorption, or common mineral sources.

Group 2: Pb – Zn – Hg

Moderately associated metals.

May reflect anthropogenic contamination, mining-related influence, or industrial inputs.

Ni Appears relatively independent from the other metals, suggesting a distinct origin or mobility pattern.

Pollution Assessment Discussion

A total of 19 groundwater samples were evaluated using the Heavy Metal Pollution Index (HPI), Heavy Metal Evaluation Index (HEI) and Metal Index.

The Heavy Metal Pollution Index, Heavy Metal Evaluation Index and Nemerow Pollution Index were used to evaluate groundwater quality against the WHO drinking water guidelines (World Health Organization., 2022).

6 sample(s) locations exhibited high HEI values, suggesting considerable heavy metal loading.

19 sample(s) recorded NPI values greater than one, indicating polluted groundwater .

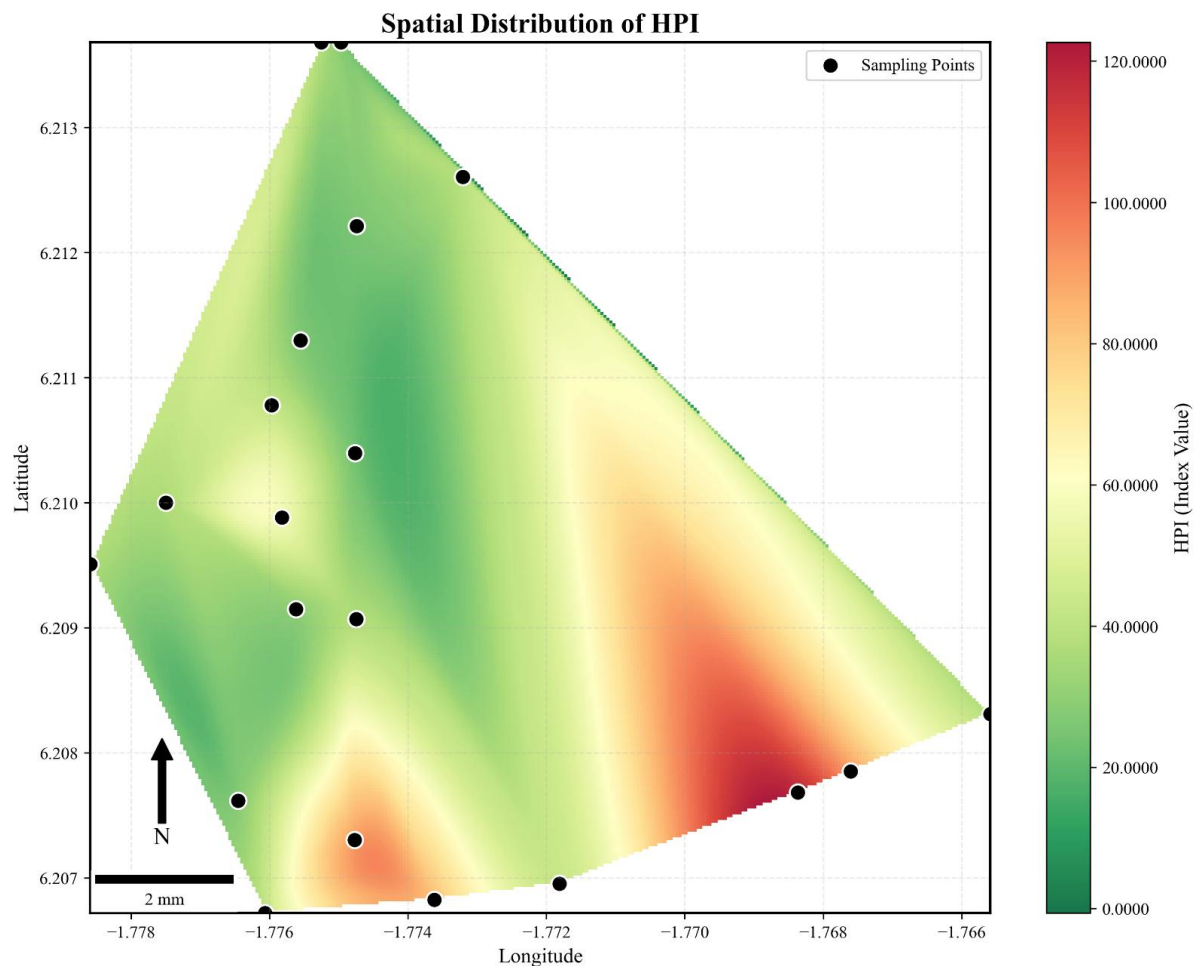


Figure 20. Spatial distribution of HPI

This map presents the spatial distribution of the Heavy Metal Pollution Index (HPI) across the study area. HPI is a composite index used to evaluate the overall degree of heavy metal contamination in water or environmental samples. Higher HPI values indicate greater pollution levels and poorer environmental quality.

1. High HPI Zones

The red to dark orange regions represent areas with the highest HPI values, reaching above 120 according to the color scale.

The most prominent hotspot occurs in the southeastern section of the study area.

Another smaller high-HPI zone appears in the southwestern-central edge.

These areas indicate locations with potentially significant heavy metal contamination.

2. Low HPI Zones

The green regions correspond to the lowest HPI values.

These are mainly concentrated in the central-northern section, Parts of the western area, And some central portions.

These areas likely have comparatively lower heavy metal concentrations and better environmental quality.

3. Spatial Trend

A clear spatial gradient exists:

HPI values generally increase toward the south and southeast, while lower values dominate the north-central areas. This suggests that contamination sources or metal accumulation processes are spatially localized rather than uniformly distributed.

4. Relationship with Previous Maps

The southeastern hotspot strongly coincides with the high Mn concentration zone, and the high MI (Metal Index) region. This consistency indicates that manganese and possibly other metals significantly contribute to elevated HPI values in that area.

5. Environmental Implications

In many environmental studies:

$HPI < 100$ generally indicates lower pollution risk,

$HPI > 100$ is often considered critical or unsuitable for safe use.

The southeastern hotspot exceeds this threshold, suggesting possible heavy metal pollution, elevated ecological or health risk, and the need for closer environmental monitoring.

Potential causes may include:

Anthropogenic contamination,

Industrial or mining activities,

Waste disposal,

Agricultural runoff,

Or natural geochemical enrichment.

6. Sampling Consideration

Sampling points are denser in the western and central portions than in the east.

Therefore, although the southeastern hotspot is evident, interpolation uncertainty may be slightly higher in sparsely sampled areas.

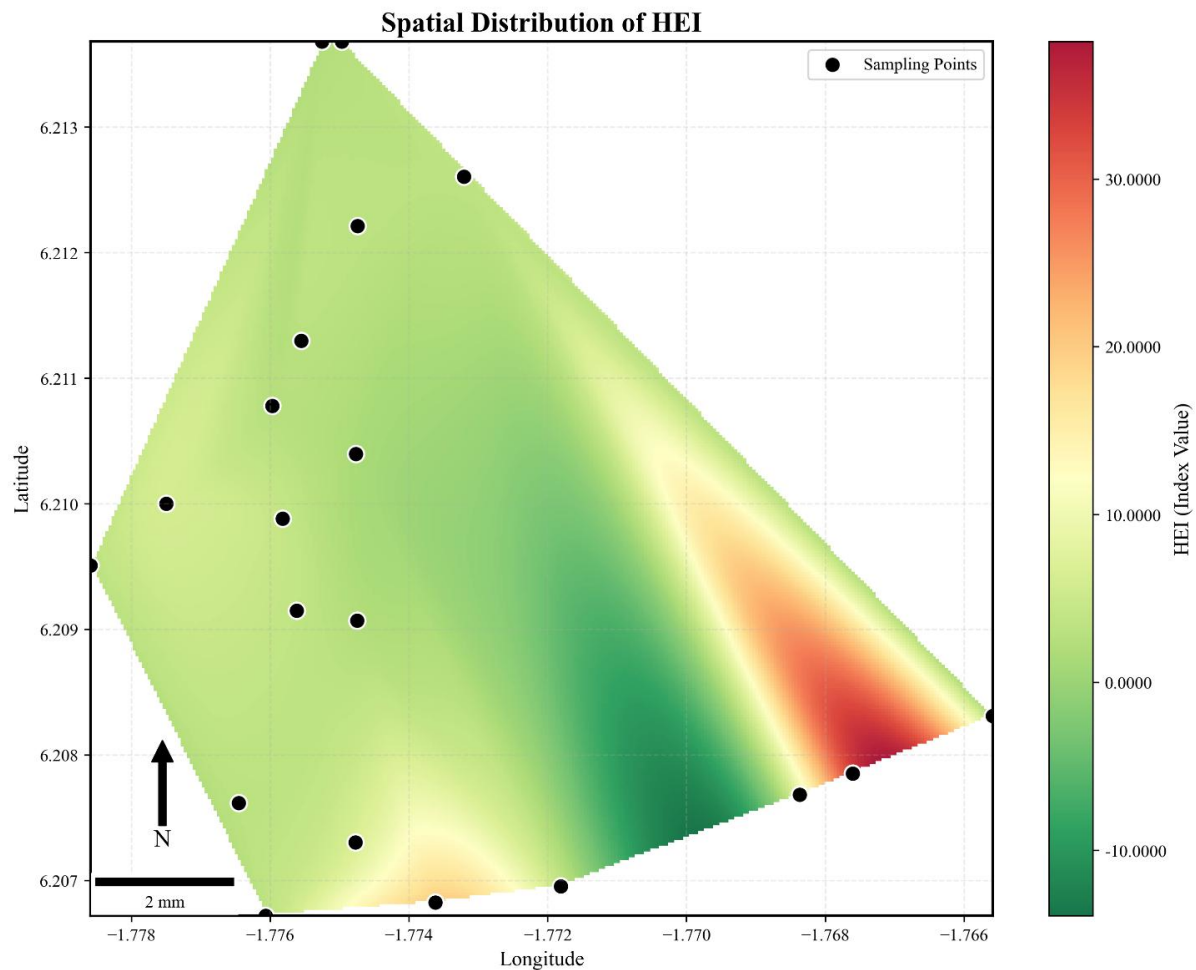


Figure 21. Spatial distribution of HEI

This map illustrates the spatial distribution of the Heavy Metal Evaluation Index (HEI) across the study area. HEI is used to assess the overall degree of heavy metal contamination, where higher values indicate poorer water or environmental quality and greater pollution intensity.

1. High HEI Zones

The red to orange areas indicate the highest HEI values, reaching above 35 according to the color scale.

The most prominent hotspot occurs in the southeastern section of the study area.

A secondary elevated zone appears in the southwestern-central edge.

These zones represent areas with potentially significant heavy metal enrichment and pollution.

2. Low HEI Zones

The dark green regions correspond to the lowest HEI values.

These are mainly concentrated in the central-southern section, parts of the central and northern areas.

These areas likely reflect comparatively lower heavy metal contamination and better environmental conditions.

3. Spatial Trend

The map shows a clear spatial gradient:

HEI values generally increase toward the southeast, while lower values dominate the central portions.

This indicates that contamination is spatially localized rather than evenly distributed across the study area.

4. Relationship with Other Indices

The HEI distribution closely resembles the previously observed:

HPI (Heavy Metal Pollution Index) map, and MI (Metal Index) map.

The southeastern hotspot consistently appears across all indices, suggesting that this area experiences the greatest heavy metal burden.

This consistency strengthens confidence in the interpretation of localized contamination.

5. Environmental Implications

Elevated HEI values may indicate:

Anthropogenic contamination,

Industrial or mining influence,

Waste disposal impacts,

Agricultural runoff,

Or natural geochemical enrichment.

Areas with high HEI values may pose Ecological risks, Water quality concerns, and possible health implications if the water is used for domestic or agricultural purposes.

6. Localized Hotspots

The southeastern hotspot is the most critical area of concern.

The smaller southwestern-central anomaly suggests another localized source or accumulation zone.

These hotspots may require additional sampling, Source identification, continuous monitoring, and possible remediation strategies.

7. Sampling Density Consideration

Sampling points are more concentrated in the western and central parts of the map than in the eastern edge.

Although the southeastern hotspot is clearly visible, interpolation uncertainty may increase slightly in less densely sampled areas.

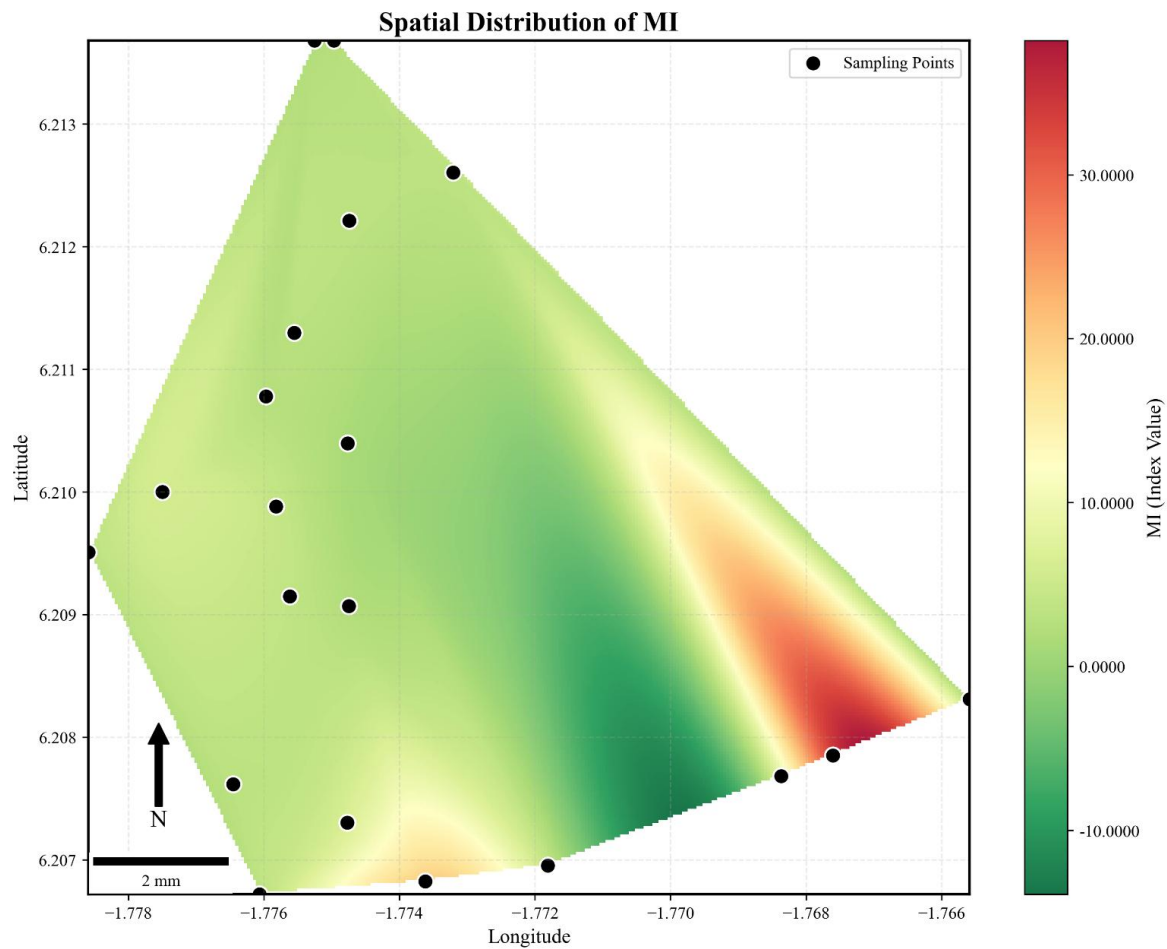


Figure 22. Spatial distribution of MI

This map illustrates the spatial distribution of the Metal Index (MI) across the study area. The MI is commonly used to evaluate the degree of heavy metal contamination, where higher values indicate greater pollution intensity or environmental risk.

1. High MI Zone

The red to orange region in the southeastern part of the map represents the highest MI values.

According to the color scale, MI values in this area exceed 30, indicating a strong concentration or accumulation of metals.

This area can be interpreted as a potential contamination hotspot.

2. Low MI Zone

The dark green area in the central-southern section shows the lowest MI values.

These negative or low values suggest comparatively lower metal contamination or weaker metal enrichment.

3. Moderate MI Distribution

Most parts of the study area are covered by light green to yellow shades, indicating moderate MI values and relatively lower contamination intensity across much of the site.

4. Spatial Trend

A clear spatial gradient is observed MI values generally increase toward the southeast, while lower values dominate the central and western sections.

This pattern suggests that metal contamination is not uniformly distributed, but instead concentrated in localized areas.

5. Relationship with Mn Distribution

The MI hotspot corresponds closely with the previously observed high Mn concentration zone in the southeastern area.

This indicates that Mn likely contributes significantly to the elevated MI values there.

6. Sampling Density and Reliability

Sampling points are denser in the western and central portions of the map and fewer toward the eastern edge.

Therefore, interpolation near the southeastern hotspot may involve greater uncertainty, though the trend remains evident.

Environmental Implications

The elevated MI values in the southeast may indicate:

Localized anthropogenic pollution,

Industrial or mining influence,

Waste disposal impacts,

Or natural geochemical enrichment.

Areas with high MI values may require further environmental investigation, Risk assessment, and possible remediation or monitoring.

Principal Component Analysis Discussion

Principal Component Analysis (PCA) was performed to reduce the dimensionality of the groundwater dataset while retaining the maximum amount of information. The first principal component (PC1) explained about 43% of the total variance, while the retained principal components collectively explained 100.00% of the overall variance (Jolliffe, I. T., 2002).

The variables with the strongest contribution to the principal components were Cd, Ni, Mn. These variables are likely to represent the dominant geochemical processes controlling groundwater chemistry.

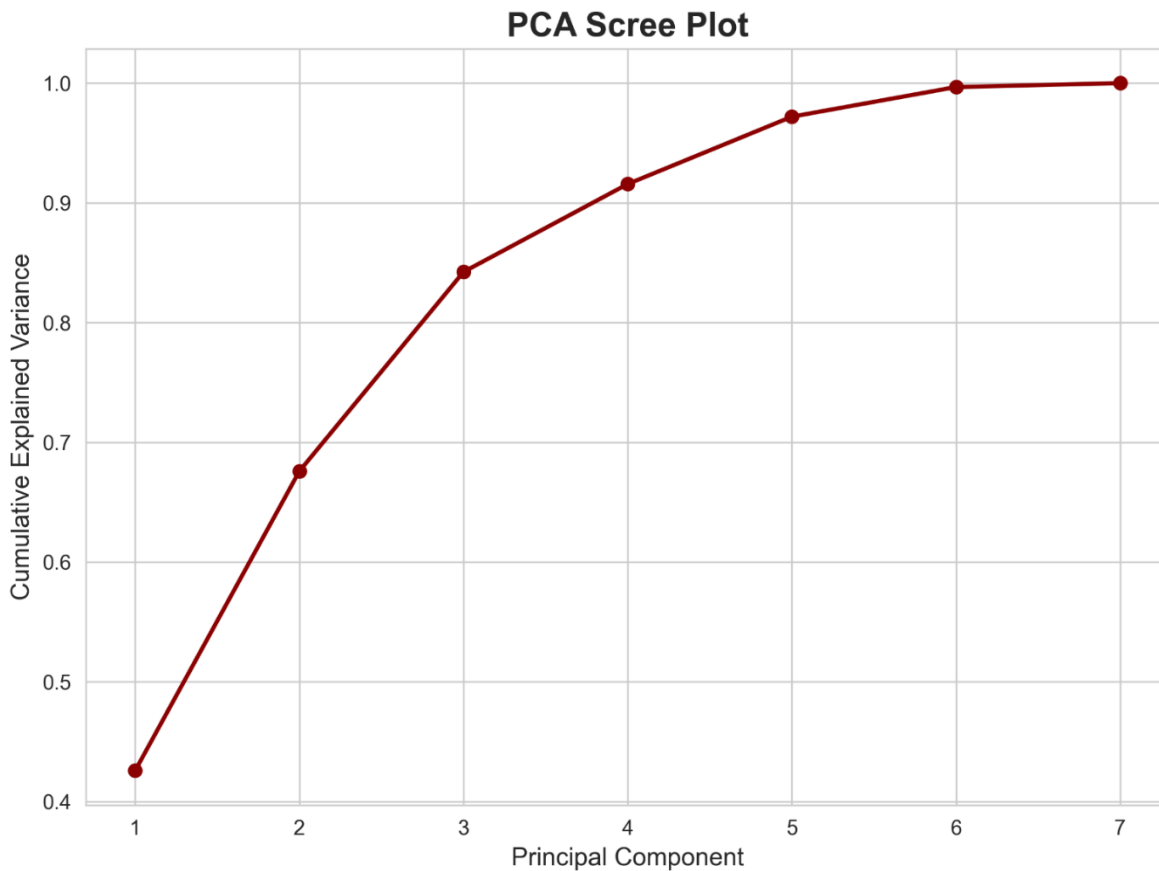


Figure 23. The PCA scree plot

The PCA Scree Plot shows the **cumulative explained variance** contributed by each principal component (PC). It is used to determine how many principal components should be retained for interpretation in Our groundwater heavy metal study.

Principal Component Analysis (PCA) reduces many correlated variables (heavy metals) into a smaller number of components that explain most of the variability in the dataset.

In Our case, it was used to

1. identify possible pollution sources,
2. group metals with similar behaviour,
3. simplify interpretation of groundwater contamination.

UNDERSTANDING THE SCREE PLOT

Axes

X-axis represents Principal Components (PC1, PC2, PC3...)

Each component is a linear combination of the original metals.

Y-axis represents cumulative explained variance.

This shows how much total information in the dataset is explained as more components are added.

INTERPRETATION OF OUR PLOT

1. PC1 Explains the Largest Portion of Variance

The first principal component (PC1) explains approximately 43% of the total variance.

This means one dominant process strongly influences groundwater chemistry, many metals vary together, or/ and a common source or geochemical control exists.

Possible sources could be rock weathering, mining activities and groundwater-rock interaction.

2. PC1 + PC2 Explain About 68%

The first two components together explain approximately 68% of the dataset variability.

This indicates two major controlling factors dominate groundwater chemistry.

PC1 could be due to geogenic influence and PC2 could be anthropogenic activities.

Environmental Meaning

This suggests groundwater chemistry is controlled mainly by natural geological processes, localized human activities.

3. PC1–PC3 Explain About 84%

The cumulative variance after PC3 is approximately 84%

This means the first three components capture most of the information in the dataset, and/ or additional components contribute very little new information.

PC1, PC2 and PC3, thereby, can be confidently be retained for interpretation because this is commonly accepted in environmental PCA studies.

4. Elbow Point of the Scree Plot

The curve begins to flatten after PC3. This flattening is called the "elbow point"

The elbow indicates diminishing returns from additional components, later PCs mostly contain noise or minor variability. Thus, PCs after PC3 are less environmentally significant.

This means PC5–PC7 add very little information, and they are not major controlling processes.

These components often represent random variation, analytical noise and localized minor effects.

PCA in this study then, suggests that **groundwater chemistry is controlled mainly by 3 major factors:**

1. Geogenic processes
2. Anthropogenic activities
3. Hydrochemical/mineral interactions

And the dominance of the first few PCs indicates:

1. strong relationships among metals,
2. common controlling mechanisms,
3. structured contamination patterns rather than random pollution.

This supports source apportionment, pollution identification, and the multivariate interpretation.

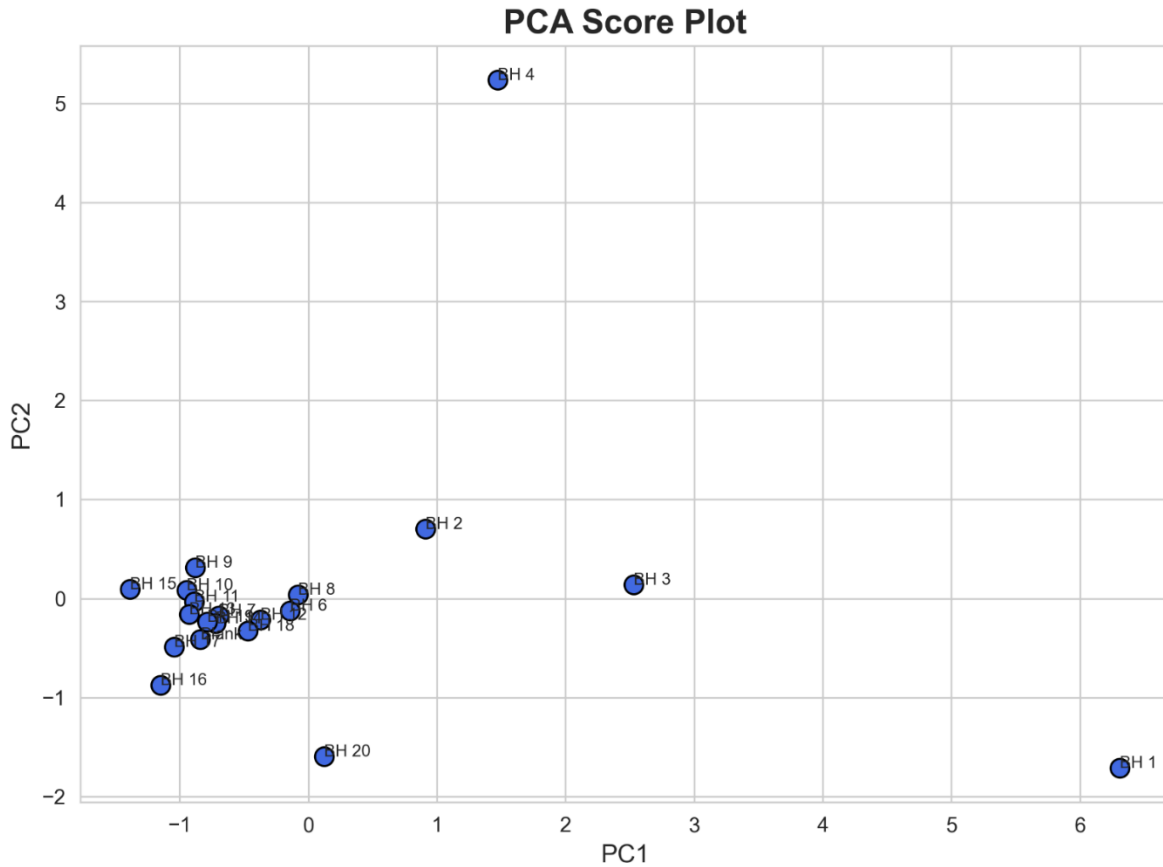


Figure 24. PCA score plot

The PCA score plot revealed that most groundwater samples clustered closely around the origin, indicating relatively homogeneous hydrochemical characteristics and suggesting dominant geogenic control within the aquifer system. However, BH1 and BH4 were distinctly separated from the main cluster, indicating anomalous heavy metal compositions and localized hydrochemical variability. The strong separation of BH1 along PC1 is likely associated with elevated iron concentrations, while BH4 exhibited strong influence along PC2, suggesting localized enrichment from secondary contamination processes. The observed clustering pattern indicates that groundwater quality within the study area is generally similar, with only a few locations exhibiting distinct contamination signatures.

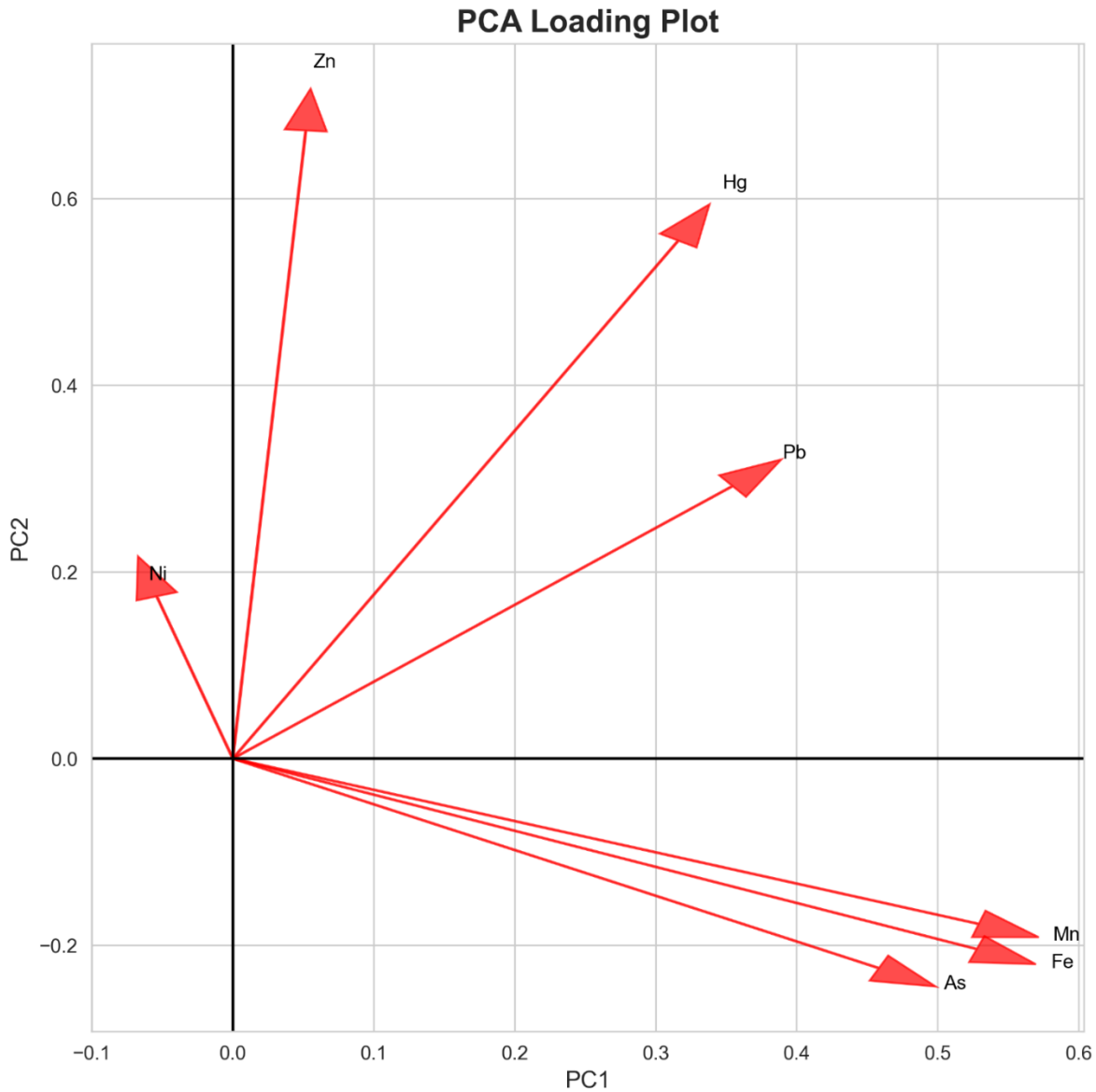


Figure 25. PCA loading plot

The PCA loading plot revealed distinct associations among the analyzed heavy metals. Fe, Mn, and As exhibited strong positive loadings along PC1, suggesting a common geogenic origin associated with mineral weathering and groundwater-rock interaction. Pb and Hg also showed positive associations, indicating possible anthropogenic contributions related to mining activities and metal contamination. Zn loaded strongly on PC2, suggesting localized enrichment and distinct hydrochemical behavior. In contrast, Ni exhibited weak negative loading on PC1, implying a separate geochemical source and limited contribution to the dominant hydrochemical processes. Overall, the PCA results indicate that groundwater quality within the study area is controlled by both natural geological processes and localized anthropogenic influences.

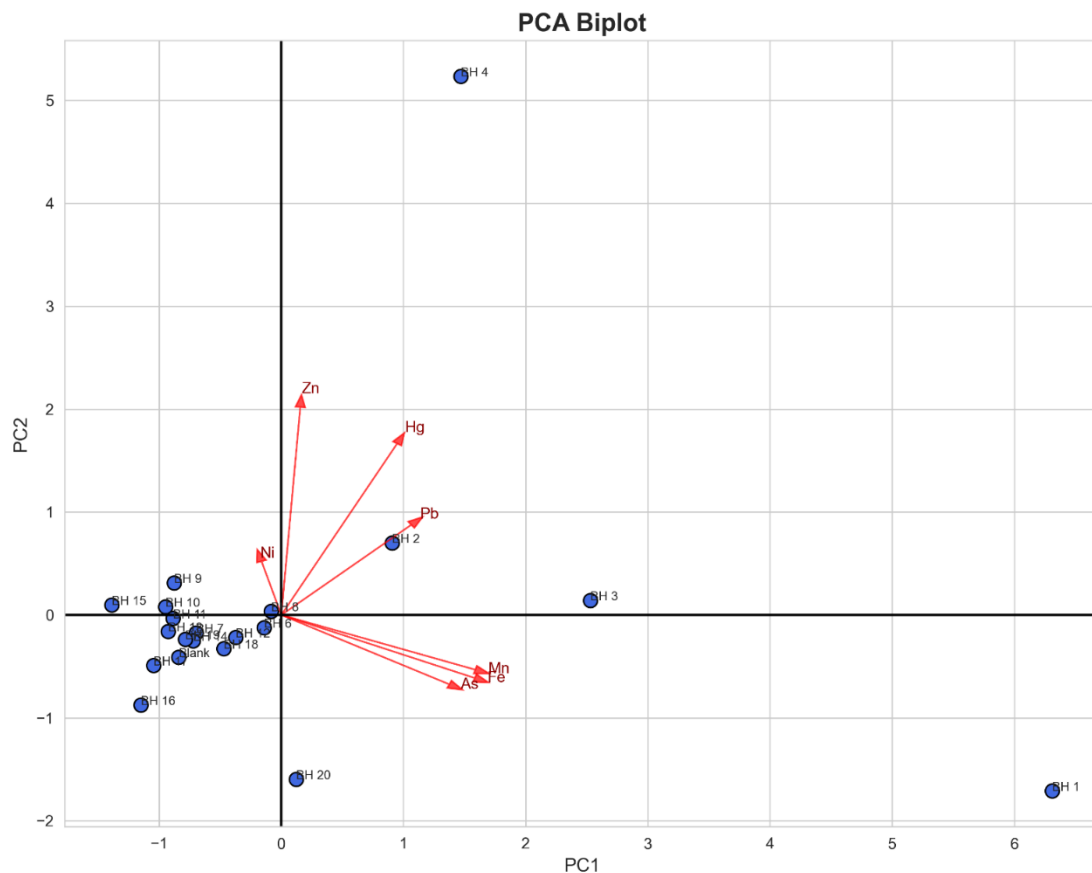


Figure 26. PCA biplot

The PCA biplot revealed distinct associations between groundwater samples and heavy metals within the study area. BH1 showed strong association with Fe, Mn, and As, suggesting dominant geogenic influence associated with mineral weathering and groundwater-rock interaction. BH2 was closely associated with Pb and Hg, indicating possible anthropogenic contamination linked to mining activities and metal pollution. BH4 exhibited strong association with Zn, suggesting localized zinc enrichment and distinct hydrochemical behavior. Most groundwater samples clustered near the origin, indicating relatively homogeneous hydrochemical characteristics and limited contamination across much of the study area. Overall, the PCA biplot demonstrates that groundwater quality is controlled by both natural geological processes and localized anthropogenic influences.

Spatial Distribution Discussion

Spatial interpolation was performed using the Inverse Distance Weighting (IDW) method (Li, J., & Heap, A. D., 2011). The generated concentration maps indicate that the distribution of heavy metals is spatially variable across the study area, suggesting that groundwater quality is influenced by local geological conditions and human activities rather than being uniformly distributed.

In this studies, calculations and maps where done in GIS softwares such as ArcGIS and QGIS, and were further confirmed with python programming language using some of its most renowned libraries for GeoScience spatial investigation.

The spatial distribution of polluted samples shows that 0 out of 19 sampling locations exceeded the Heavy Metal Pollution Index threshold. These locations deserve priority attention for routine groundwater monitoring and possible remediation.

Areas showing elevated concentrations of multiple heavy metals may indicate common contamination sources such as mining activities, weathering of mineralized rocks, agricultural inputs or domestic waste disposal. The interpolation maps provide a useful visual tool for identifying groundwater quality hotspots and supporting environmental management decisions.

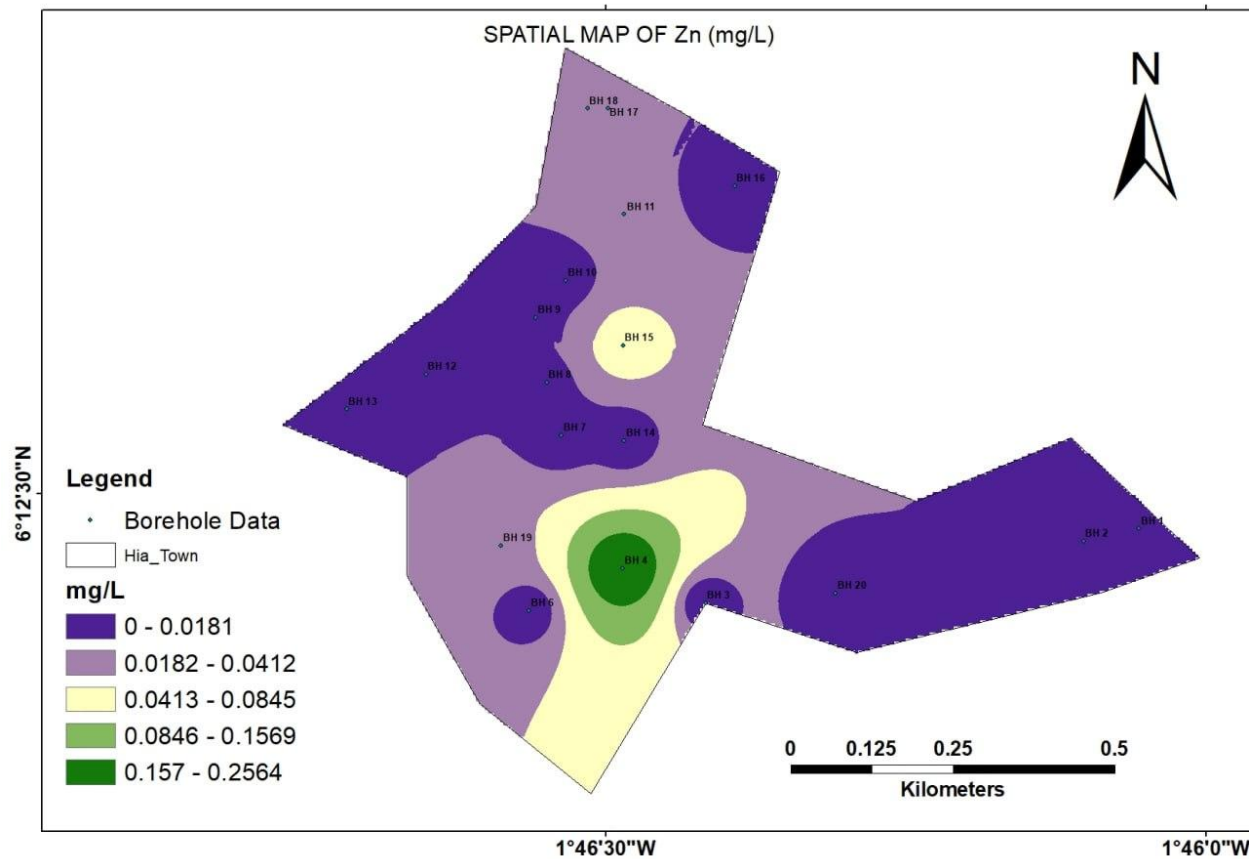


Figure 27. The spatial distribution map of Zn

The spatial distribution map of Zn reveals considerable spatial variability across the study area. Elevated Zn concentrations were observed in the southern-central section of the area, forming a distinct hotspot zone. This localized enrichment may be attributed to geogenic weathering processes or possible anthropogenic activities within the vicinity. The remaining portions of the study area exhibited relatively low Zn concentrations, suggesting limited widespread contamination. The observed spatial heterogeneity indicates that Zn occurrence is controlled by localized hydrogeochemical conditions and possible point-source contamination.

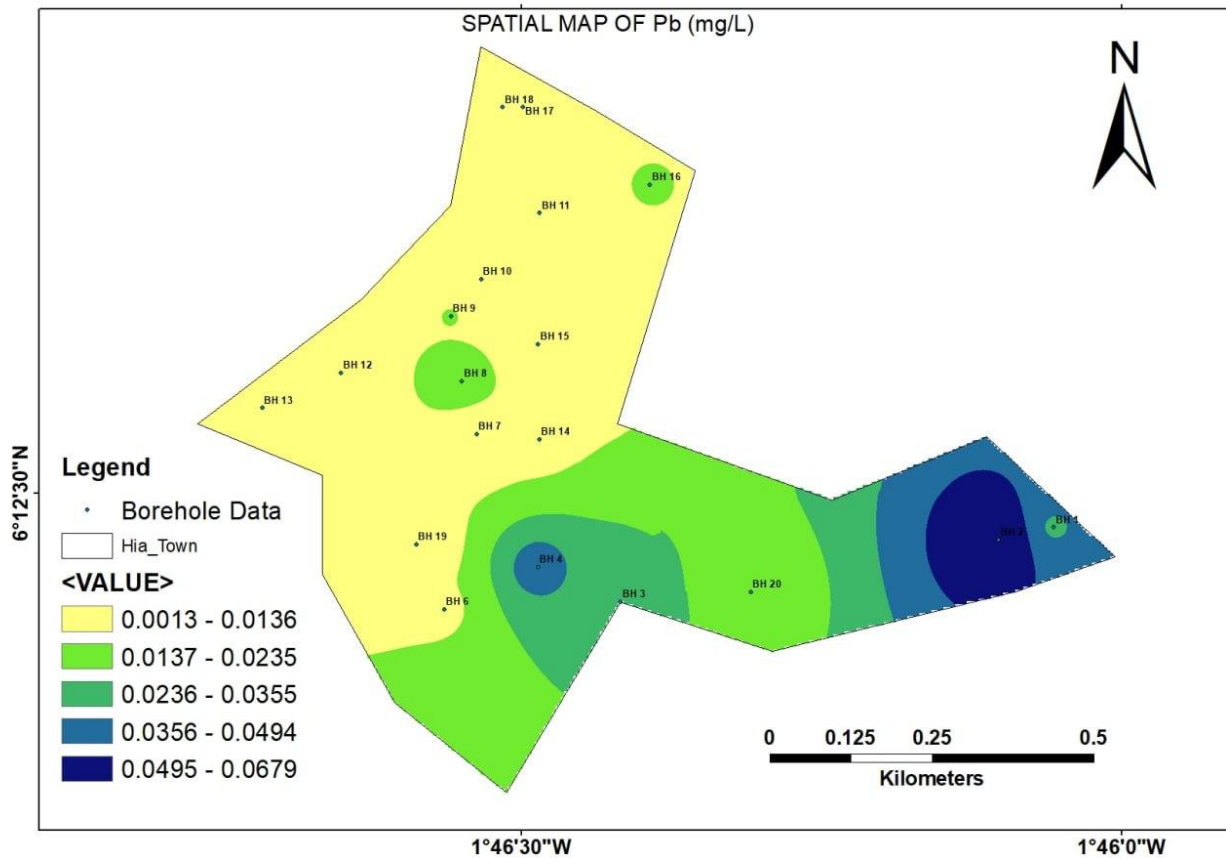


Figure 28. spatial distribution map of Pb

The spatial distribution map of Pb revealed considerable spatial variability across the study area, with elevated concentrations occurring predominantly in the southeastern portion. The observed hotspot suggests localized Pb enrichment likely associated with anthropogenic activities such as mining, metal leaching, or waste disposal. In contrast, the northwestern and western regions exhibited relatively low Pb concentrations, indicating limited contamination and more stable groundwater conditions. The heterogeneous distribution pattern suggests that Pb contamination is controlled by localized hydrogeochemical and anthropogenic processes rather than uniform regional background conditions.

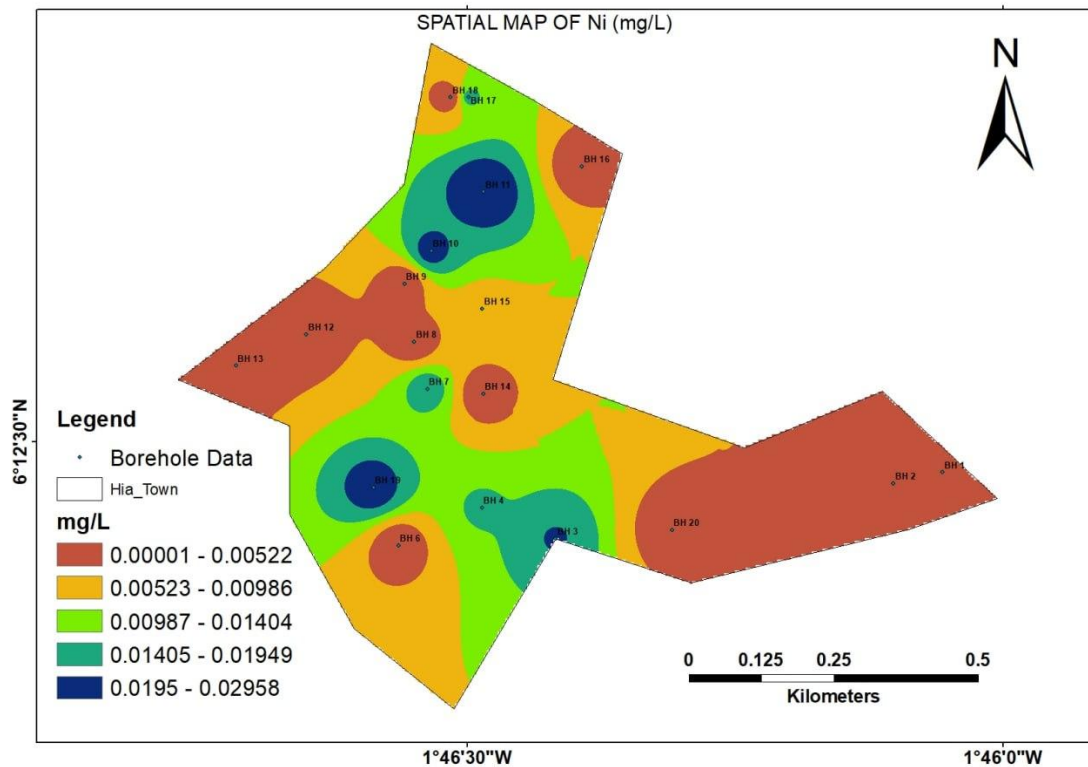


Figure 29. spatial distribution map of Ni

The spatial distribution map of Ni revealed heterogeneous concentration patterns across the study area, with relatively elevated concentrations occurring mainly within the northwestern and southwestern portions. In contrast, the central and eastern regions exhibited comparatively lower Ni levels. The localized enrichment zones suggest that Ni distribution is controlled by spatially variable hydrogeochemical processes, including mineral weathering and possible anthropogenic contributions. The absence of widespread high concentrations indicates that Ni contamination is localized rather than regionally extensive. Furthermore, the distinct spatial behavior of Ni compared to Pb and Hg supports the PCA results, which suggested that Ni may originate from different geochemical or contamination sources.

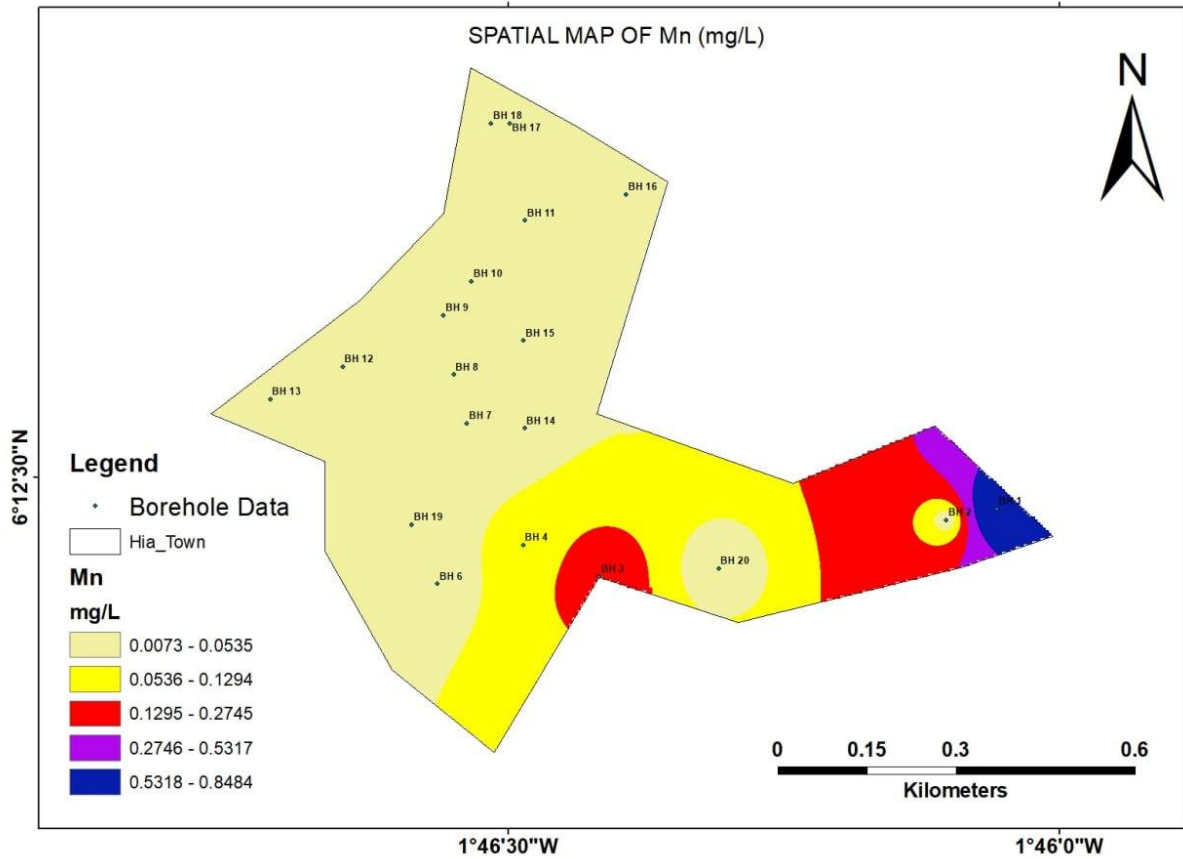


Figure 30. spatial distribution map of Mn

The map indicates that Mn is not uniformly distributed across the area. Instead, it exhibits a distinct hotspot in the southeast, while the central-southern area shows comparatively depleted concentrations. This spatial variability suggests that localized environmental controls strongly influence Mn distribution.

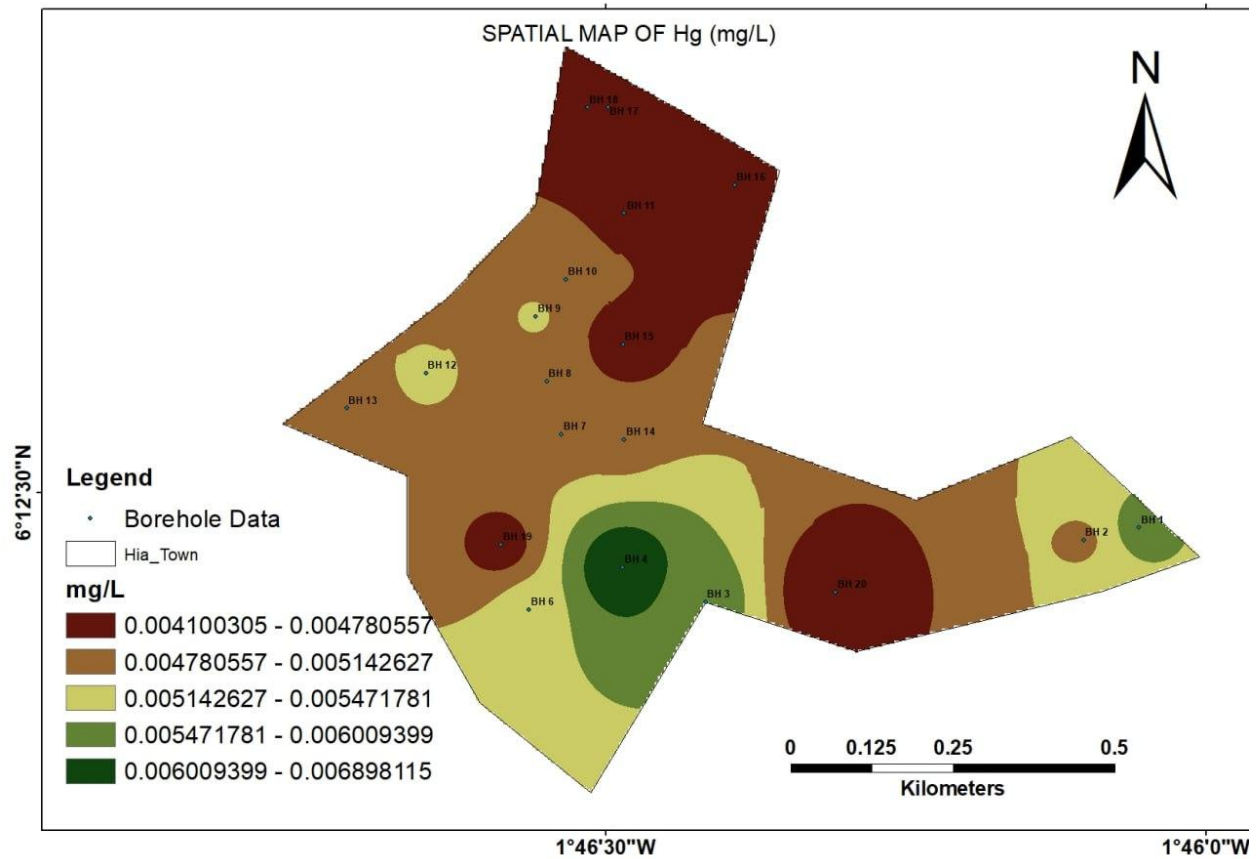


Figure 31. spatial distribution map of Hg

The Hg spatial distribution map reveals that mercury concentrations are generally moderate to low across most of the study area, but a pronounced localized hotspot occurs in the southwestern-central section. This pattern differs from the Mn, MI, and HPI distributions, suggesting that Hg contamination may be controlled by different environmental or anthropogenic factors. The hotspot indicates a potentially important zone for further environmental monitoring and contamination assessment.

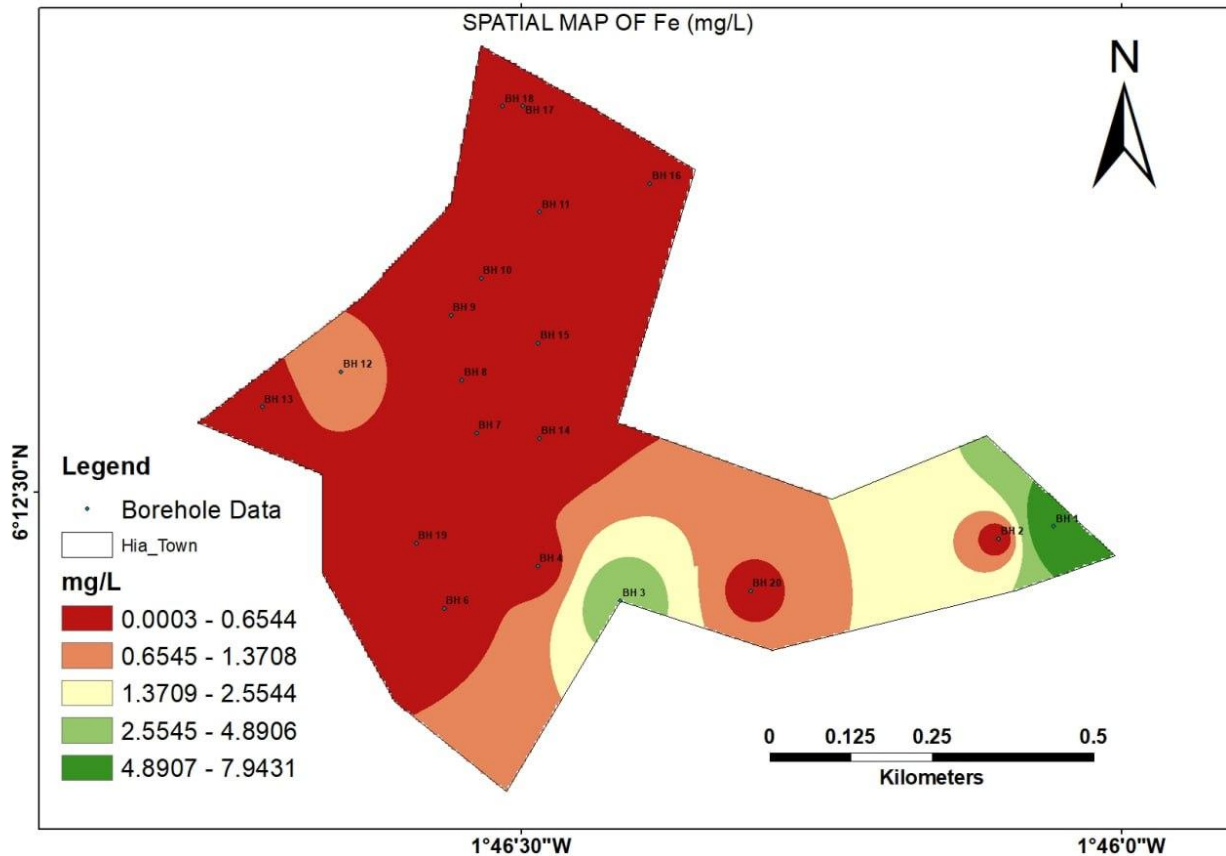


Figure 32. spatial distribution map of Fe

The spatial distribution map of Fe (Iron) shows clear spatial variability across the study area, indicating that Fe concentrations are not uniformly distributed.

High Fe concentrations are indicated in the eastern to southeastern section of the study area (yellow to bright green colors). The highest hotspot appears around longitude -1.768 to -1.766 and latitude 6.208 – 6.209 . This suggests localized enrichment of iron in that zone.

Moderate Fe concentrations occur in the southern-central area, where colors transition through green and teal.

Low Fe concentrations are observed in the central to southwestern region (purple to dark blue colors), especially around:

longitude -1.771 to -1.770

latitude 6.207 – 6.209

Spatial Trend

There is a noticeable gradient where Fe concentrations generally; increase from west to east, and peak toward the southeastern edge of the area.

Possible Environmental Implications

The elevated Fe concentrations in the eastern/southeastern area may be linked to natural geological mineralization, iron-rich parent materials, weathering processes, or anthropogenic activities such as mining, waste disposal, or runoff accumulation.

Relationship with Other Metals

The Fe hotspot pattern appears similar to the spatial patterns previously observed for HPI, and HEI,

suggesting that Fe may contribute significantly to the overall heavy metal pollution intensity in the area.

The map indicates that Fe contamination/enrichment is spatially heterogeneous, the southeastern zone is the most enriched area, and the central region contains comparatively lower Fe levels.

This suggests that monitoring and possible remediation efforts should focus more on the eastern/southeastern portion of the study area.

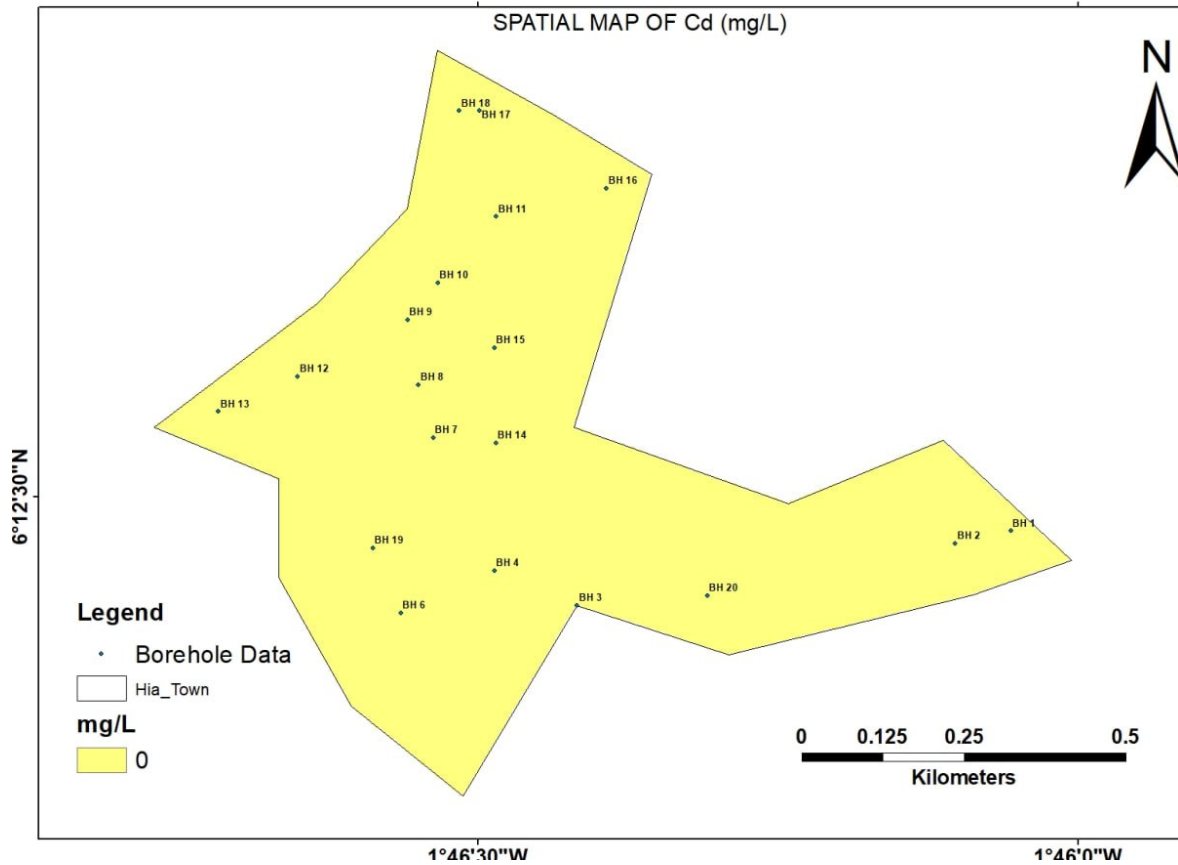


Figure 33. spatial distribution map of Cd

Figure 34. spatial distribution map of Cd

The spatial distribution map of Cd indicates very limited spatial variability across the study area. Cd concentrations were consistently low and remained below detectable or significant levels in most groundwater samples. The absence of distinct hotspot zones suggests minimal Cd contamination within the aquifer system. Minor variations observed on the interpolated surface are likely artifacts of the interpolation process resulting from the extremely low variability in the dataset. Overall, the findings indicate that Cd does not constitute a significant groundwater pollution concern in the study area.

The Cd results collectively indicate Cadmium is not a major groundwater contamination issue in the study area.

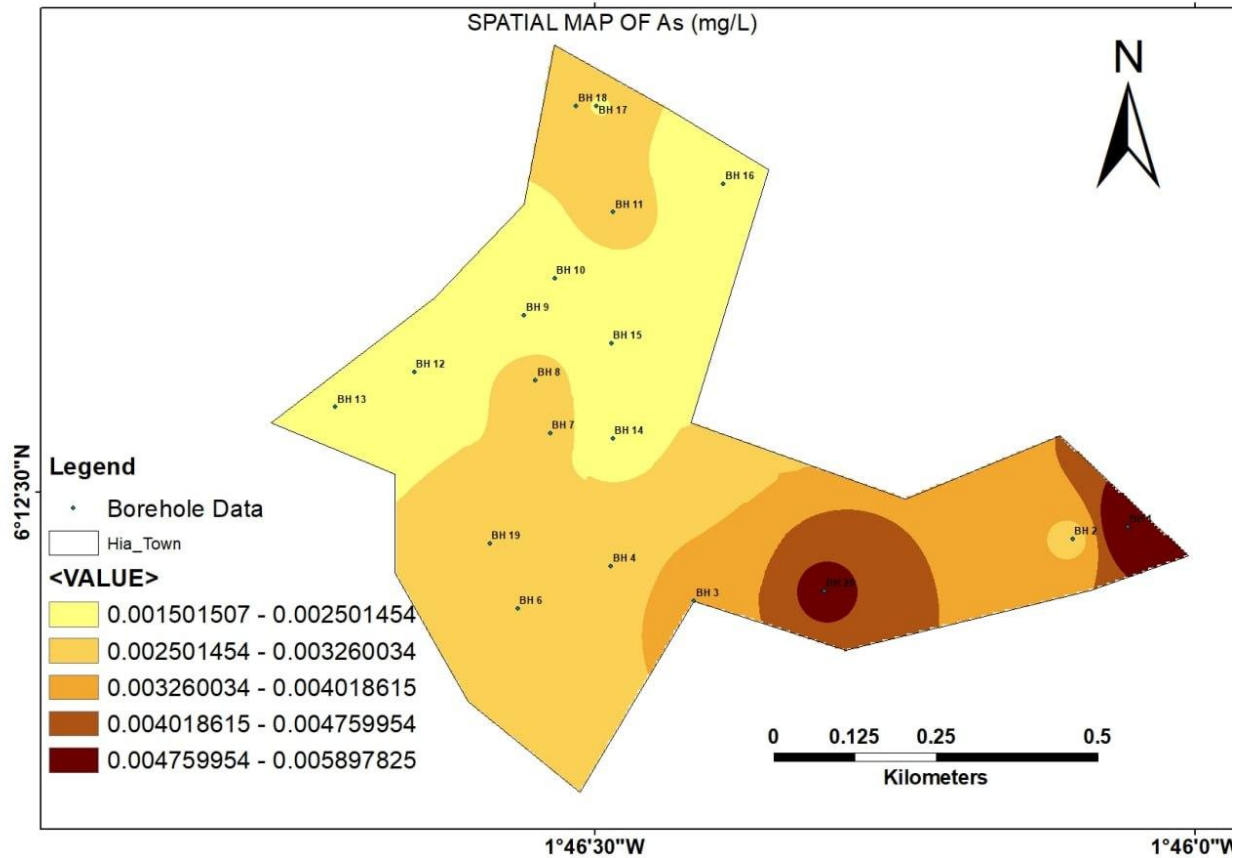


Figure 35. spatial distribution map of As

The spatial distribution map of As reveals significant spatial variability across the study area, with elevated concentrations occurring predominantly within the southeastern section. The identified hotspot zone suggests localized arsenic enrichment possibly associated with geogenic weathering processes or anthropogenic activities. Lower concentrations were observed in the northern and central parts of the study area, indicating reduced arsenic influence within those regions. The observed spatial trend suggests that arsenic distribution is controlled by localized hydrogeochemical conditions and possible groundwater flow dynamics.

Arsenic enrichment is localized and concentrated within specific portions of the aquifer.

Cluster Membership Discussion

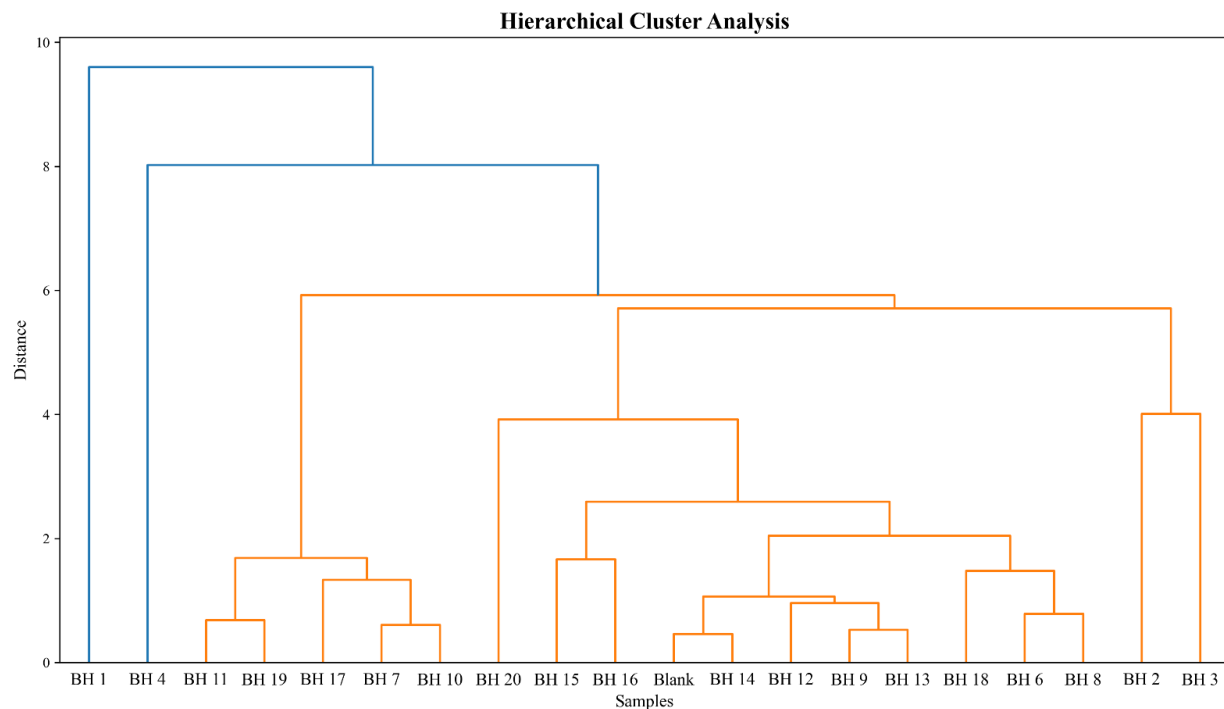


Figure 36. The hierarchical cluster analysis Dendrogram

The hierarchical cluster analysis identifies clear groupings among the sampling locations based on heavy metal characteristics. BH 1 and BH 4 are the most chemically distinct samples, indicating localized anomalies or unique contamination conditions. Most other samples form coherent clusters with strong internal similarity, suggesting shared geochemical controls or contamination sources. Overall, the dendrogram confirms substantial spatial heterogeneity in heavy metal distribution across the study area.

The vertical axis (“Distance”) represents the degree of dissimilarity:

Lower linkage distances indicate samples with very similar chemical compositions.

Higher linkage distances indicate samples that are more dissimilar.

1. Presence of Distinct Sample Groups

The dendrogram reveals several major clusters, indicating that the sampling locations can be grouped according to similarities in heavy metal composition.

Broadly, the samples separate into a few isolated/distinct samples, and several closely related clusters.

2. BH 1 Is Highly Distinct

BH 1 separates from all other samples at the highest linkage distance (around 9.5–10).

This means BH 1 has the most unique heavy metal signature in the dataset.

BH 1 may represent a localized contamination source, strong geochemical anomaly, different hydrochemical conditions, or unusually high/low metal concentrations compared to other sites.

This sample deserves special attention because it behaves independently from the rest of the dataset.

3. BH 4 Also Forms an Isolated Cluster

BH 4 joins the rest of the clusters at a relatively high distance (~8).

This indicates that BH 4 is also chemically distinct, though less extreme than BH 1.

BH 4 may experience different contamination processes, be influenced by unique lithology/geochemistry, or represent another localized pollution zone.

4. Strongly Similar Sample Pairs

Several samples merge at very low linkage distances, indicating very similar heavy metal characteristics.

Highly Similar Pairs Include:

BH 11 – BH 19

BH 7 – BH 10

Blank – BH 14

BH 9 – BH 13

BH 6 – BH 8

These paired samples likely share similar contamination sources, occur under similar environmental conditions, or belong to the same hydrogeochemical zone.

The close clustering suggests strong compositional similarity.

5. Larger Cluster Structures

Cluster A

Includes BH 11, BH 19, BH 17, BH 7, BH 10

This cluster suggests similar heavy metal concentrations, comparable geochemical behavior, possibly a common source or transport mechanism.

Cluster B

Includes Blank, BH 14, BH 12, BH 9, BH 13

This group appears relatively homogeneous and may represent background or low-contamination conditions, or similar environmental settings.

The clustering of the Blank with BH 14 suggests BH 14 may contain very low contamination levels close to background conditions.

Cluster C

Includes BH 18, BH 6, BH 8, BH 2, BH 3

This cluster forms another chemically related group, though BH 2 and BH 3 join at slightly larger distances, indicating moderate variability within the group.

6. Intermediate Samples

Some samples such as:

BH 15,

BH 16,

BH 20

Act as transitional samples linking larger clusters together.

This suggests mixed geochemical characteristics, intermediate contamination levels, or overlapping environmental influences.

7. Environmental Implications

The clustering pattern indicates that heavy metal contamination is spatially heterogeneous, influenced by localized factors, and not uniformly distributed across the study area.

The isolated behavior of BH 1 and BH 4 suggests potential hotspots or unique hydrochemical environments.

Meanwhile, clustered samples likely reflect shared contamination pathways, similar lithology, similar groundwater flow conditions, or common anthropogenic influences.

CONCLUSIONS

This study evaluated groundwater quality using 19 groundwater samples collected within the study area. Descriptive statistics, pollution indices, multivariate statistical techniques and spatial interpolation were integrated to assess heavy metal contamination.

The pollution assessment showed that 6 sampling location(s) exceeded the recommended Heavy Metal Pollution Index (HPI) threshold, indicating that groundwater quality is not spatially uniform.

Principal Component Analysis indicated that the first principal component explained 43% of the overall variance, suggesting that a limited number of controlling processes account for much of the groundwater chemistry.

Overall, the integrated statistical and spatial analyses indicate that groundwater quality is controlled by both natural geochemical processes and localized anthropogenic activities. The generated maps and pollution indices provide a scientific basis for groundwater monitoring and sustainable resource management.

RECOMMENDATIONS

Establish routine groundwater quality monitoring programmes for all sampling locations.

Prioritize locations with elevated Heavy Metal Pollution Index values for detailed investigation.

Investigate possible anthropogenic sources of contamination including mining, agriculture and waste disposal activities.

Increase sampling density to improve spatial interpolation accuracy.

Conduct seasonal groundwater monitoring to evaluate temporal variations in heavy metal concentrations.

Complement chemical analyses with hydrogeological and isotopic investigations to better understand contaminant pathways.

Develop groundwater protection strategies based on identified contamination hotspots.

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APPENDIX

TABLES

Table 10. Summary of the groundwater dataset.

S/N	Sample ID	Latitude	Longitude	Cd	Fe	Mn	Ni	Pb	Zn	As	Hg
1	Blank	6.20831	-1.765585	< 0.00001	0.0074	< 0.00001	< 0.00001	0.0118	< 0.00001	0.0024	0.0048
2	BH 1	6.207853	-1.767603	< 0.00001	7.949	0.849	0.0004	0.0345	< 0.00001	0.0059	0.0056
3	BH 2	6.207683	-1.76837	< 0.00001	0.1324	0.0258	< 0.00001	0.0679	< 0.00001	0.003	0.0051
4	BH 3	6.206825	-1.773623	< 0.00001	4.1875	0.2355	0.0201	0.0339	< 0.00001	0.0039	0.0057
5	BH 4	6.207303	-1.77477	< 0.00001	0.2544	0.065	0.0149	0.0426	0.2567	0.0027	0.0069
6	BH 6	6.20672	-1.776067	< 0.00001	0.3281	0.0401	< 0.00001	0.0124	< 0.00001	0.0029	0.0053
7	BH 7	6.20915	-1.775618	< 0.00001	0.589	0.0162	0.0184	0.0042	< 0.00001	0.0026	0.005
8	BH 8	6.209883	-1.77582	< 0.00001	0.2132	0.0095	< 0.00001	0.023	0.0111	0.003	0.0051
9	BH 9	6.21078	-1.775975	< 0.00001	0.1046	0.0198	< 0.00001	0.014	0.0131	0.0015	0.0052
10	BH 10	6.211297	-1.775553	< 0.00001	0.445	0.0128	0.0232	0.0018	0.0112	0.0023	0.005
11	BH 11	6.212212	-1.774745	< 0.00001	0.4858	0.0187	0.0296	0.0013	0.026	0.0028	0.0047
12	BH 12	6.21	-1.777498	< 0.00001	1.2055	0.0247	< 0.00001	0.0086	< 0.00001	0.0022	0.0052
14	BH 13	6.20951	-1.778595	< 0.00001	0.3902	0.0125	< 0.00001	0.0103	< 0.00001	0.0017	0.005
15	BH 14	6.209071	-1.774748	< 0.00001	< 0.00001	0.0451	< 0.00001	0.0106	< 0.00001	0.0022	0.005
16	BH 15	6.210396	-1.774765	< 0.00001	< 0.00001	0.0503	0.0074	0.0026	0.0678	0.0017	0.0044
17	BH 16	6.212603	-1.773211	< 0.00001	0.3127	0.0113	< 0.00001	0.0144	< 0.00001	0.002	0.0042
18	BH 17	6.21368	-1.77497	< 0.00001	0.2641	0.052	0.0159	0.0048	0.0184	0.0024	0.0044
19	BH 18	6.21368	-1.775255	< 0.00001	< 0.00001	0.0406	0.0009	0.0107	0.0402	0.0032	0.0046
20	BH 19	6.207618	-1.776455	< 0.00001	0.4367	0.0073	0.0253	0.0036	0.0326	0.0032	0.0045
21	BH 20	6.206955	-1.771813	< 0.00001	0.4366	0.0086	0.0014	0.0143	< 0.00001	0.005	0.0041

Table 11. Cleaned groundwater dataset.

S/N	Sample ID	Latitude	Longitude	Cd	Fe	Mn	Ni	Pb	Zn	As	Hg
2	BH 1	6.207853	-1.7676	0.000005	7.949	0.849	0.0004	0.0345	0.000005	0.0059	0.0056
3	BH 2	6.207683	-1.76837	0.000005	0.1324	0.0258	0.000005	0.0679	0.000005	0.003	0.0051
4	BH 3	6.206825	-1.77362	0.000005	4.1875	0.2355	0.0201	0.0339	0.000005	0.0039	0.0057
5	BH 4	6.207303	-1.77477	0.000005	0.2544	0.065	0.0149	0.0426	0.2567	0.0027	0.0069
6	BH 6	6.20672	-1.77607	0.000005	0.3281	0.0401	0.000005	0.0124	0.000005	0.0029	0.0053
7	BH 7	6.20915	-1.77562	0.000005	0.589	0.0162	0.0184	0.0042	0.000005	0.0026	0.005
8	BH 8	6.209883	-1.77582	0.000005	0.2132	0.0095	0.000005	0.023	0.0111	0.003	0.0051
9	BH 9	6.21078	-1.77598	0.000005	0.1046	0.0198	0.000005	0.014	0.0131	0.0015	0.0052
10	BH 10	6.211297	-1.77555	0.000005	0.445	0.0128	0.0232	0.0018	0.0112	0.0023	0.005
11	BH 11	6.212212	-1.77475	0.000005	0.4858	0.0187	0.0296	0.0013	0.026	0.0028	0.0047
12	BH 12	6.21	-1.7775	0.000005	1.2055	0.0247	0.000005	0.0086	0.000005	0.0022	0.0052
14	BH 13	6.20951	-1.7786	0.000005	0.3902	0.0125	0.000005	0.0103	0.000005	0.0017	0.005
15	BH 14	6.209071	-1.77475	0.000005	0.000005	0.0451	0.000005	0.0106	0.000005	0.0022	0.005
16	BH 15	6.210396	-1.77477	0.000005	0.000005	0.0503	0.0074	0.0026	0.0678	0.0017	0.0044
17	BH 16	6.212603	-1.77321	0.000005	0.3127	0.0113	0.000005	0.0144	0.000005	0.002	0.0042
18	BH 17	6.21368	-1.77497	0.000005	0.2641	0.052	0.0159	0.0048	0.0184	0.0024	0.0044
19	BH 18	6.21368	-1.77526	0.000005	0.000005	0.0406	0.0009	0.0107	0.0402	0.0032	0.0046
20	BH 19	6.207618	-1.77646	0.000005	0.4367	0.0073	0.0253	0.0036	0.0326	0.0032	0.0045
21	BH 20	6.206955	-1.77181	0.000005	0.4366	0.0086	0.0014	0.0143	0.000005	0.005	0.0041

Table 12. Transformed Groundwater dataset

S/N	Sample ID	Latitude	Longitude	Cd	Fe	Mn	Ni	Pb	Zn	As	Hg
2	BH 1	6.207853	-1.7676	0.000005	2.597929	-0.16979	-6.95588	-2.93062	-17.0889	-26.1758	-12389.5
3	BH 2	6.207683	-1.76837	0.000005	-1.64729	-9.17152	-10.1812	-2.40621	-17.0889	-38.1159	-14862.8
4	BH 3	6.206825	-1.77362	0.000005	1.670924	-2.02774	-3.68197	-2.94382	-17.0889	-32.9765	-11970
5	BH 4	6.207303	-1.77477	0.000005	-1.18958	-5.32747	-3.9463	-2.77048	-1.40943	-40.3869	-8252.94
6	BH 6	6.20672	-1.77607	0.000005	-0.99357	-7.14026	-10.1812	-3.66841	-17.0889	-38.833	-13790.8
7	BH 7	6.20915	-1.77562	0.000005	-0.50095	-11.7931	-3.76025	-4.38262	-17.0889	-41.2313	-15446.8
8	BH 8	6.209883	-1.77582	0.000005	-1.31966	-15.5485	-10.1812	-3.23062	-5.07534	-38.1159	-14862.8
9	BH 9	6.21078	-1.77598	0.000005	-1.79764	-10.596	-10.1812	-3.5842	-4.86656	-55.6366	-14311.6
10	BH 10	6.211297	-1.77555	0.000005	-0.74455	-13.3416	-3.55446	-4.8978	-5.064	-44.0927	-15446.8
11	BH 11	6.212212	-1.77475	0.000005	-0.66986	-10.9262	-3.33659	-5.08607	-4.02167	-39.5888	-17423.5
12	BH 12	6.21	-1.7775	0.000005	0.190613	-9.39465	-10.1812	-3.91716	-17.0889	-45.1764	-14311.6
14	BH 13	6.20951	-1.7786	0.000005	-0.85389	-13.5066	-10.1812	-3.7955	-17.0889	-51.9848	-15446.8
15	BH 14	6.209071	-1.77475	0.000005	-4.3929	-6.66284	-10.1812	-3.77597	-17.0889	-45.1764	-15446.8
16	BH 15	6.210396	-1.77477	0.000005	-4.3929	-6.24123	-4.55489	-4.67873	-2.89002	-51.9848	-19810
17	BH 16	6.212603	-1.77321	0.000005	-1.03141	-14.2288	-10.1812	-3.56453	-17.0889	-47.5868	-21687.1
18	BH 17	6.21368	-1.77497	0.000005	-1.1614	-6.11682	-3.88915	-4.29802	-4.44403	-43.0785	-19810
19	BH 18	6.21368	-1.77526	0.000005	-4.3929	-7.08873	-6.31	-3.76957	-3.50011	-36.7854	-18168.3
20	BH 19	6.207618	-1.77646	0.000005	-0.7604	-17.7567	-3.47715	-4.47911	-3.74944	-36.7854	-18962.2
21	BH 20	6.206955	-1.77181	0.000005	-0.76059	-16.3526	-5.95127	-3.5694	-17.0889	-28.7193	-22728.4

Table 13. Normality Tests

	Metal	Shapiro p	D'Agostino p	Jarque-Bera p	Anderson	Interpretation
0	Cd	nan	nan	nan	nan	Constant values
1	Fe	0.0000	nan	0.0000	4.1760	Not Normally Distributed
2	Mn	0.0000	nan	0.0000	4.6988	Not Normally Distributed
3	Ni	0.0004	nan	0.2699	1.9693	Not Normally Distributed
4	Pb	0.0009	nan	0.0015	1.4106	Not Normally Distributed
5	Zn	0.0000	nan	0.0000	3.7382	Not Normally Distributed
6	As	0.0125	nan	0.0181	0.8668	Not Normally Distributed
7	Hg	0.0378	nan	0.0122	0.6027	Not Normally Distributed

Table 14. Outlier Detection

	Metal	Lower Limit	Upper Limit	Outliers	Outlier (%)
0	Cd	0.0000	0.0000	0	0.0000
1	Fe	-0.2661	0.9043	3	15.7900
2	Mn	-0.0399	0.1003	2	10.5300
3	Ni	-0.0257	0.0429	0	0.0000
4	Pb	-0.0168	0.0400	2	10.5300
5	Zn	-0.0333	0.0555	2	10.5300
6	As	0.0008	0.0045	2	10.5300
7	Hg	0.0036	0.0062	1	5.2600

BOXPLOTS

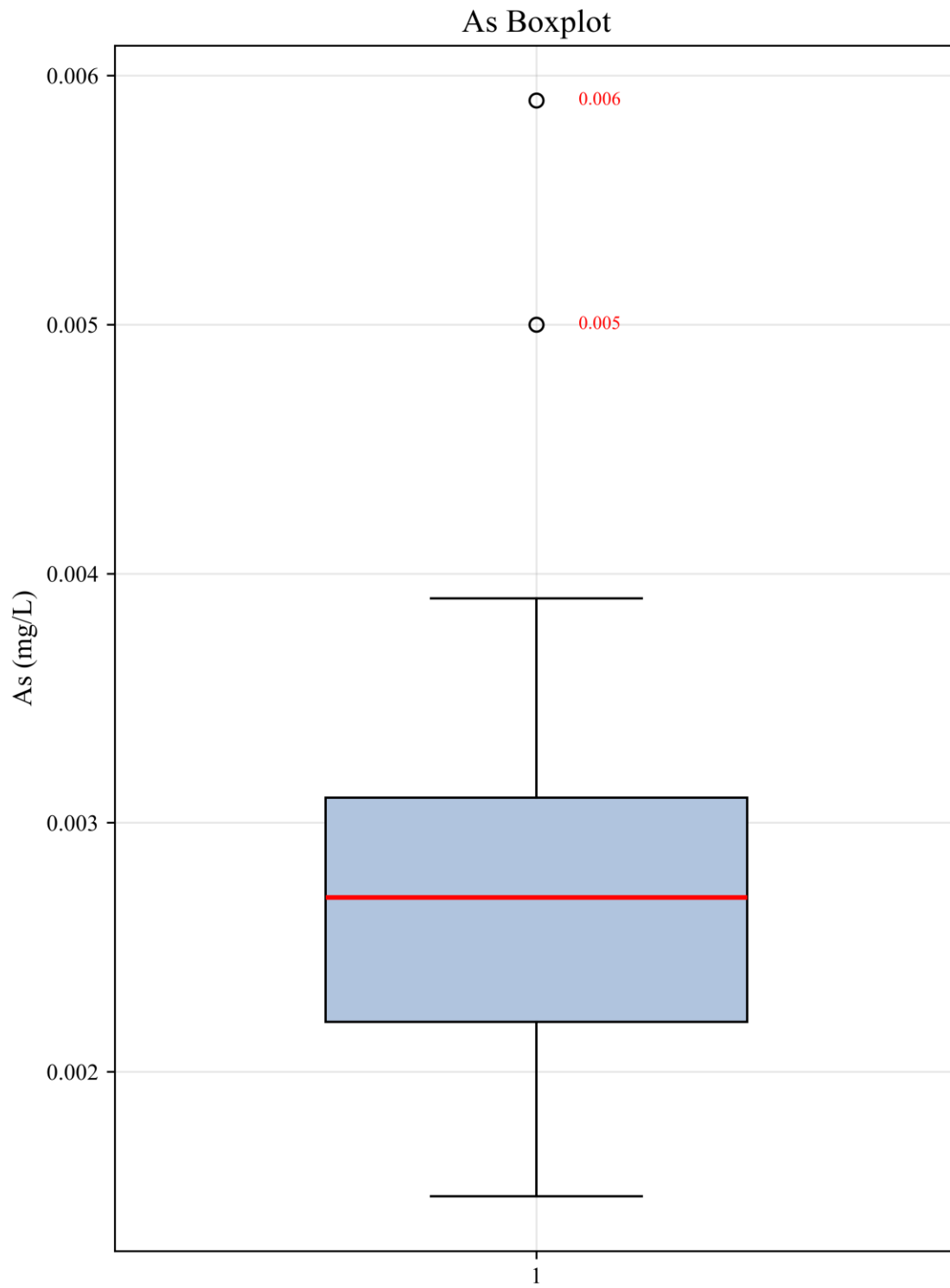


Figure 37. Boxplot illustrating the distribution of Arsenic (As).

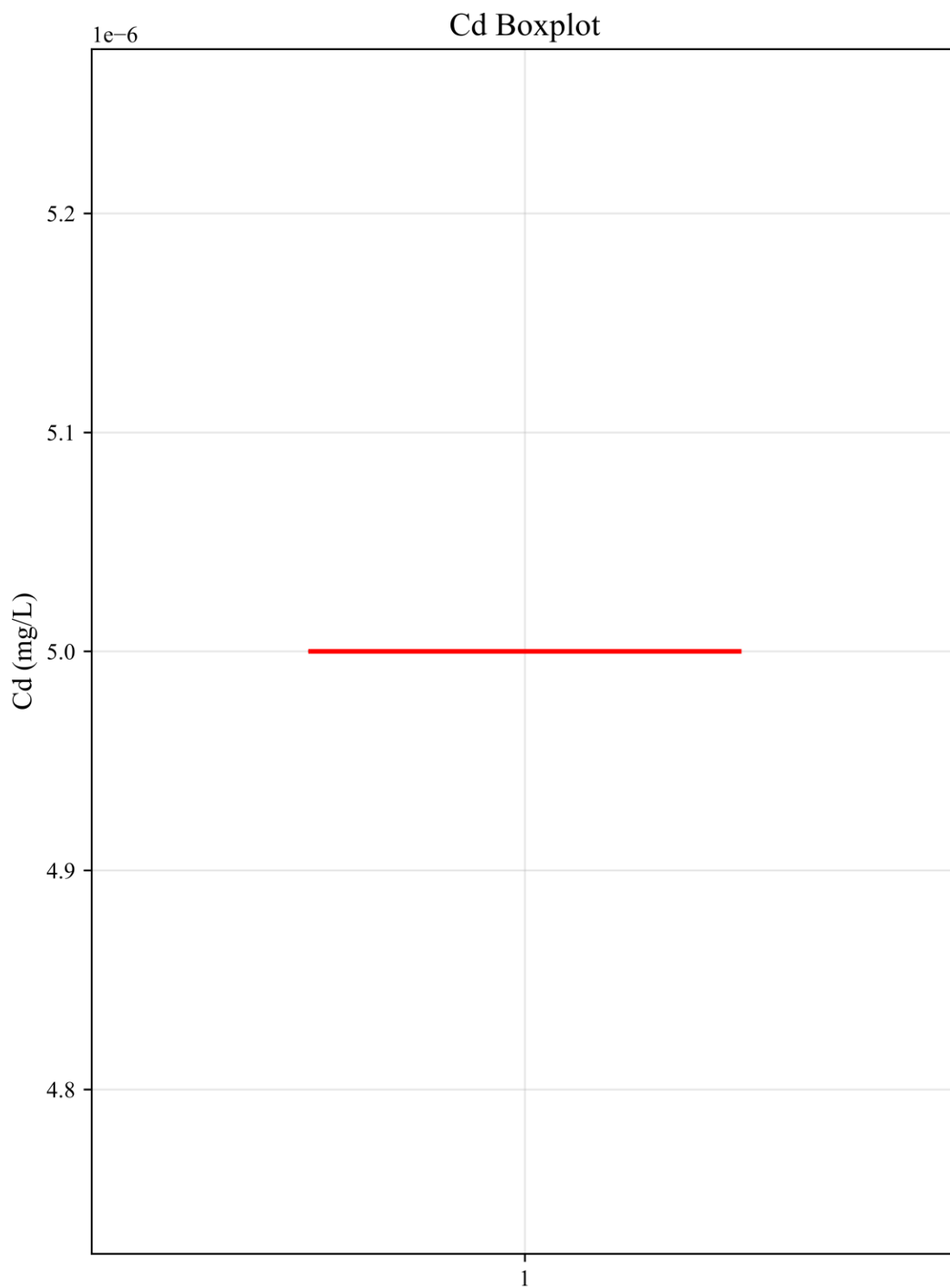


Figure 38. Boxplot illustrating the distribution of Cadmium (Cd).

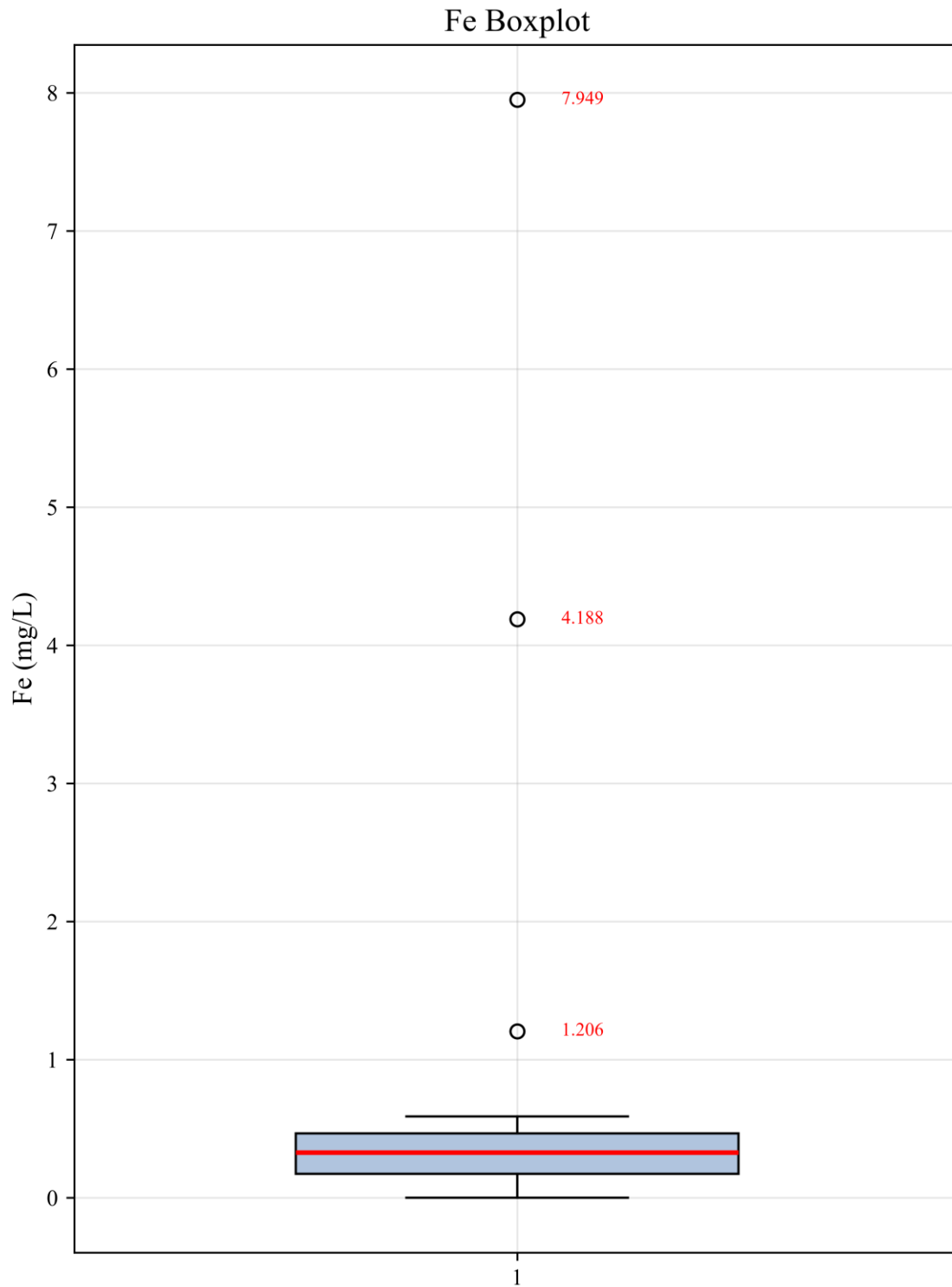


Figure 39. Boxplot illustrating the distribution of Iron (Fe).

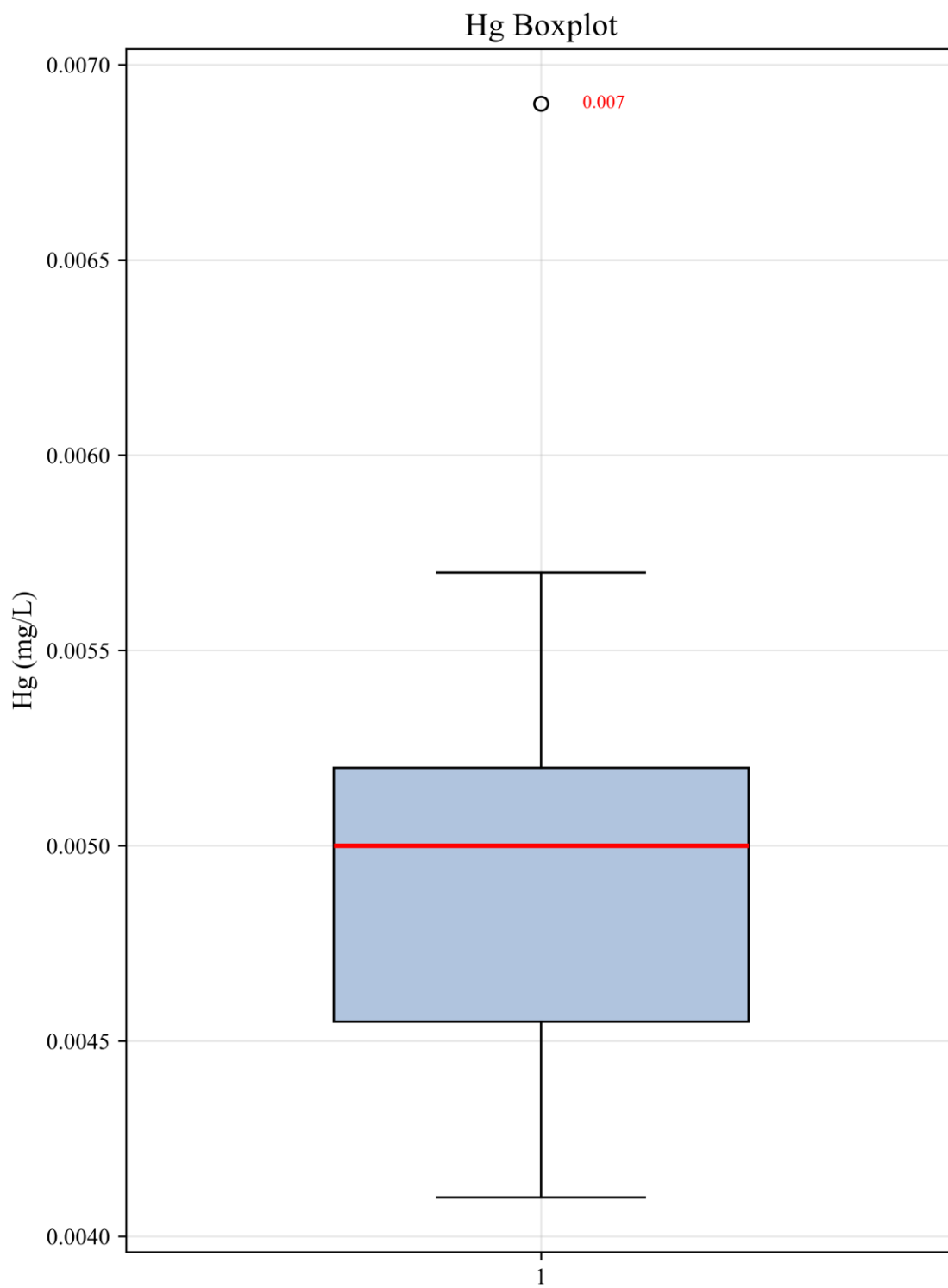


Figure 40. Boxplot illustrating the distribution of Mercury (Hg).



Figure 41. Boxplot illustrating the distribution of Manganese (Mn).

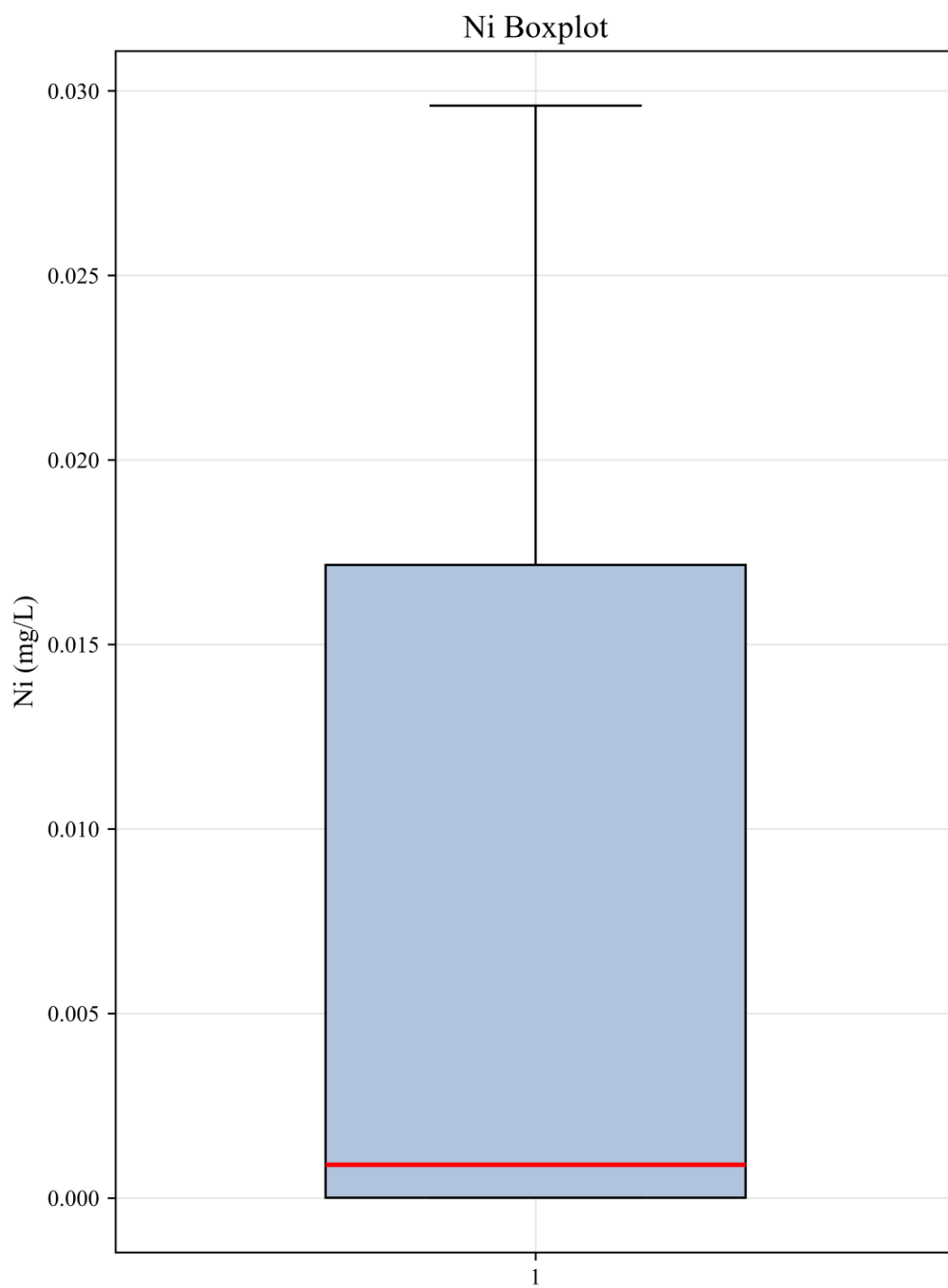


Figure 42. Boxplot illustrating the distribution of Nickel (Ni).

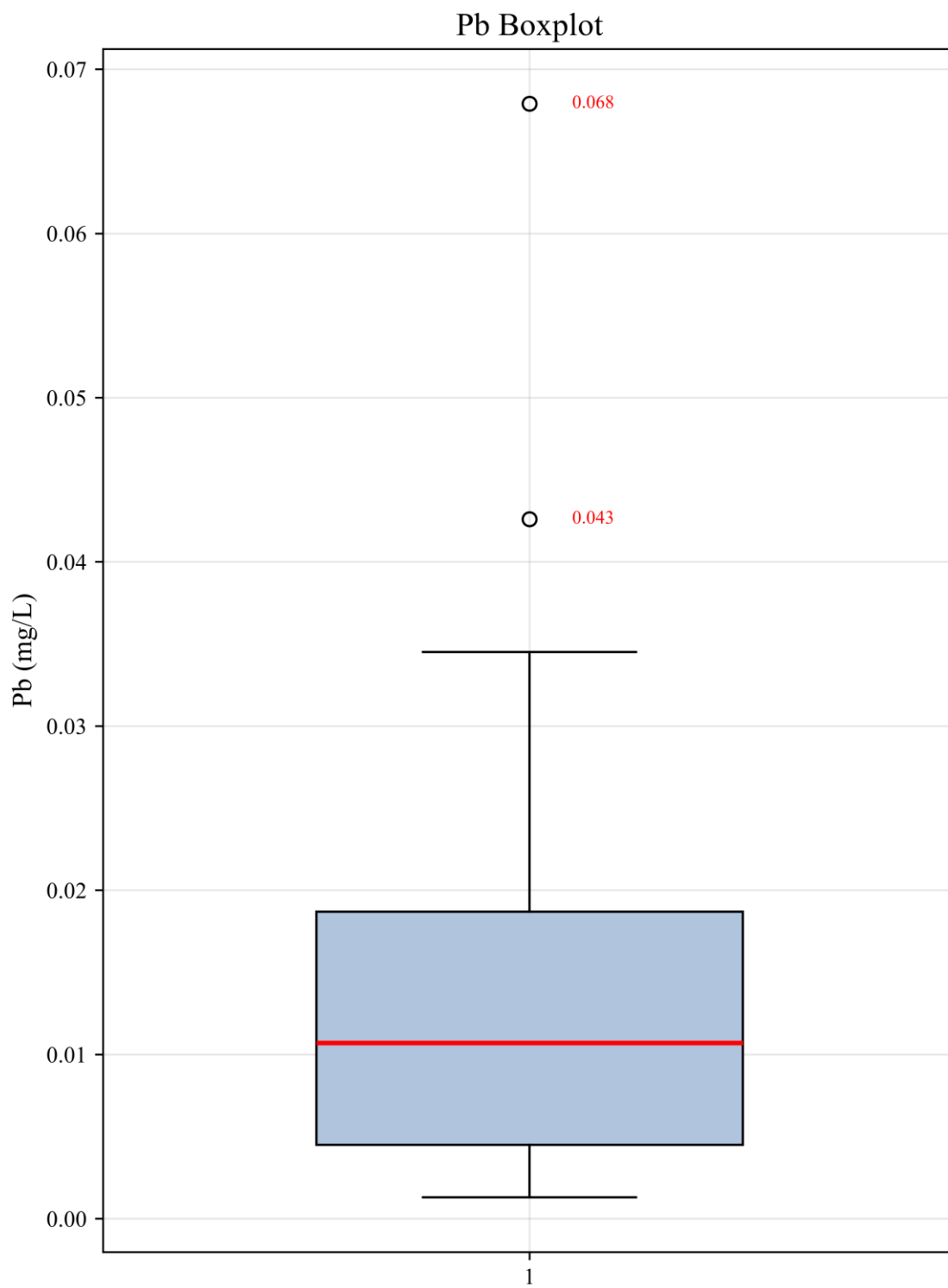


Figure 43. Boxplot illustrating the distribution of Lead (Pb).

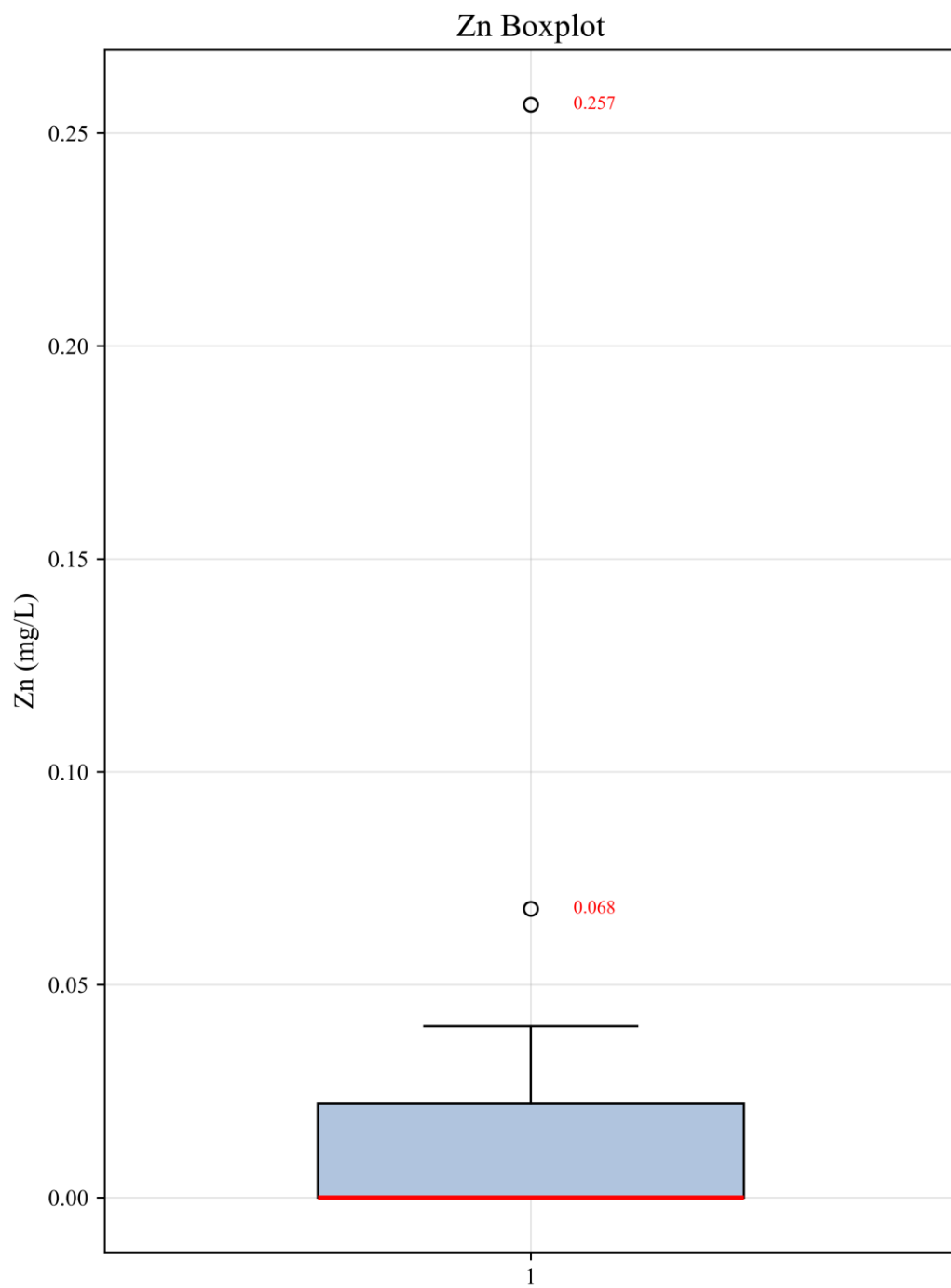


Figure 44. Boxplot illustrating the distribution of Lead (Zn).

Q-Q PLOTS OF SOME INFLUENTIAL METALS IN THE STUDY

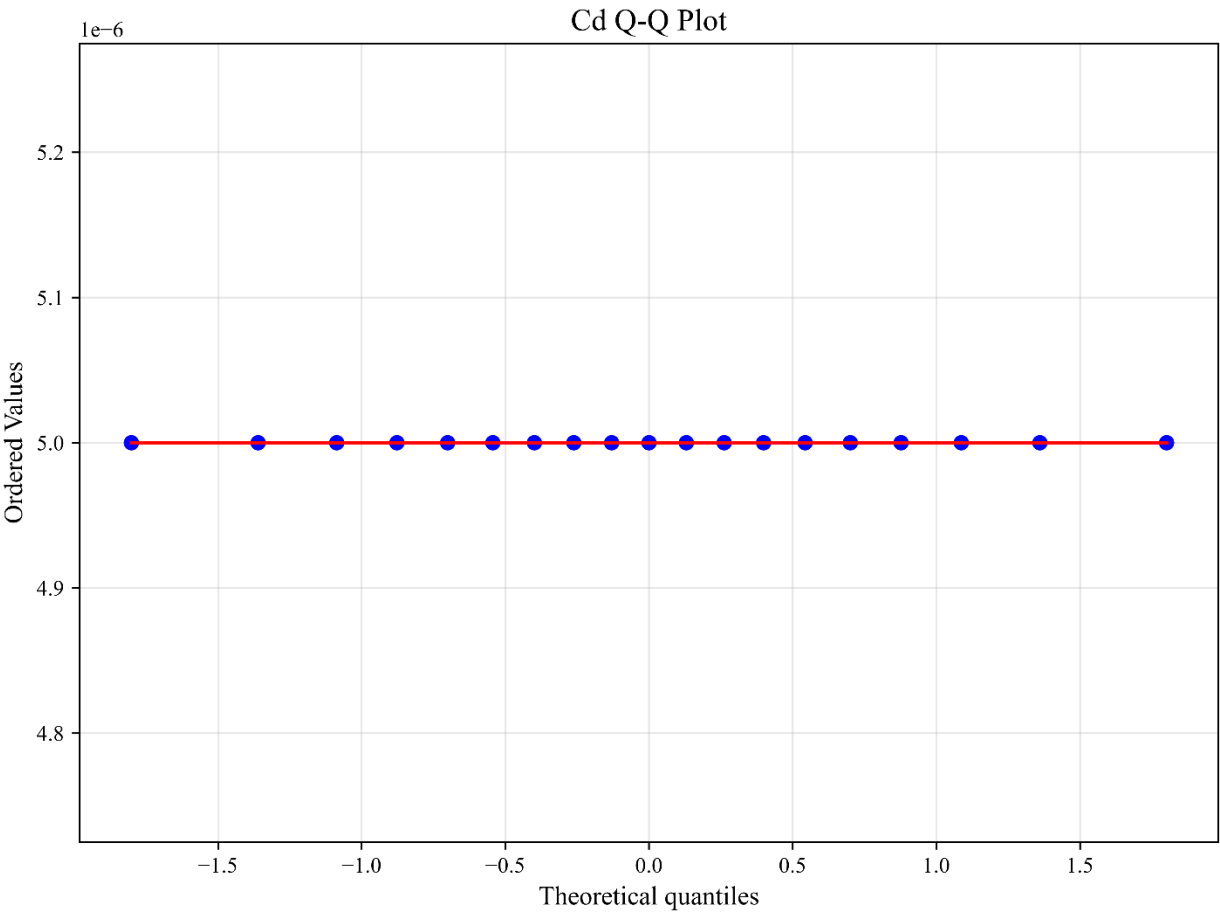


Figure 45. Q-Q plots of Cd

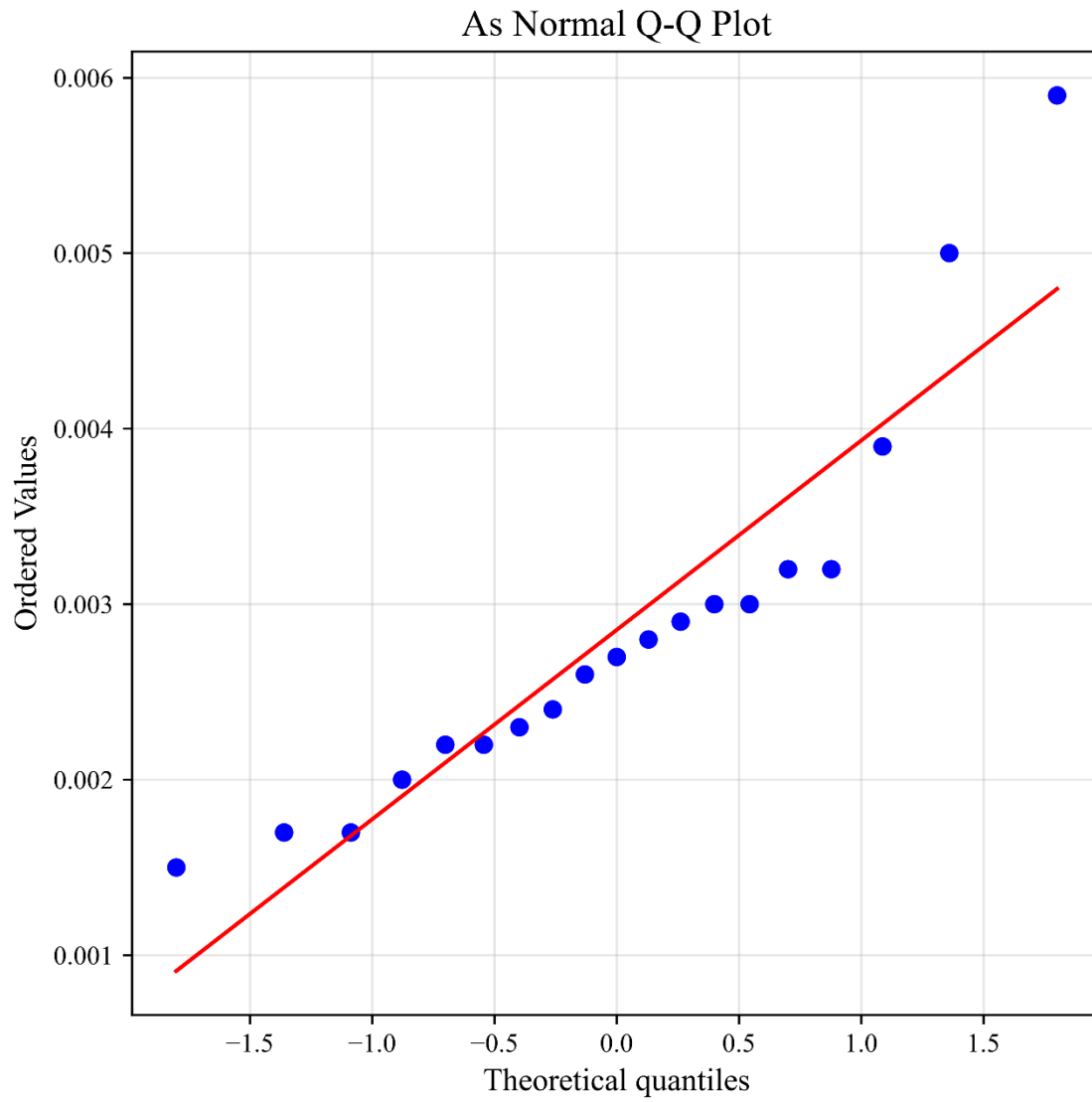


Figure 46. Q-Q plot of As

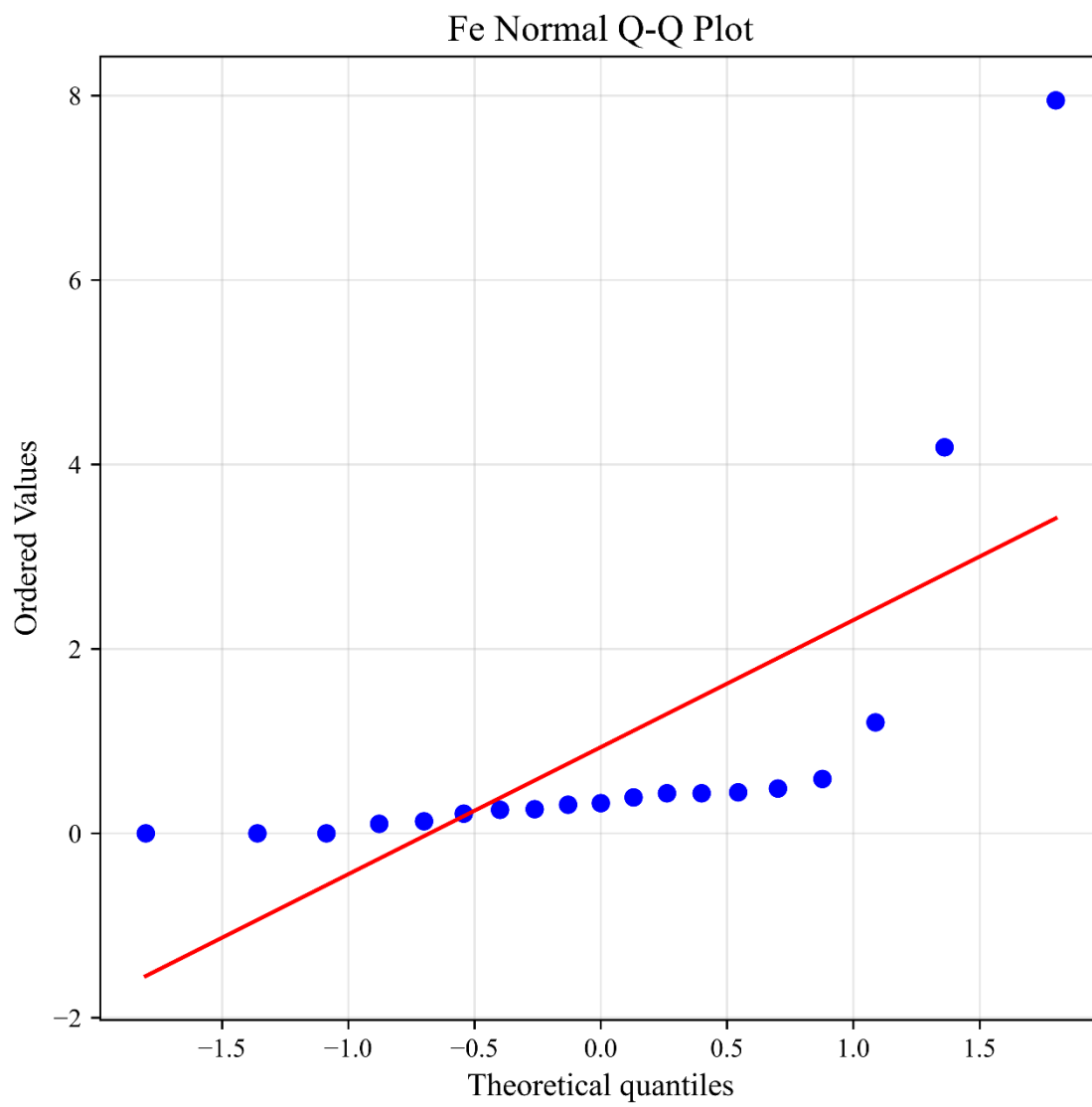


Figure 47. *Q-Q plot of Fe*

VIOLIN PLOTS OF SOME INFLUENTIAL METALS IN THE STUDY

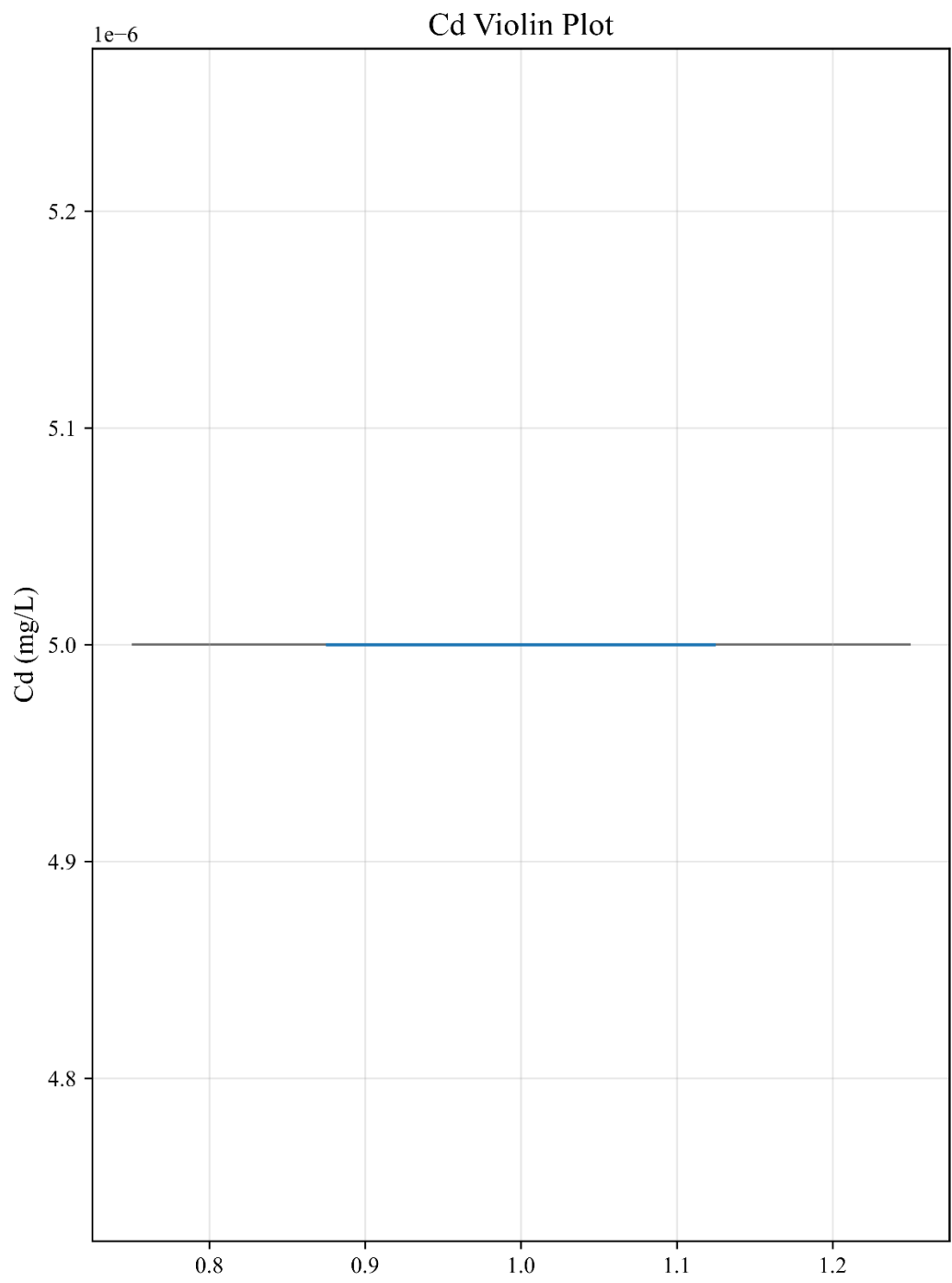


Figure 48. Violin plot of Cd

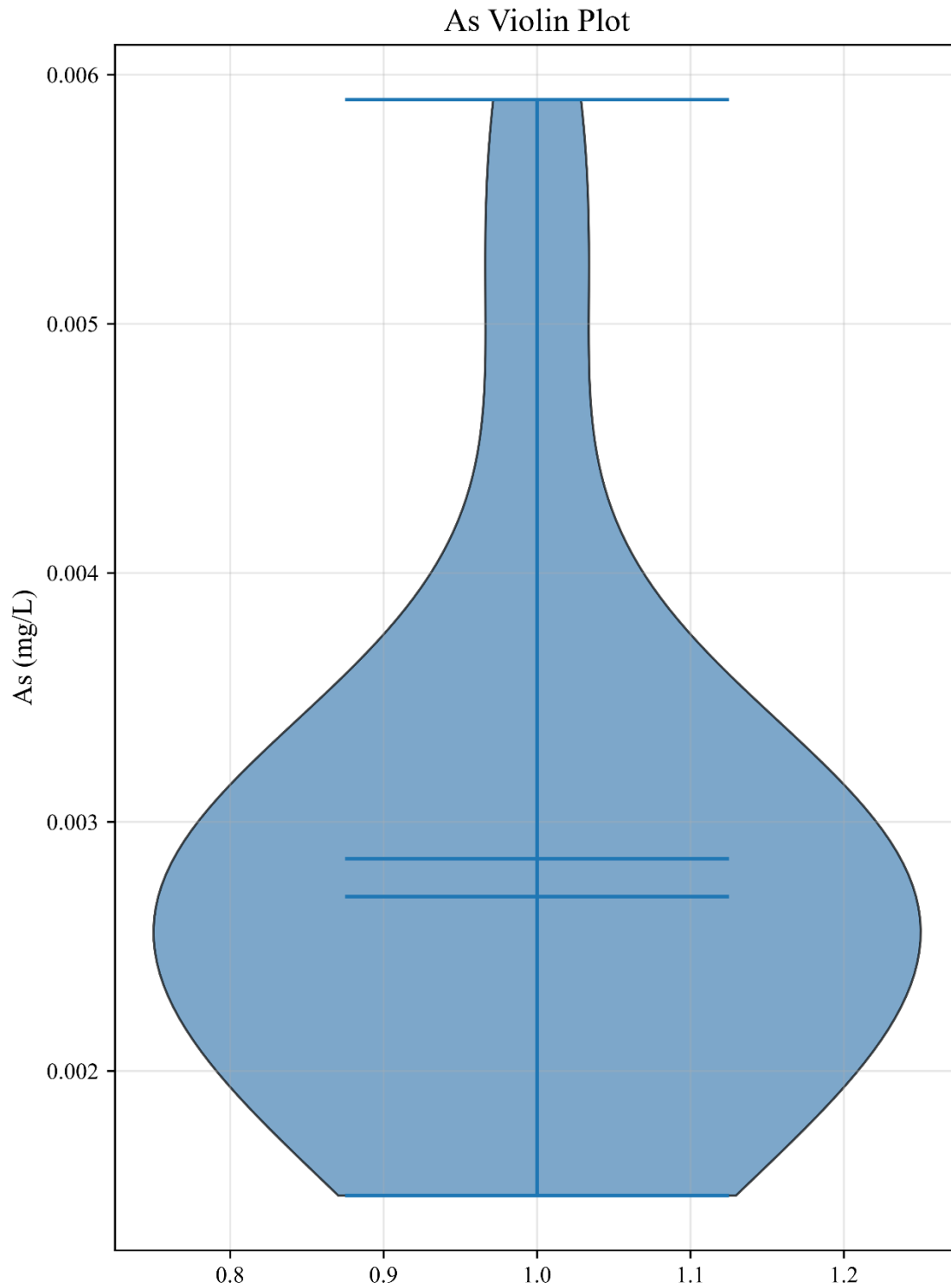


Figure 49. Violin plot of As

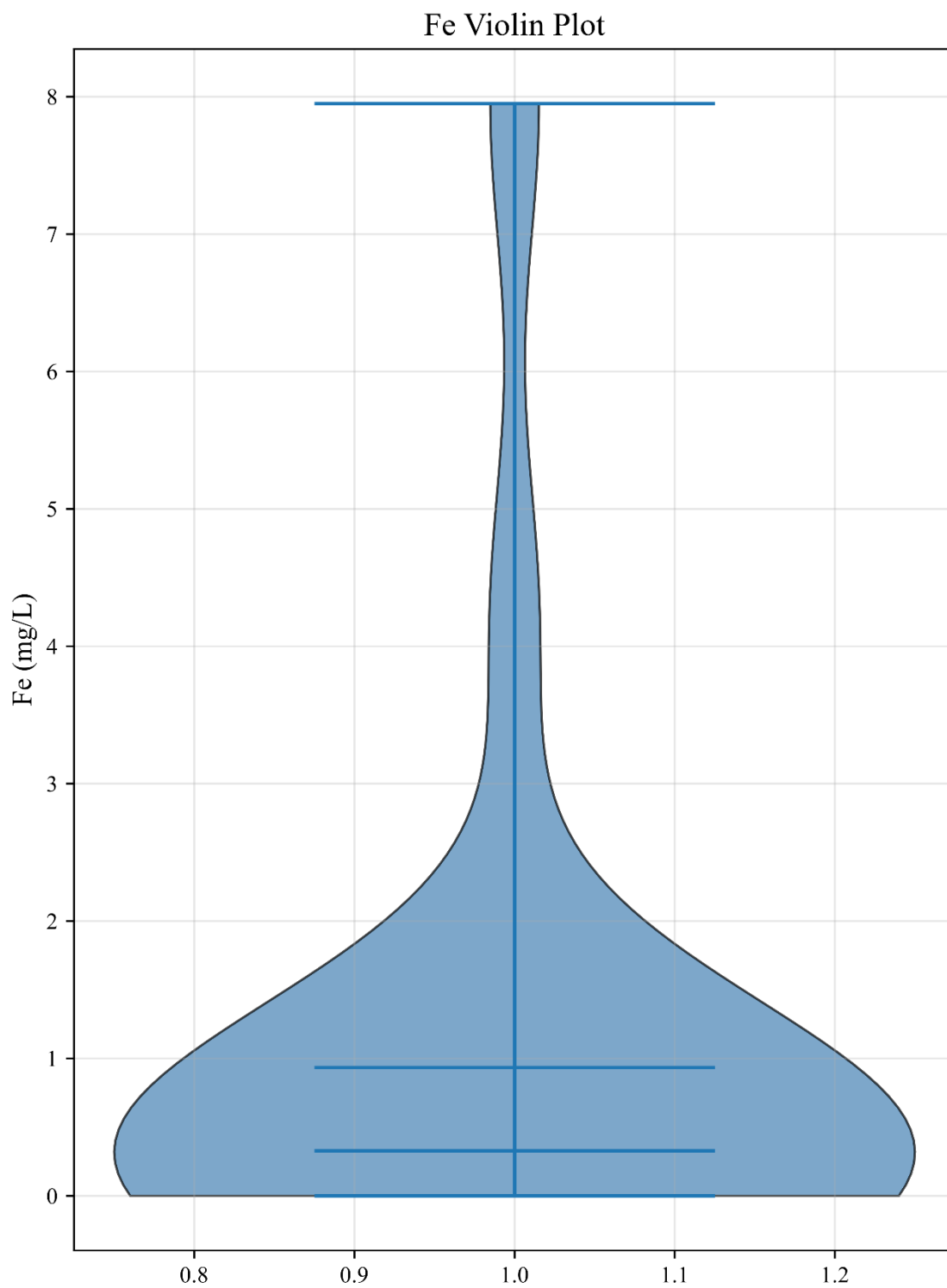


Figure 50. Violin plot of Fe

BOREHOLE LOCATIONS

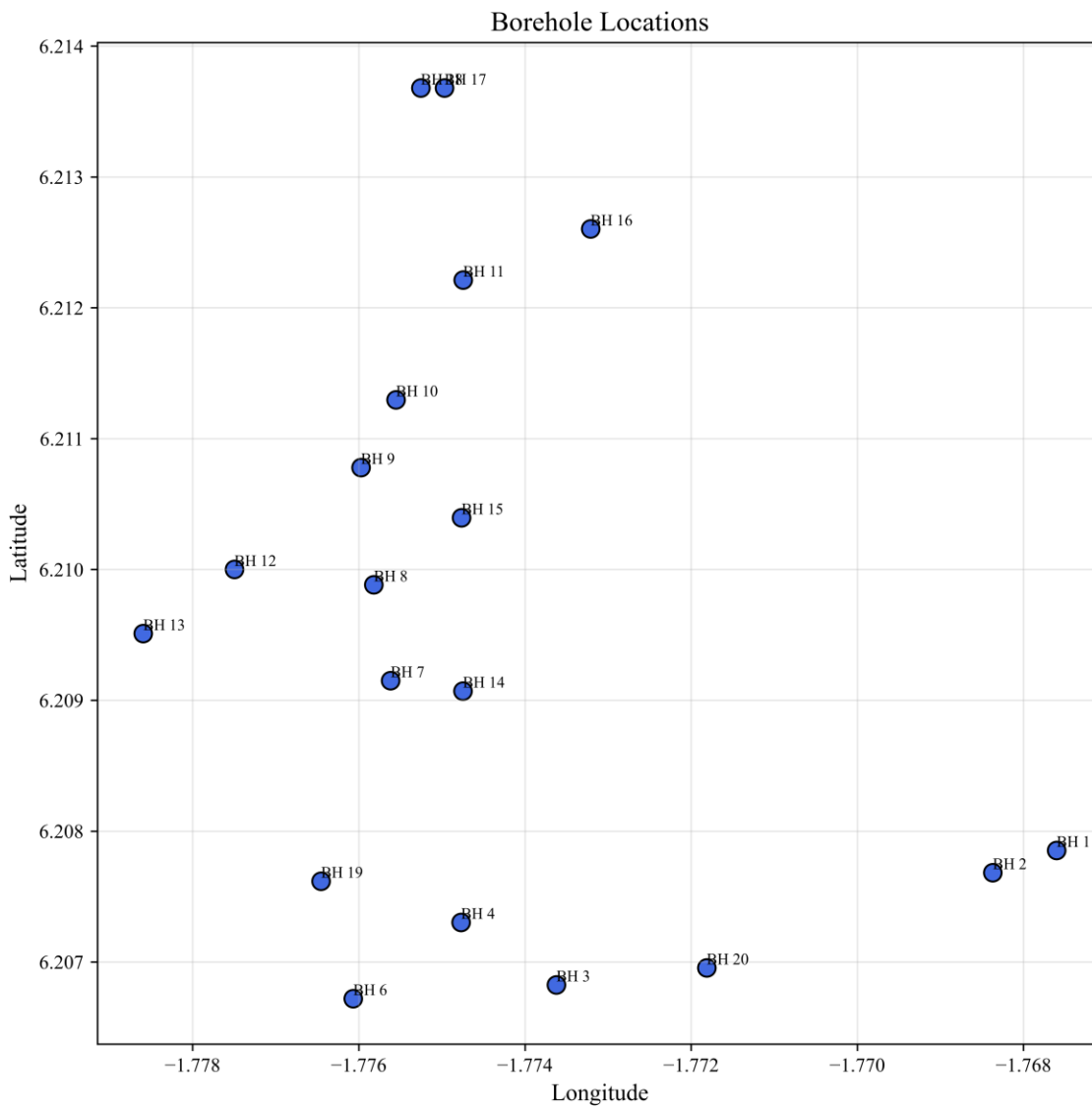


Figure 51. Borehole locations

SITE VISIT



Figure 52. Site image 1



Figure 53. Site image 2



Figure 54. Site image 3